MSc Thesis — Working Draft

End-to-End Calibration: Theory and Empirical Evaluation in Biomedical Context

Suvinava Basak
TU Braunschweig

Supervisor: Prof. Dr. Nicole Mücke
Institute for Statistics and Foundations of Machine Learning
TU Braunschweig

Draft, September 2026



Institute for Statistics and
Foundations of Machine Learning

Master's Thesis

End-to-end Calibration:

Theory and Empirical Evaluation in Biomedical Context

by

Suvinava Basak
Matriculation Nr.: 5413021

Master's Program in Data Science
Carl-Friedrich-Gauß-Fakultät
Technische Universität Braunschweig

October 15, 2026

Examiners : **Prof. Dr. Nicole Mücke**
Institute for Statistics & Foundations of Machine Learning
Technische Universität Braunschweig

Prof. Dr. Tim Kacprowski
Head of Data Science Department in Biomedicine
Technische Universität Braunschweig

Supervisor : **Prof. Dr. Nicole Mücke**
Institute for Statistics & Foundations of Machine Learning
Technische Universität Braunschweig

Technische Universität Braunschweig

Master's Thesis – Task sheet for Suvinava Basak (5413021)

Title (English):	End-to-end Calibration: Theory and Empirical Evaluation in Biomedical Context
Title (German):	End-zu-End-Kalibrierung: Theorie und empirische Evaluierung im biomedizinischen Kontext
Duration:	6 months (15.04.2026 – 15.10.2026)
First examiner:	Prof. Dr. Nicole Mücke
Second examiner:	Prof. Dr. Tim Kacprowski
Supervisor:	Prof. Dr. Nicole Mücke

Overall Research Question:

Can calibration, when embedded into the training objective as a structural constraint, be improved by solving the optimization problem

$$\min_{\theta\in\Theta} \widehat{\mathcal{C}}(f_\theta) + \lambda\widehat{\mathcal{R}}(f_\theta) = \min_{\theta\in\Theta}(1-\lambda)\widehat{\mathcal{C}}(f_\theta) + \lambda\widehat{\mathcal{E}}(f_\theta),$$

while maintaining statistical accuracy (by means of choosing $\lambda > 0$ appropriately)?

Tasks:

- Does augmenting ERM with a calibration objective act as an implicit regularizer, and how does it reshape the bias–variance trade-off relative to predictive-only training followed by post-hoc calibration?
- Can the effect of end-to-end calibration on generalization be characterized within statistical learning theory?
- In simulation-supervised settings with a known true posterior, how do end-to-end calibrated models compare to baselines on exact measures of calibration, refinement, and generalization across model capacity and regularization strength?
- In GRN inference, does end-to-end calibration yield more trustworthy and causally meaningful probabilistic predictions than predictive-only training, without sacrificing the scalability that motivates the use of deep models?

Optional task: What does "calibrated" mean under distribution shift? How does it impact marginal calibration and conditional calibration? Does enforcing calibration during training improve robustness under covariate shift?

Declaration

I hereby confirm that the present master thesis was composed by myself and that the work contained herein is my own. I also confirm that I have prepared this Master's thesis independently and solely by using the sources and resources indicated. All passages and ideas taken directly or indirectly from published or unpublished works, including material obtained from electronic sources and the Internet, have been properly acknowledged and cited in accordance with the principles of good scientific practice.

I further declare that this thesis has not been submitted, either in whole or in part, to any other university or institution for the purpose of obtaining an academic degree or any other qualification.

The electronic version submitted is identical in content to the printed version, if applicable.

I am aware that any false declaration or violation of the principles of academic integrity may result in academic and legal consequences.



..... Braunschweig, 15.10.2026

(Signature) *(Place, Date)*

Abstract

Modern machine learning models achieve remarkable predictive performance, yet their probabilistic outputs are often poorly calibrated, leading to overconfident and unreliable decisions. While post-hoc calibration techniques can adjust predictions after training, they leave the underlying hypothesis class and learning dynamics unchanged. This thesis investigates **End-to-End Calibrated Learning** as a principled alternative: calibration objectives are integrated directly into the training process and treated as structural constraints on the hypothesis space.

We hypothesize that embedding calibration into empirical risk minimization acts as an implicit regularization mechanism, shaping the bias–variance trade-off. Using simulation-supervised settings with controlled ground truth, we aim to theoretically and empirically analyze how end-to-end calibration influences generalization bounds.

As a challenging application domain, we consider the inference of Gene Regulatory Networks (GRNs) from high-dimensional gene expression data. Uncovering directed regulatory relationships is essential for understanding dynamic cellular processes, yet the data is fraught with elevated noise and dropouts. While Bayesian Networks (BNs) offer a rigorous graphical paradigm for modeling pseudo-temporal dynamics and allowing for strict causal reasoning, classical Bayesian structural learning becomes computationally intractable at genome scale. Deep learning provides the necessary scalability, but standard models often optimize for purely predictive metrics (like F1-scores) while failing to capture true causal validity or provide reliable uncertainty estimates. This setting provides a concrete testbed to evaluate whether end-to-end calibration can bridge the gap: ensuring that scalable deep learning can produce the trustworthy, causally-valid probabilistic inference.

Keywords: • Calibration • Deep Learning • Generalization Error • Gene Regulatory Network • Trustworthy AI.

Contents

Acronyms ix

List of Notations x

List of Figures xiv

List of Tables xvi

1 Introduction 1
- 1.1 From Accurate Predictions to Trustworthy Probabilities . . . 1
- 1.2 The Limits of Post-hoc Calibration . . . 2
- 1.3 End-to-End Calibrated Learning . . . 3
- 1.4 Gene Regulatory Network Inference as a Testbed . . . 4
- 1.5 Problem Statement, Research Questions, and Hypotheses . . . 5
- 1.6 Contributions . . . 6
- 1.7 Thesis Outline . . . 7

2 Background and Literature Review 8
- 2.1 Notation and Problem Setup . . . 8
- 2.2 What It Means for a Model to be Calibrated . . . 9
- 2.3 Calibration Under Distribution Shift . . . 11
- 2.4 Calibration Metrics and Calibration–Refinement Decomposition . . . 11
  - 2.4.1 Expected calibration error (ECE) and reliability diagrams . . . 12
  - 2.4.2 The bias of binned estimators . . . 13
  - 2.4.3 Other calibration metrics . . . 13
  - 2.4.4 Proper Losses . . . 15
  - 2.4.5 Calibration-Refinement Decomposition . . . 16
- 2.5 Post-hoc Calibration methods . . . 18
  - 2.5.1 Platt scaling . . . 18
  - 2.5.2 Temperature scaling . . . 19
  - 2.5.3 Beta calibration . . . 19
  - 2.5.4 Histogram binning . . . 19
  - 2.5.5 Isotonic regression . . . 20
  - 2.5.6 Bayesian Binning into Quantiles (BBQ) . . . 20
  - 2.5.7 Dirichlet calibration . . . 20
  - 2.5.8 A shared limitation . . . 21
- 2.6 Calibration During Training (End-to-End) . . . 21
  - 2.6.1 Auxiliary calibration losses . . . 22

2.6.2 Other methods to improve calibration during training . . . 25
2.6.3 Calibration, accuracy, and architecture . . . 26
2.7 Statistical Learning Theory Foundations . . . 26
2.7.1 Empirical risk minimization and the generalization gap . . . 26
2.7.2 Rademacher complexity bounds . . . 26
2.7.3 PAC-Bayesian bounds . . . 27
2.7.4 Calibration as a constraint on the hypothesis space . . . 27
2.8 Gene Regulatory Network Inference . . . 28
2.8.1 Biological background . . . 28
2.8.2 Data and its pathologies . . . 28
2.8.3 Classical and statistical approaches . . . 28
2.8.4 Machine-learning and deep-learning approaches . . . 29
2.8.5 Benchmarking, simulation, and evaluation . . . 30
2.9 Synthesis and Positioning . . . 30

3 End-to-End Calibration as a Structural Constraint 31
3.1 The Composite Calibration-Regularized Objective . . . 31
3.2 The Calibration–Refinement Decomposition, Made Precise . . . 32
3.3 From Penalized to Constrained ERM . . . 33
3.4 Generalization via Rademacher Complexity . . . 33
3.5 A PAC-Bayesian View . . . 34
3.6 Interpretation: Calibration as Capacity Control . . . 35
3.7 Robustness Under Covariate Shift . . . 35
3.8 Assumptions, Scope, and Open Problems . . . 36

4 Simulation-Supervised Experiments 38
4.1 Overview and Objectives . . . 38
4.2 Experimental Setup . . . 40
4.2.1 Data-Generating Process . . . 40
4.2.2 Model and Hypothesis Class . . . 40
4.2.3 Training Objective and Weight Schemes . . . 42
4.2.4 Baselines . . . 43
4.2.5 Evaluation Metrics . . . 44
4.2.6 Implementation and Reproducibility . . . 45
4.3 Results . . . 46
4.3.1 Training Dynamics of the Calibration–Refinement Decomposition . . . 46
4.3.2 Calibration–Refinement Trade-off Across Weights . . . 48
4.3.3 Effect on Generalisation and on Variance Across Seeds . . . 52
4.3.4 Effect of Model Capacity . . . 56
4.3.5 Comparison with Post-hoc Calibration . . . 56
4.3.6 Comparison with Focal Loss . . . 59
4.3.7 Ablations and Sensitivity Analysis . . . 61
4.4 Discussion . . . 66
4.4.1 The Central Question in the Light of the Controlled Experiments . . . 66
4.4.2 RQ1: Calibration as an Implicit Regulariser . . . 68

4.4.3 RQ2: What the Experiments Imply for the Theory 70
4.4.4 RQ3: Summary of the Controlled Comparison 70
4.4.5 Limitations 71
4.4.6 Implications for the GRN Experiments 73

Bibliography **74**

Acronyms

NN	Neural Network
ANN	Artificial Neural Network
ML	Machine Learning
DCA	Difference of Confidence and Accuracy
MSE	Mean Squared Error
MAE	Mean Absolute Error
ERM	Empirical Risk Minimization
BN	Bayesian Network
KDE	Kernel Density Estimation
ECE	Expected Calibration Error
MCE	Maximum Calibration Error
ECE_{em}	Equal-mass ECE
SCE	Static Calibration Error
ACE	Adaptive Calibration Error
TACE	Threshold Adaptive Calibration Error
cwCE	Class-wise ECE
$cwCE_{em}$	Class-wise Equal-mass ECE
RMSCE	Root Mean Squared Calibration Error
KSCE	Kolmogorov–Smirnov Calibration Error
NLL	Negative Log-Likelihood
DECE	Differentiable Expected Calibration Error
MMCE	Maximum Mean Calibration Error
RKHS	Reproducing Kernel Hilbert Space
RLCR	Reinforcement Learning with Calibration Rewards
GRN	Gene Regulatory Network

List of Notations

Symbols are grouped by the section of the thesis in which they are introduced. Where the draft currently uses a symbol inconsistently, this is noted with a footnote marker (†). These instances correspond to the open notation-collision items from the current draft version and should be resolved in the corresponding main `.tex` files before finalizing the thesis report.

Data, Random Variables, and Sample

X, Y	Input and label random variables; $(X, Y) \sim \mathcal{D}$.
$\mathcal{X}$	Input space, $X \in \mathcal{X}$.
$\mathcal{Y}$	Label space. Preferred convention: $Y \in \mathcal{Y} = \{e_1, \dots, e_K\} \subset \Delta_K$ (one-hot / simplex vertices).†
$\mathcal{D}$	Joint data-generating distribution, with density $\pi(x, y) = \pi(y \mid x)\,\pi(x)$.
$\pi(\cdot)$	Density function. Used for the joint density $\pi(x, y)$, the conditional $\pi(y \mid x)$, the marginal $\pi(x)$, and the posterior-as-function $\pi(\cdot \mid x)$.†
Δ_K	Probability simplex, $\Delta_K := \{p \in [0,1]^K : \mathbf{1}^\top p = 1\}$.
K	Number of classes.
e_k	Standard basis (one-hot) vector; the k-th vertex of Δ_K.
n	Sample size. Also appears as m in some sections (e.g. Rademacher complexity subscripts).†
$S = \{(x_i, y_i)\}_{i=1}^n$	Training sample, drawn i.i.d. from D^n.

Predictors and Hypothesis Class

f	Probabilistic classifier, $f : \mathcal{X} \to \Delta_K$.
$f(x) = (f_1(x), \dots, f_K(x))$	Predicted probability vector at input x.
$\hat{y}(x)$	Predicted class, $\arg\max_k f_k(x)$.
$\hat{p}(x)$	Predicted confidence, $\max_k f_k(x)$.
$\eta(x)$	True posterior / conditional class-probability vector, $\eta(x) := \mathbb{E}[Y \mid X = x]$.
$f^\star$	Bayes-optimal predictor. Preferred convention: retain η as primary and state $f^\star = \eta$ explicitly on first use.†
$\mathcal{F}$	Hypothesis class of probabilistic classifiers.

θ, Θ	Model parameters and parameter space; f_θ denotes the parameterized predictor.
$P := f(X)$	Prediction random variable. Note: P is also used for the probability measure $\mathbb{P}(\cdot)$ and, in Ch. 3, for the PAC-Bayes prior — three distinct uses of the same symbol.†
$\gamma(p)$	Calibration function, $\gamma(p) := \mathbb{E}_D[Y \mid P = p]$.

Loss Functions, Scoring Rules, and Calibration–Refinement Decomposition

$\ell(p, y)$	Per-example loss function. Codomain stated inconsistently across the draft ($\mathbb{R}_+$ in Section 2.1 vs. $\mathbb{R} \cup \{\infty\}$ in Definition 2.1).†
$\ell(p, q) := \mathbb{E}_{Y \sim q}[\ell(p, Y)]$	Conditional expectation extension of the loss to a target distribution q.
$\mathcal{E}(f)$	Population risk.
$\hat{\mathcal{E}}(f)$	Empirical risk.
$S(p, y)$	Scoring rule (proper if Eq. 2.13 holds). S is also used for sample.†
$d_\ell(p, q)$	Bregman divergence associated with ℓ ("ℓ-divergence"), $d_\ell(p, q) := \ell(p, q) - \ell(q, q)$.
$e_\ell(q)$	Entropy associated with ℓ ("ℓ-entropy"), $e_\ell(q) := \ell(q, q)$.
$\mathcal{C}_\ell(f)$	Population calibration error (calibration term of the decomposition, Eq. 2.14).
$\mathcal{R}_\ell(f)$	Population refinement error.
$U_\ell(Y)$	Irreducible uncertainty term (Eq. 2.15), independent of f.
$S_\ell(f)$	Sharpness / resolution term (Eq. 2.15).

Calibration Metrics (Evaluation-Time, Non-Differentiable)

ECE	Expected Calibration Error (Eq. 2.6-2.7).
ECE_{prec}	Precision-aware variant of binned calibration error (Eq. 2.8).
MCE	Maximum Calibration Error, worst per-bin gap (Eq. 2.9).
ECE_{em}	Equal-mass (adaptive-binning) ECE.
SCE	Static Calibration Error, class-wise fixed-width binning (Eq. 2.10).
ACE	Adaptive Calibration Error, class-wise equal-mass binning (Eq. 2.11).
$\text{TACE}(t)$	Thresholded Adaptive Calibration Error, threshold t (Eq. 2.12).
RMSCE	Root-Mean-Squared Calibration Error (L_2 per-bin gap).
KSCE	Kolmogorov–Smirnov Calibration Error (binning-free).
B_m	m-th bin of a partition $\mathcal{B} = \{B_1, \ldots, B_M\}$ of $[0, 1]$.
$\text{conf}(B_m)$, $\text{acc}(B_m)$, $\text{prec}(B_m)$	Mean confidence, accuracy, precision of predictions in bin B_m.

M	Number of (fixed-width) bins.
R	Number of adaptive (equal-mass) ranges.
t	Threshold parameter of TACE.

Differentiable Calibration Estimators (Training-Time)

$w_{j,i}$	Soft bin-membership weight of sample i to bin j (Eq. 2.23).
c_j	Fixed center of soft bin j.
σ	Kernel bandwidth (soft-binning / MMCE kernels).
$\mathcal{L}_{\mathrm{DECE}}$	Differentiable (kernel-smoothed / soft-binned) ECE loss (Eq. 2.24).
$\kappa(c_i, c_j)$	Differentiable kernel function (typically Laplacian).
$\mathcal{L}_{\mathrm{MMCE}}$	Maximum Mean Calibration Error loss (Eq. 2.26).
r_i	Correctness indicator, $r_i = \mathbf{1}[\hat{y}_i = y_i]$.
$CE_p(f)$	L_p canonical calibration error (Eq. 2.27).
κ_{Dir}	Dirichlet kernel used for KDE over the simplex (Eq. 2.29).
α_i	Dirichlet concentration parameters, $\alpha_i = f(x_i)/h + 1$.
h	KDE bandwidth parameter ($h \to 0$ as $n \to \infty$).
$\widehat{CE_p}(f)$	KDE-based empirical estimator of $CE_p(f)$ (denoted $\mathrm{ECE}_{\mathrm{KDE}}$ in the text).

Statistical Learning Theory

$\hat{\mathfrak{R}}_m(\mathcal{G})$	Empirical Rademacher complexity of function class $\mathcal{G}$ on a sample of size m (Eq. 2.32).
σ_i	i.i.d. Rademacher (± 1) random variables.
δ	Confidence parameter in generalization bounds (bound holds w.p. $\geq 1 - \delta$).
P, Q	PAC-Bayes prior and posterior distributions over hypotheses (Subsection 2.7.3, Eq. 2.34). Distinct from $P := f(X)$ above.†
$\mathrm{KL}(Q\|P)$	Kullback–Leibler divergence between posterior and prior.

Composite Calibration-Regularized Objective (Chapter 3)

$\mathcal{L}_\lambda(f_\theta)$	Composite training objective, $\mathcal{L}_\lambda = (1-\lambda)\hat{\mathcal{C}} + \lambda\hat{\mathcal{E}}$ (Eq. 3.1).
$\lambda \in (0, 1]$	Calibration/refinement trade-off weight. $\lambda = 1$ recovers plain proper-loss ERM.
$\hat{\mathcal{C}}(f)$	Empirical, differentiable calibration surrogate (instantiated as Soft-ECE, MMCE, or $\mathrm{ECE}_{\mathrm{KDE}}$).

$\hat{\mathcal{R}}(f) := \hat{\mathcal{E}}(f) - \hat{\mathcal{C}}(f)$	Empirical refinement, defined as the residual.
$\hat{f}_\lambda$	Calibration-regularized ERM estimator, $\hat{f}_\lambda \in \arg\min_\theta \mathcal{L}_\lambda(f_\theta)$.
$\mu = (1-\lambda)/\lambda$	Equivalent penalty weight in the penalized (Lagrangian) formulation.
$\mathcal{F}_c$	Calibration-restricted hypothesis class, $\mathcal{F}_c := \{f \in \mathcal{F} : \hat{\mathcal{C}}_\ell(f) \leq c\}$ (Eq. 3.3).
$c = c(\mu)$	Constraint level corresponding to penalty weight μ; non-increasing in μ.
$\ell \circ \mathcal{F}$	Induced loss class, $\{(x, y) \mapsto \ell(f(x), y) : f \in \mathcal{F}\}$.
$D_{\text{train}}, D_{\text{test}}$	Training and test distributions (Section 2.3, 3.7); equal under the i.i.d. assumption, may differ under covariate shift.

† *Marks a symbol whose usage is currently inconsistent across the draft. Needs to be reviewed.*

List of Figures

1.1 Confidence histograms and Reliability Diagrams 2
1.2 A conceptual schematic of the GRN inference pipeline 5

4.1 Approximation error of $\sigma(P_m)$ relative to η as a function of the degree m, for the noiseless Legendre projection of $g = \sigma^{-1} \circ \eta$ (no training data involved). Dotted vertical lines mark the primary capacities $m \in \{10, 100, 1000\}$. 41
4.2 Calibration–refinement decomposition of the cross-entropy over 1,200 epochs, for $m \in \{10, 100, 1000\}$ (single seed). Top block: predictive-only baseline ($\lambda = 1$). Bottom block: end-to-end training with $\lambda = 0.1$. Within each block, the upper row is the training set and the lower row the test set. The figure is cropped from the full grid, which covers $\lambda \in \{0, 0.1, 0.5, 1\}$ for both the CE and the Brier base loss and is available on GitHub. 47
4.3 Confidence histograms (upper rows) and reliability diagrams (lower rows) at epochs 1, 50, 100, 200, 500 and 1,200 for the CE base loss (single seed). Top block: $\lambda = 1$. Bottom block: $\lambda = 0.1$. Shaded areas show the gap to the diagonal. The legend in the last panel of each block gives the final ECE. The snapshot at epoch 500 replaces the epoch 400 listed in the experiment plan. Cropped from the full figure with $\lambda \in \{0, 0.1, 0.5, 1\}$ and both base losses, available on GitHub. 49
4.4 Learned function $p_\theta(x)$ (solid) against the true posterior $\eta(x)$ (dashed) at the final epoch, CE base loss, for $m \in \{10, 100, 1000\}$ (rows) and $\lambda \in \{0, 0.1, 0.5, 1\}$ (columns), single seed. The corresponding figure for the Brier base loss is available on GitHub. 50
4.5 MSE_η, top-label ECE and Brier score at the final epoch as functions of λ (21-point grid, single seed). Colour encodes capacity and line style the base loss. 51
4.6 Hard-binned Brier decomposition (total = calibration + refinement) at the final epoch as a function of λ, for the same runs as Figure 4.5. Every model is scored with the Brier decomposition, whichever loss it was trained with. 51
4.7 Distribution over 30 seeds of test CE, Brier score, MSE_η and ECE for the CE base loss, grouped by λ, with one box per capacity (median, interquartile range, whiskers without outliers). 54
4.8 Test ECE of the raw baseline, the four post-hoc methods and end-to-end training ($\lambda = 0.1$) across the full capacity grid, for the CE (top) and Brier (bottom) base losses (single seed, single 80/20 split). The baseline is trained on 80 points and the maps are fitted on the remaining 20; the end-to-end model uses all 100 points. 58
4.9 Bias (left; mean $\pm$ standard deviation over 30 seeds) and across-seed variance (right) of TS-based and isotonic refinement estimators relative to the ground-truth refinement, for CE baseline models. 60

4.10 ECE (top) and MSE_η (bottom) for focal loss with $\gamma \in \{0, 1, 2, 5\}$ and for end-to-end training ($\lambda = 0.05$, CE base loss), mean $\pm$ standard deviation over 30 seeds. 61
4.11 Ablation A: MSE_η (top) and ECE (bottom) for the baseline, end-to-end training at $\lambda \in \{0.05, 0.1, 0.5\}$ (dark) and plain CE at the matched learning rate 0.5λ (light), over 30 seeds. 63
4.12 Ablation B: MSE_η, ECE and Brier calibration term for the additive objective $\widetilde{E} + \beta\,\widetilde{C}_{\mathrm{soft}}$ at the fixed learning rate 0.5 (mean $\pm$ standard deviation, 30 seeds). $\beta = 9$ and $\beta = 19$ correspond to $\lambda = 0.1$ and $\lambda = 0.05$ at a correspondingly reduced learning rate. 65
4.13 Ablation C: distribution of the validation-selected λ over 30 seeds (left), and test MSE_η (middle) and ECE (right) for end-to-end training with validation-selected λ and for CE with validation early stopping, trained on 80 points or refitted on all 100 (mean $\pm$ standard deviation). 67
4.14 Ablation D: implied convex weight $\lambda_{\mathrm{eff}} = w_E/(w_C + w_E)$ (left, logarithmic scale) and log-variance s_C of the calibration term (right) during training under uncertainty weighting (mean and 10–90% band over 30 seeds). The dashed line marks the lower clipping bound. 69

List of Tables

2.1 Notions of calibration 10
2.2 Calibration metrics 15
2.3 Divergences and entropy from loss functions ℓ 17
2.4 Post-hoc calibration methods 21
2.5 Differentiable calibration estimators 25
2.6 Families of GRN inference methods 29

4.1 Experiment questions of Track 1, the experiment that answers each of them, and where the result is reported. "1 seed" refers to data seed 1 and initialisation seed 42; "30 seeds" to data seeds $s = 1, \ldots, 30$ with initialisation seed $10000 + s$. 39
4.2 Mean and across-seed variance of MSE_η at $m = 1000$ (CE base loss, 30 seeds). Decreasing λ means increasing weight on the calibration term. 53
4.3 Final-epoch test metrics, CE base loss, mean $\pm$ standard deviation over 30 seeds. Markers on the $\lambda = 0.1$ rows give the result of a paired Wilcoxon test against the baseline $\lambda = 1$, Bonferroni-corrected over the three capacities ($^{*}\, p < 0.05$, $^{**}\, p < 0.01$, n.s. not significant). All marked differences favour $\lambda = 0.1$. 55
4.4 Test ECE before and after post-hoc calibration, and for end-to-end training ($\lambda = 0.1$), by base loss and capacity (single seed, single 80/20 split). 59
4.5 Refinement estimators relative to the ground-truth refinement (30 seeds, CE baseline, Brier-based estimators, 20 calibration points). Bias is mean $\pm$ standard deviation of (estimate $-$ ground truth); variance is the across-seed variance of the estimate. The last column tests $|\mathrm{bias}_{\mathrm{TS}}|$ against $|\mathrm{bias}_{\mathrm{iso}}|$ (paired Wilcoxon, Bonferroni over three capacities). 60
4.6 Focal loss and end-to-end training ($\lambda = 0.05$), mean $\pm$ standard deviation over 30 seeds. The best mean per capacity and column is in bold. Paired Wilcoxon tests of end-to-end training against each γ (Bonferroni over 12 comparisons per metric) are significant at $p < 0.01$ for ECE and MSE_η in all cases except ECE at $m = 1000$, $\gamma = 1$ (n.s.). 62
4.7 Ablation A: end-to-end training against plain CE at the matched learning rate 0.5λ (mean $\pm$ standard deviation, 30 seeds). Markers on end-to-end rows compare with the control directly below (paired Wilcoxon, Bonferroni over nine comparisons per metric). $^{*}/^{**}$: significantly better; $^{\dagger}/^{\dagger\dagger}$: significantly worse; unmarked: not significant. 64
4.8 Ablation B: additive penalty at learning rate 0.5 (mean $\pm$ standard deviation, 30 seeds). $\beta = 0$ is the CE baseline. Markers compare each $\beta > 0$ with $\beta = 0$ (paired Wilcoxon, Bonferroni over 15 comparisons per metric); ††: significantly worse at $p < 0.01$; unmarked: not significant. 66

4.9 Ablation C: validation-based pipelines (20 validation points, criterion: Brier score; mean $\pm$ standard deviation, 30 seeds). Markers on end-to-end rows compare with the early-stopping row of the same version (80 points or refit; paired Wilcoxon, Bonferroni over six comparisons per metric). The last two rows per capacity are references: the oracle $\lambda = 0.1$, and plain CE trained for all 1,200 epochs, both on 80 points. 68
4.10 Ablation D: uncertainty weighting compared with the baseline and end-to-end training at $\lambda = 0.1$ (mean $\pm$ standard deviation, 30 seeds). ††: significantly worse than both references at $p < 0.01$ (paired Wilcoxon, Bonferroni over six comparisons per metric). 69
4.11 Summary of the answers to the Track 1 experiment questions. 72

1 Introduction

1.1 From Accurate Predictions to Trustworthy Probabilities

Over the past decade, machine learning models (and deep neural networks in particular) have achieved remarkable predictive accuracy across domains ranging from image recognition to natural language understanding. Yet accuracy alone is an incomplete description of a model's usefulness. In any setting where a prediction feeds into a downstream decision under uncertainty, what matters is not only *whether* the model is right on average, but *how much its stated confidence can be trusted*. A model that assigns a probability of 0.9 to an event should be correct about ninety percent of the time it makes such a claim; when this correspondence holds, the model is said to be *calibrated*.

Calibration is a prerequisite for rational decision-making, especially in clinical or biomedical context. A clinician deferring borderline cases to further testing, an autonomous controller choosing when to cede control to a human, and a scientist deciding which predicted interactions are worth the cost of experimental validation all rely on the numerical confidence attached to a prediction, not merely on the predicted label [1, 2]. If those confidences are systematically too high, the decision system inherits the model's overconfidence and acts on evidence that is weaker than it appears.

Unfortunately, modern neural networks are frequently miscalibrated, despite often being accurate. In an influential study, Guo *et al.* [1] showed that contemporary deep architectures, despite being far more accurate than their predecessors, are substantially more overconfident: their average predicted confidence considerably exceeds their empirical accuracy. The phenomenon was attributed to factors such as increased depth and width, weight decay, and the use of normalization layers, all of which improve discrimination while distorting the probability scale. Subsequent work has refined this picture. Minderer *et al.* [3] revisited the question for newer architectures and found that some recent, non-convolutional models are in fact among the best calibrated, suggesting that architectural design is itself a determinant of calibration. Broad benchmark studies have since catalogued how calibration interacts with accuracy, robustness, and architecture across tens of thousands of models [4]. The collective message is twofold: miscalibration is neither universal nor inevitable, but it is common enough, and consequential enough, that it must be treated as a first-class objective rather than an afterthought.

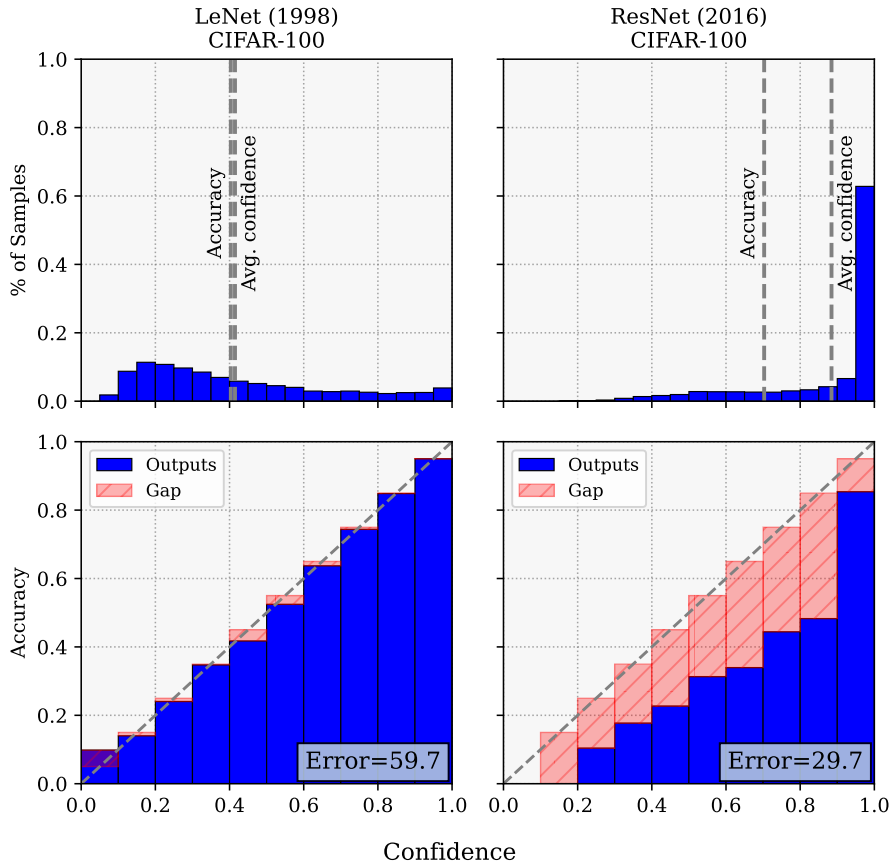


Figure 1.1: Confidence histograms (top) and reliability diagrams (bottom) for a 5-layer LeNet (left) and a 110-layer ResNet (right) on CIFAR-100, reproduced from [1].

1.2 The Limits of Post-hoc Calibration

The dominant response to miscalibration has been *post-hoc calibration*: train a model in the usual way to maximize predictive performance, then adjust its outputs afterwards using a small held-out dataset. Classical instances include Platt scaling, which fits a sigmoid to a classifier's scores [5]; histogram binning and isotonic regression, which map scores to empirical frequencies [6]; Bayesian Binning into Quantiles (BBQ), a non-parametric method that averages over binning schemes [7]; and temperature scaling, a single-parameter simplification of Platt scaling that Guo *et al.* [1] found to be surprisingly effective for deep networks. These methods are attractive precisely because they are model-agnostic and inexpensive: they treat the trained predictor as a black box and reshape only its output probabilities.

This modularity, however, is also their central limitation. Post-hoc recalibration operates on a frozen model and therefore leaves the *hypothesis class* and the *learning dynamics* unchanged. It rescales the probabilities a model already produces, but it cannot alter the representation the model learned, the decision boundary it settled on, or the inductive biases that shaped them. Three consequences

follow. *First*, since it does not change the representation, the expressive power of the correction is limited. Temperature scaling, for instance, can only apply a monotone, global transformation and cannot repair calibration that varies across regions of input space. *Second*, the corrected calibration is fragile under distribution shift, because a transformation fit on one distribution need not transfer to another [3]. *Third*, and most fundamentally, calibration is decoupled from representation learning. Since it is treated as an afterthought of training, the model is never given any incentive during training to organize its internal computation in a way that yields meaningful probabilities.

A recent line of work makes this tension explicit. Proper losses such as cross-entropy admit a decomposition into a *calibration* term and a *refinement* (or sharpness) term. Berta *et al.* [8] demonstrate, both theoretically and empirically, that these two terms are not minimized simultaneously during training. Selecting a model by validation loss thus lands on a compromise that is optimal for neither. Their proposed remedy is to optimize refinement during training and to handle calibration afterwards. While effective, this reaffirms that, under standard training, calibration and the learned representation pull in different directions. It is natural to ask the converse question: *what happens if calibration is enforced during the training via a differentiable penalty folded into the training objective*, so that the learner is shaped by it from the start?

1.3 End-to-End Calibrated Learning

This thesis investigates **end-to-end calibrated learning**: the integration of calibration objectives directly into the training process, where they act not as a post-processing correction but as a *structural constraint on the hypothesis space*. Rather than choosing a predictor by predictive risk alone and then recalibrating, we minimize a composite objective in which a calibration penalty participates in empirical risk minimization, thereby influencing which hypothesis is selected.

Several ingredients make this concrete. The principal technical obstacle is that the standard calibration metric, the expected calibration error (ECE), estimated by binning predictions, is piecewise constant and hence has zero gradient almost everywhere, so it cannot be optimized directly. A series of works has proposed differentiable surrogates that overcome this. The Difference of Confidence and Accuracy (DCA) term penalizes the gap between mean confidence and accuracy as an auxiliary loss [2]; the Maximum Mean Calibration Error (MMCE) recasts calibration in a reproducing kernel Hilbert space, yielding a trainable measure [9]; soft (continuous) binning replaces hard bin assignments with smooth weights, making the calibration error itself differentiable [10]; and consistent kernel-density estimators of the L_p canonical calibration error provide low-bias, differentiable penalties suitable for direct risk regularization [11]. Beyond supervised classification, reinforcement-learning approaches augment a correctness reward with a bounded proper scoring rule (such as the Brier score) to train models that are simultaneously accurate and calibrated [12]; notably, the boundedness of the scoring rule is a deliberate design choice, since unbounded rules such as the log-loss can be dominated by a few confidently wrong predictions.

The conceptual claim of this thesis is that incorporating such a calibration objective into empirical risk minimization functions as a form of *implicit regularization*. By requiring that the model's confidences agree with empirical frequencies, the calibration term restricts the set of admissible

solutions: it excludes hypotheses that fit the training labels while making over- or under-confident probabilistic claims. In this view, calibration is an *inductive bias* that reshapes the bias–variance trade-off, much as weight decay or early stopping do, and it should therefore be analyzable through the lens of generalization theory. We hypothesize that, under suitable conditions, end-to-end calibration reduces the effective capacity exploited by the learner and can be related to tighter generalization guarantees. Establishing the precise form of this relationship, and testing it where ground truth is known, is the theoretical motivation of this thesis.

We deliberately situate the empirical study in *simulation supervised* settings: data generating processes in which the true conditional distribution (and hence the perfectly calibrated predictor) is known by construction. Such settings are indispensable here because calibration quality and generalization gaps can only be measured exactly when the ground truth is available, which is never the case for real data alone.

1.4 Gene Regulatory Network Inference as a Testbed

To stress-test these ideas in a realistic and scientifically meaningful setting, we turn to the inference of *gene regulatory networks* (GRNs). A GRN is a directed graph whose nodes are genes and whose edges encode regulatory influence: typically the action of a transcription factor on the expression of a target gene. Reconstructing these networks from high-throughput expression measurements is a central problem in computational biology, because the wiring of regulatory circuits underlies cellular differentiation, response to stimuli, and disease [13, 14]. The task is also a natural home for probabilistic reasoning: edges are uncertain, evidence is indirect, and the practical question is almost always *which predicted interactions are confident enough to pursue experimentally*: a question that is meaningful only if the predicted probabilities are calibrated.

GRN inference is demanding for three reasons that make it an ideal testbed. *First*, the data are high-dimensional and severely corrupted. Modern single-cell RNA-sequencing assays measure thousands of genes but suffer from elevated technical noise, sparsity, and "dropout" events in which expressed transcripts are recorded as zero; tellingly, simulation studies show that such technical noise can dominate the accuracy of network inference [15]. *Second*, the field offers a rigorous but computationally forbidding classical paradigm. Bayesian networks model the joint distribution of gene expression as a directed graphical model and support principled reasoning about conditional independence and, under assumptions, causal structure and pseudo-temporal dynamics [13]. However, learning the structure of a Bayesian network from data is, in general, NP-hard, and exact structure learning does not scale to the thousands of genes assayed in genome-scale studies [16]. *Third*, the methods that *do* scale tend to optimize the wrong thing. Tree-ensemble regressors such as GENIE3 [17] and its gradient-boosted successor GRNBoost2 [18], information-theoretic scores [19], and a growing family of deep models (convolutional classifiers of gene-pair co-expression [20] and graph attention networks [21]) are evaluated almost exclusively on *ranking-based predictive metrics* such as the areas under the ROC and precision–recall curves [14, 22]. These metrics reward correct ordering of candidate edges but say nothing about whether the associated probabilities are trustworthy, nor whether the recovered edges reflect genuine regulatory (causal) structure rather than mere statistical association.

This sets up the gap that the thesis targets. Deep learning supplies the scalability that classical Bayesian structure learning lacks, but standard deep models, trained for predictive accuracy and judged by predictive metrics, provide neither calibrated uncertainty nor any guarantee of causal validity. GRN inference is therefore a concrete arena in which to ask whether end-to-end calibration can close the gap: whether a scalable, deep predictor, trained with calibration as a structural objective, can produce the trustworthy, causally meaningful probabilistic inferences that the application actually requires.

Crucially for the methodological aims of this thesis, GRN inference comes equipped with controlled, simulation-supervised benchmarks. The DREAM5 challenge provides standardized expression datasets with known gold-standard networks and established evaluation protocols [22]; the BEELINE framework systematizes the evaluation of single-cell GRN methods against curated and synthetic ground truth [14]; and the SERGIO simulator generates single-cell expression data from a user-specified GRN, modelling the stochastic and noisy nature of transcription [15]. These resources allow us to construct settings in which the true network, and hence the perfectly calibrated predictor, is known, so that calibration and generalization can be measured rather than merely estimated.

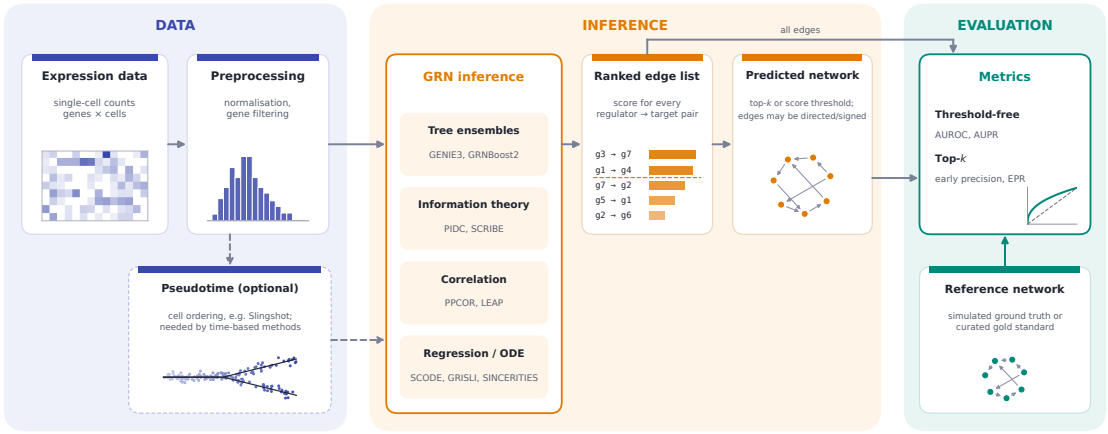


Figure 1.2: A conceptual schematic of the GRN inference pipeline: expression matrix (genes × cells) → learned model → predicted edge probabilities → comparison against a gold-standard reference network. This shows calibration governs which predicted edges are "trustworthy".

1.5 Problem Statement, Research Questions, and Hypotheses

The thesis is organized around a single overarching question:

> *Can calibration, when embedded into the training objective as a structural constraint, improve the generalization and trustworthiness of scalable probabilistic predictors—and can this effect be both characterized theoretically and validated empirically in*

simulation-supervised settings, including gene regulatory network inference?

We decompose this into the following research questions.

RQ1 (Mechanism). Does augmenting empirical risk minimization with a calibration objective act as an implicit regularizer, and how does it reshape the bias–variance trade-off relative to predictive-only training followed by post-hoc calibration?

RQ2 (Theory). Can the effect of end-to-end calibration on generalization be characterized within statistical learning theory, for example, by relating the calibration-constrained hypothesis space to complexity-based or PAC-Bayesian generalization bounds?

RQ3 (Controlled validation). In simulation-supervised settings with a known true posterior, how do end-to-end calibrated models compare to baselines on exact measures of calibration, refinement, and generalization across model capacity and regularization strength?

RQ4 (Application). In GRN inference, does end-to-end calibration yield more trustworthy and causally meaningful probabilistic predictions than predictive-only training, without sacrificing the scalability that motivates the use of deep models?

These questions are guided by the central **hypothesis** that embedding calibration into empirical risk minimization constrains the hypothesis space in a way that behaves like regularization, and that this constraint can, under appropriate conditions, improve generalization while producing reliable uncertainty estimates.

1.6 Contributions

This thesis makes the following contributions.

1. **A unifying framing of end-to-end calibration as a structural constraint.** We consolidate differentiable calibration objectives [2, 9, 10, 11] and reward-based proper-scoring-rule formulations [12] into a single perspective in which calibration acts as an inductive bias on the hypothesis space, and we articulate why bounded proper scoring rules are preferable at training time.

2. **A theoretical analysis linking calibrated training to generalization.** We study how a calibration constraint interacts with classical generalization machinery (complexity-based and PAC-Bayesian bounds) and with the calibration–refinement decomposition of proper losses [8], clarifying the sense in which calibration can tighten the bias–variance trade-off.

3. **A controlled, simulation-supervised study.** Using data-generating processes with a known true posterior, we empirically characterize how end-to-end calibration affects calibration error, refinement, and the generalization gap as a function of model capacity and penalty strength, isolating effects that cannot be measured on real data.

4. **An application case study in GRN inference.** We apply end-to-end calibrated learning to scalable GRN inference and evaluate it on simulation-supervised and challenge benchmarks [22, 14, 15], assessing not only predictive ranking but the trustworthiness and structural validity of the inferred probabilities.

1.7 Thesis Outline

The remainder of this thesis is organized as follows. **Chapter 2** reviews the necessary background and related work: the formal definitions and notions of calibration, estimators and their biases, proper scoring rules and the calibration–refinement decomposition, post-hoc and in-training calibration methods, the statistical learning theory used to reason about generalization, and the gene regulatory network inference literature. **Chapter 3** develops the theoretical analysis of end-to-end calibration as a structural constraint and its connection to generalization. **Chapter 4** presents the simulation-supervised experiments with known ground truth. **Chapter 5** presents the gene regulatory network case study. **Chapter 6** concludes with a discussion of limitations and directions for future work.

2 Background and Literature Review

This chapter lays the conceptual and technical foundation for the rest of the thesis. It moves from general principles to the specific application domain. Sections 2.1–2.4 introduce the theory of calibration, covering its formal definition and metrics, different methods to measure it, and its relationship to proper scoring rules and the calibration–refinement decomposition. Section 2.5 reviews traditional post-hoc recalibration, including a catalog of the main methods and a comparative summary, while section 2.6 reviews in-training (end-to-end) calibration. These approaches frame the central research question addressed in this thesis. Section 2.7 presents the statistical learning theory needed to analyze generalization and to examine the hypothesis that calibration can serve as a structural constraint on learning. Finally, section 2.8 introduces gene regulatory network (GRN) inference as an application domain, and section 2.9 brings these threads together to identify the research gaps that motivate the remainder of the thesis.

2.1 Notation and Problem Setup

We consider a supervised classification setting.

Convention: Throughout the rest of the thesis, uppercase letters denote random variables and the corresponding lowercase letters their realizations (or generic arguments of a function); a deterministic function may be applied to either, so $f(x)$ is the prediction at a fixed input x while $P = f(X)$ is the (random) prediction at the random input X.

Let (X, Y) be random variables with joint distribution $\mathcal{D}$ with density $\pi(x, y) = \pi(y \mid x)\,\pi(x)$, where $X \in \mathcal{X}, Y \in \mathcal{Y} \subset \Delta_K$, and

$$\Delta_K := \{p \in [0,1]^K \ : \ \mathbf{1}^\top p = 1\}$$

denote the probability simplex.

A *probabilistic classifier* is a measurable function $f : \mathcal{X} \to \Delta_K$ mapping inputs to the probability simplex, so that for an input x, it returns $f(x) = (f_1(x), \ldots, f_K(x))$ with $f_k(x) \geq 0$ and $\sum_k f_k(x) = 1$ for $k = 1, \ldots, K$.

We write $\hat{y}(x) = \arg\max_k f_k(x)$ for the predicted class at x and $\hat{p}(x) = \max_k f_k(x)$ for the associated *confidence*. We also let $\ell : \Delta_K \times \mathcal{Y}_K \to \mathbb{R}_+$ be a loss function. We define its conditional expectation extension as

$$\ell(p, q) := \mathbb{E}_{Y \sim q}[\ell(p, Y)] \quad \text{for } p \in \Delta_K,\ q \in \Delta_K.$$

The (population) risk of a classifier f is defined as

$$\mathcal{E}(f) := \mathbb{E}_{(X,Y)\sim D}\big[\ell(f(X), Y)\big] = \mathbb{E}_X\big[\ell(f(X), \eta(X))\big],$$

where $\eta(x) := \mathbb{E}[Y \mid X = x] \in \Delta_K$ denotes the *conditional class-probability vector* or the *true posterior*.

Given a training sample $S = \{(x_i, y_i)\}_{i=1}^n \sim \mathcal{D}^n$ (drawn i.i.d.), learning selects f from a hypothesis class $\mathcal{F}$ by minimizing an empirical risk

$$\widehat{\mathcal{E}}(f) \;=\; \frac{1}{n}\sum_{i=1}^{n} \ell\big(f(x_i), y_i\big), \qquad \mathcal{E}(f) \;=\; \mathbb{E}_{(X,Y)\sim\mathcal{D}}\big[\ell(f(X), Y)\big], \tag{2.1}$$

where ℓ is the defined loss function and $\mathcal{E}(f)$ is the population (true) risk.

The Bayes-optimal probabilistic predictor is $f^\star(x) = \eta(x) = \pi(\cdot \mid x)$, the *true posterior*; it is the unique minimizer of any *strictly proper* loss (explained in detail in Section 2.4.4). The central problem explored in this thesis is that accuracy, governed by $\arg\max_k f_k(X)$, and the quality of the full probability vector $f(X)$ are distinct objectives, and that standard training optimizes the former far more reliably than the latter.

2.2 What It Means for a Model to be Calibrated

Informally, a model is calibrated if its stated probabilities match observed frequencies: a requirement formalized for sequential forecasters by Dawid's notion of the well-calibrated Bayesian [23]. For a probabilistic classifier f we define the prediction random variable $P := f(X)$. The classifier f is said to be *(perfectly) calibrated* if

$$\mathbb{E}[Y \mid P] = P \quad \text{almost surely.}$$

Equivalently, for all $p \in \Delta_K$ with $\mathbb{P}(P = p) > 0$,

$$\mathbb{P}(Y = e_k \mid P = p) = p_k \quad \text{for all } k = 1, \ldots, K.$$

The notion admits several formalizations of increasing strictness [24, 11].

Confidence (top-label) calibration. The weakest commonly used notion concerns only the predicted class and its confidence [1]. A model is confidence-calibrated if

$$\mathbb{P}\big(Y = \hat{y}(X) \,\big|\, \hat{p}(X) = p\big) \;=\; p, \qquad \forall\, p \in [0, 1]. \tag{2.2}$$

That is, among all inputs assigned confidence p, a fraction p are classified correctly. This is the notion underlying the expected calibration error and reliability diagrams used by Guo *et al.* [1].

Table 2.1: Notions of calibration in increasing order of strictness. Each notion conditions on a different statistic of the predicted vector $f(X)$ and implies all weaker notions above it. The conditioning event becomes higher-dimensional from top to bottom, which is why consistent estimation of canonical calibration is the hardest. The comparison has been adapted from [11].

Notion	Conditions on	Formal condition	Hierarchy	Typical estimator
Confidence (top-label)	Top confidence $\hat{p}(X) = \max_k f_k(X)$	$\mathbb{P}\big(Y = \hat{y}(X) \mid \hat{p}(X) = p\big) = p$	Weakest; implied by the two below	Binned ECE; reliability diagram [1, 7]
Marginal (class-wise)	Each class score $f_k(X)$	$\mathbb{P}\big(Y = e_k \mid f_k(X) = p\big) = p,\ \forall k$	Implies confidence; implied by canonical	Class-wise ECE; Dirichlet diagnostics [24, 26]
Canonical (distribution)	Full vector $f(X) = \mathbf{p}$	$\mathbb{P}\big(Y = e_k \mid f(X) = \mathbf{p}\big) = p_k,\ \forall k$	Strongest; implies both above	Kernel/KDE canonical error $\mathrm{ECE}_{\mathrm{KDE}}$ [11, 25]

Marginal (class-wise) calibration. A stronger requirement is that each class probability be calibrated one-vs-rest [24]: for every class k,

$$\mathbb{P}\big(Y = e_k \,\big|\, f_k(X) = p\big) \;=\; p, \qquad \forall\, p \in [0, 1]. \tag{2.3}$$

Canonical (distribution) calibration. The strongest notion requires the *entire* probability vector to be calibrated [11, 25]:

$$\mathbb{P}\big(Y = e_k \,\big|\, f(X) = \mathbf{p}\big) \;=\; p_k, \qquad \forall\, \mathbf{p} \in \Delta_K,\ \forall k. \tag{2.4}$$

Canonical calibration implies marginal calibration, which implies confidence calibration; the converses do not hold.

Estimating canonical calibration is statistically challenging because it requires conditioning on points in the $(K-1)$-dimensional probability simplex. Consequently, most practical estimators focus on weaker notions of calibration; while consistent, low-bias estimators for canonical calibration have only recently become available as a significant methodological advance [11].

These three notions, the statistic each conditions on, their implication hierarchy, and the estimator typically used for each are summarised in Table 2.1.

It is essential, and a recurring theme of this thesis, that calibration is *necessary but not sufficient* for a good probabilistic model. A model that always outputs the marginal base rates $\mathbb{P}(Y = k)$ is perfectly calibrated yet useless, because it is not *sharp*: it fails to discriminate between inputs. The complementary virtue, resolution or sharpness, is formalized in Section 2.4.5.

Note: *Calibration* is defined with respect to the distribution of predictions, not directly with respect to X. This definition does not require optimality of risk. A model can be perfectly calibrated and still inaccurate.

2.3 Calibration Under Distribution Shift

The calibration notions of Table 2.1 are defined with respect to a single fixed distribution $\mathcal{D}$. In real life, often the test distribution $\mathcal{D}_{\text{test}}$ may differ from the training distribution $\mathcal{D}_{\text{train}}$, a phenomenon broadly known as *distribution shift*. Ovadia *et al.* [27] and Minderer *et al.* [3] study this setting empirically at scale. They evaluated both predictive accuracy and calibration (via ECE) as models are exposed to progressively corrupted or naturally out-of-distribution data, and found that miscalibration under shift is severe and not well predicted by in-distribution calibration quality.

A more narrowly structured and analytically tractable special case is *covariate shift*, in which (A) only the marginal distribution of the input changes between training and test data, while (B) the conditional label distribution is preserved. This means that while the relationship between the input features and the target variable doesn't change, the way the input features are distributed does. Mathematically, we can denote it as below:

$$\begin{aligned} \mathbb{P}_{\text{train}}(X) &\neq \mathbb{P}_{\text{test}}(X) \\ \mathbb{P}_{\text{train}}(Y \mid X = x) &= \mathbb{P}_{\text{test}}(Y \mid X = x) \end{aligned} \tag{2.5}$$

Here the first equation refers the input distribution change, while the second one indicates the conditional distribution stability, two criteria for covariate shift. In the notation of section 2.1, this states that $\pi_{\text{train}}(\cdot \mid x) = \pi_{\text{test}}(\cdot \mid x)$ for all x, i.e. $\eta(x)$ is invariant across train and test, even though the population of inputs X on which η is evaluated shifts. Following Shimodaira [28], Sugiyama [29] and Sugiyama & Kawanabe [30], this decomposition of "what changes" and "what doesn't" is what makes covariate shift the setting most directly analyzable, as opposed to the more general dataset shift considered by Ovadia *et al.* [27] and Minderer *et al.* [3], which permits both $\mathbb{P}(X)$ and $\mathbb{P}(Y \mid X)$ to change.

Covariate shift interacts differently with each level of the calibration hierarchy, mentioned in Table 2.1. Confidence calibration (Eq. 2.2) can degrade under covariate shift even when $\pi(\cdot \mid x)$ is exactly preserved, because the marginal distribution of the confidence score $\hat{p}(X)$ shifts along with $\mathbb{P}(X)$, changing which inputs populate each confidence bin. Marginal (class-wise) calibration (Eq. 2.3) is affected identically per class. Canonical calibration (Eq. 2.4), depending on the full vector $f(X)$, is generally hardest to preserve, since it requires the conditional law of Y given the entire prediction vector to remain stable even as the input population feeding each region of the simplex changes. In summary, calibration under covariate shift is not the invariance of $\pi(\cdot \mid x)$, which the covariate-shift assumption already assumes, it is the invariance of the calibration function $\gamma(p)$ (introduced in subsection 2.4.5), as the distribution feeding it shifts.

2.4 Calibration Metrics and Calibration–Refinement Decomposition

Having defined what it means for a model to be calibrated (Section 2.2), we now turn to how calibration is measured and how it relates to the losses used to train a model in the first place.

We proceed in five steps. Section 2.4.1 introduces the expected calibration error (ECE) and the reliability diagram, the standard tools for measuring the weakest (confidence) notion of calibration from Table 2.1. Section 2.4.2 discusses the statistical bias of the binned ECE estimator and section 2.4.3 surveys other named calibration metrics beyond ECE, for completeness. This motivates a more principled question, addressed in section 2.4.4: *what properties should a loss function have if it is to be trusted as an evaluation criterion for probabilistic predictions at all?* We introduce proper scoring rules to answer this. Section 2.4.5 then shows that any proper loss admits an exact calibration–refinement decomposition, which both explains why ECE and the total training loss are measuring related but distinct things, and supplies the theoretical object that Chapter 3 regularizes directly.

2.4.1 Expected calibration error (ECE) and reliability diagrams

The most widely reported scalar summary is the *expected calibration error* (ECE), the expected gap between confidence and accuracy [7, 1]:

$$\mathrm{ECE} \;=\; \mathbb{E}_{\hat{p}}\Big[\,\big|\,\mathbb{P}(Y=\hat{y}\mid\hat{p})-\hat{p}\,\big|\,\Big]. \tag{2.6}$$

Because the conditioning variable $\hat{p}$ is continuous, ECE is estimated via discrete hard binning and then comparing per-bin accuracy and confidence.

Let $\mathcal{B}=\{B_1,\dots,B_M\}$ be a partition of $[0,1]$ into M non-overlapping bins. For a dataset of n predictions $\{(p_i,y_i)\}_{i=1}^{n}$, each sample i is assigned to bin B_m if $p_i\in B_m$. Within each bin, we define:

- $\mathrm{conf}(B_m)=\dfrac{1}{|B_m|}\sum_{i\in B_m}p_i$: the mean predicted confidence,
- $\mathrm{acc}(B_m)=\dfrac{1}{|B_m|}\sum_{i\in B_m}\mathbf{1}[\hat{y}_i=y_i]$: the empirical accuracy (fraction of correct predictions),
- $\mathrm{prec}(B_m)=\dfrac{\sum_{i\in B_m}\mathbf{1}[\hat{y}_i=y_i\wedge y_i=1]}{\sum_{i\in B_m}\mathbf{1}[\hat{y}_i=1]}$: the empirical precision (fraction of positive predictions that are true positives), with the convention that $\mathrm{prec}(B_m)=0$ when the denominator vanishes.

The estimated ECE is then

$$\widehat{\mathrm{ECE}} \;=\; \sum_{m=1}^{M}\frac{|B_m|}{n}\,\big|\,\mathrm{acc}(B_m)-\mathrm{conf}(B_m)\,\big|, \tag{2.7}$$

where $\mathrm{acc}(B_m)$ and $\mathrm{conf}(B_m)$ are the average accuracy and confidence of the predictions falling in bin B_m. A *reliability diagram* (Figure 1.1) plots $\mathrm{acc}(B_m)$ against $\mathrm{conf}(B_m)$; perfect calibration lies on the diagonal, and systematic deviation below it indicates overconfidence [1, 31].

In the binary classification setting with severe class imbalance, accuracy alone can be misleading. We therefore also track a *precision-aware* variant of the binned calibration error:

$$\mathrm{ECE}_{\mathrm{prec}} = \sum_{m=1}^{M} \frac{|B_m|}{n} \big|\mathrm{prec}(B_m) - \mathrm{conf}(B_m)\big|, \tag{2.8}$$

which measures how well the model's confidence aligns with the fraction of true positives among its positive predictions, a stricter and more informative criterion for imbalanced binary problems.

Hard, equal-width binning is the most common estimator of ECE but not the only one. *Adaptive (equal-mass) binning* [32] sizes each bin to contain an equal number of predictions rather than an equal-width interval of $[0, 1]$, which mitigates the sensitivity to sparsely populated bins. This equal-mass idea along with several other calibration metrics built on related but distinct constructions, are surveyed separately in Section 2.4.3. A further family of differentiable, binning-aware or binning-free estimators, designed specifically to be optimized during training rather than only evaluated post hoc, is explained in Section 2.6.1, where we introduce the soft-binned, MMCE, and $\mathrm{ECE}_{\mathrm{KDE}}$ estimators that this thesis builds on directly.

2.4.2 The bias of binned estimators

Binned ECE is convenient but statistically fraught. Its value depends on the number and placement of bins (equal-width vs equal-mass (adaptive)) [26], and the estimator is *biased*: with finite samples it can report low error for a miscalibrated model and is not a consistent estimator of the underlying calibration error [25, 33]. Vaicenavicius *et al.* [25] analyze this dependence and show that conclusions drawn from binned ECE can be artifacts of the binning scheme. These concerns matter directly for this thesis, because they distinguish the *training-time penalty* from the *evaluation metric*: an estimator suitable as a smooth loss is not necessarily suitable for unbiased evaluation, and conflating the two is a known source of error.

2.4.3 Other calibration metrics

Several other scalar summaries of calibration have been proposed, differing either in the summary statistic applied within each bin, in whether they evaluate per-class or only top-label calibration, or in whether they require binning at all. We survey these here for completeness, alongside ECE itself so that it can be directly compared against the alternatives; none of them is used as a training-time penalty in this thesis (the differentiable estimators actually used for that purpose, Soft-ECE, MMCE, and $\mathrm{ECE}_{\mathrm{KDE}}$, are introduced separately in Section 2.6.1).

The *Maximum Calibration Error* (MCE) reports the single worst per-bin gap rather than the weighted average [7]:

$$\mathrm{MCE} = \max_{m=1,\dots,M} \big|\, \mathrm{acc}(B_m) - \mathrm{conf}(B_m) \,\big| \tag{2.9}$$

Equal-mass ECE (ECE_{em}) [26] applies the same formula as Eq. **??** but with bins chosen to contain an equal number of predictions rather than an equal width of $[0, 1]$.

Beyond varying only the binning scheme applied to the single top-label confidence, a separate line of work evaluates calibration jointly across all K class probabilities rather than only the confidence of the predicted class. Nixon *et al.* [26] propose two such class-conditional extensions of ECE.

The *Static Calibration Error* (SCE) bins each class's one-vs-rest confidence separately using fixed-width bins, computes the calibration error within each bin, and averages the result across all K classes to evaluate marginal (Table 2.1) rather than only confidence calibration:

$$\text{SCE} = \frac{1}{K} \sum_{k=1}^{K} \sum_{b=1}^{B} \frac{n_{b,k}}{N} \big|\, \text{acc}(b,k) - \text{conf}(b,k) \,\big|, \tag{2.10}$$

where B is the number of bins, $\text{acc}(b,k)$ and $\text{conf}(b,k)$ are the accuracy and confidence of bin b restricted to class k, $n_{b,k}$ is the number of predictions of class k falling in bin b, and N is the total number of predictions.

The *Adaptive Calibration Error* (ACE) applies the same construction but with R adaptively chosen ranges, each containing an equal number of predictions, in place of fixed-width bins:

$$\text{ACE} = \frac{1}{KR} \sum_{k=1}^{K} \sum_{r=1}^{R} \big|\, \text{acc}(r,k) - \text{conf}(r,k) \,\big|. \tag{2.11}$$

A further refinement, the *Thresholded Adaptive Calibration Error (TACE)*, applies the same formula as ACE but first discards all predicted class probabilities below a fixed threshold t before adaptively binning the remainder into R ranges [26]:

$$\text{TACE}(t) = \frac{1}{KR} \sum_{k=1}^{K} \sum_{r=1}^{R} \big|\, \text{acc}(r,k) - \text{conf}(r,k) \,\big|, \qquad \text{computed only over } \{(i,k) : f_k(x_i) > t\}. \tag{2.12}$$

Thresholding was introduced specifically to make adaptive binning usable in high-class-count settings, where the vast majority of the $K - 1$ non-predicted class probabilities are close to zero and would otherwise dominate the adaptive ranges [26]. We note, however, that a later reproduction study finds TACE to be sensitive to both of its hyperparameters (the number of ranges and the threshold) and, in practice, not to yield a consistent ranking of models [34].

The *Root-Mean-Squared Calibration Error* (RMSCE) replaces the $L1$ per-bin gap of Eq. **??** with an $L2$ (squared) gap, which upweighs bins with large miscalibration relative to standard ECE [35]. Another binning-free alternative is the *Kolmogorov–Smirnov Calibration Error* (KSCE), which compares empirical cumulative distributions of confidence and correctness directly rather than via bins [36].

Table 2.2 compares ECE and all of these variants in a single place, listing down the underlying mechanism and whether it is class-wise or top-label.

Table 2.2: Comparison of Expected Calibration Error (ECE) and other common calibration metrics. "Binning" indicates whether a choice of bins (fixed-width or adaptive) is required. "Class-wise" indicates whether the metric evaluates per-class (marginal) rather than only top-label (confidence) calibration.

Metric	Binning	Class-wise	Consistent
ECE [7]	Fixed-width	×	×[1]
MCE [7]	Fixed-width	×	Not verified
ECE_{em} [26]	Adaptive (equal-mass)	×	Not verified
SCE[2][26]	Fixed-width	✓	Not verified
ACE[2][26]	Adaptive (equal-mass)	✓	Not verified
TACE [26]	Adaptive (equal-mass, thresholded)	✓	Not verified[3]
RMSCE [35]	Fixed-width	×	Not verified
KSCE [36]	Binning-free	×	Not verified

[1] ECE is biased and not a consistent estimator of the underlying calibration error; see the discussion in Section 2.4.2 and [25].
[2] SCE and ACE are, respectively, equivalent to what some other sources term the class-wise calibration error (cwCE) and class-wise (equal-mass) calibration error (cwCE_{em}), differing only in the choice of fixed-width versus adaptive binning [37].
[3] Ashukha *et al.* [34] report that TACE is sensitive to its threshold and bin-count hyperparameters and does not consistently rank models in practice.

Throughout this thesis, we maintain a deliberate distinction between two roles a calibration function can play. A calibration metric (ECE from subsection 2.4.1 along with other alternatives like MCE, ACE, TACE, RMSCE, KSCE introduced in this subsection 2.4.3) is used only to evaluate a trained model, is typically non-differentiable (built from hard binning or empirical CDFs), and is never optimized directly. A calibration estimator (section 2.6.1: Soft-ECE, MMCE, ECE_{KDE}) is a differentiable surrogate designed specifically to serve as the term $\widehat{\mathcal{C}}$ inside the composite training objective (Eq. 3.1), following calibration-refinement decomposition (Eq. 2.14). The two categories estimate related population quantities but are held to different standards: metrics are chosen for interpretability and evaluation-time reliability, estimators are chosen for differentiability and gradient behavior during training.

2.4.4 Proper Losses

A principled route to calibration runs through *proper scoring rules*, the decision-theoretic standard for evaluating probabilistic forecasts systematized by Gneiting and Raftery [38]. A scoring rule $S(f(x), y)$ is *proper* if the expected score is optimized by reporting the true conditional distribution, and *strictly proper* if that optimum is unique:

$$\mathbb{E}_{Y\sim q}\big[S(q,Y)\big] \;\geq\; \mathbb{E}_{Y\sim q}\big[S(p,Y)\big] \quad \text{for all } p, \text{ with equality iff } p = q. \tag{2.13}$$

Berta *et al.* [8] evaluate predictions with a loss function $\ell : \Delta_K \times \mathcal{Y}_K \to \mathbb{R}_+$. ℓ is a nonnegative

function of p and y such that $\ell(p, y)$ assesses the quality of prediction p for label y. Two of the most widely popular such examples include the *cross-entropy* [39] or negative log-loss (NLL), $\ell(p, y) = -\sum_{i=1}^{k} y_i \log(p_i)$ and *Brier score* [40], $\ell(p, y) = \|y - p\|_2^2$.

If label Y follows a categorical distribution $q \in \Delta_K$, then the expected loss can be denoted as: $\ell(p, q) = \mathbb{E}_{Y\sim q}[\ell(p, Y)]$. Berta *et al.* [8] also identifies that a requirement for the loss function ℓ is that it should be minimized when the predicted distribution p matches the true target distribution q. Losses that verify this property are called *proper*.

Following the work of Gneiting and Raftery [38] and Berta *et al.* [8], we formally introduce the definition of *proper loss function*.

Definition 2.1 (Proper loss function)**.** We consider losses $\ell : \Delta_K \times \mathcal{Y}_K \to \mathbb{R} \cup \{\infty\}$ such that for any $p \in \Delta_K, \ell(p, \cdot)$ is measurable and quasi-integrable with respect to any $q \in \Delta_K$. This allows to define $\ell(p, q) := \mathbb{E}_{Y\sim q}[\ell(p, Y)]$, taking values in $\mathbb{R} \cup \{\infty\}$. If $\ell(q, q) \leq \ell(p, q)$ for all p, q, ℓ is said to be *proper*. Equality holds iff $p = q$ and ℓ is then said to be strictly proper.

Both *cross-entropy* and *Brier score* that were introduced earlier, are strictly proper and hence minimized in expectation by the true posterior $f^\star$. They differ in a way that is central to this thesis: the log score is *unbounded*, whereas the Brier score is *bounded*. Boundedness is precisely the property exploited by the Reinforcement Learning with Calibration Rewards (RLCR) framework [12], which augments a correctness reward with a bounded proper scoring rule and proves that the resulting optimum is both accurate and calibrated. This furnishes a principled, theory-backed reason to prefer bounded proper scoring rules as calibration objectives at training time.

Not every loss used in practice is proper. The *hinge loss*, for instance, is defined directly on margins rather than on a probability report and therefore has no guarantee to minimize expectation in the form of Eq. 2.13. More relevantly to this thesis, focal loss [41], which we revisit in section 2.6.2 as an implicit calibration regularizer, is also not a proper scoring rule, since it reweighs the per-example log-loss by a confidence-dependent factor that need not vanish at the true posterior. That focal loss nonetheless tends to improve calibration empirically; that is precisely why it is treated in this thesis as an implicit, rather than principled, calibration mechanism (section 2.6.2).

2.4.5 Calibration-Refinement Decomposition

Let $d_\ell(p, q) := \ell(p, q) - \ell(q, q)$ and $e_\ell(q) := \ell(q, q)$ be the divergence (Bregman divergence) and entropy associated with loss function ℓ (we refer to these as ℓ-divergence and ℓ-entropy). For different choices of the loss function ℓ, we can recover well-known entropy and divergence functions as given in Table 2.3.

We define $\gamma(p) := \mathbb{E}_{\mathcal{D}}[Y \mid P = p]$ as the *calibration function* of a stochastic classifier f. Now, since Y is one-hot ($Y = e_k$ when the true class is k), the conditional expectation $\mathbb{E}_{\mathcal{D}}[Y \mid P = p]$ is a vector whose k-th entry is $\mathbb{E}[Y_k \mid P = p] = \mathbb{E}[\mathbb{1}(Y = e_k) \mid P = p] = \mathbb{P}[Y = e_k \mid P = p]$. Since, the expectation of an indicator equals the probability of the event. So, $\gamma(p)$, the "calibration

Table 2.3: Controlling the choice of loss function ℓ can recover well-known entropy and divergence functions.

Loss ℓ	**Divergence** d_ℓ	**Entropy** e_ℓ
Log loss	KL divergence	Shannon entropy
$-\sum_i y_i \log(p_i)$	$\sum_i q_i \log \frac{q_i}{p_i}$	$-\sum_i q_i \log q_i$
Brier score	Squared distance	Gini index
$\|y-p\|_2^2$	$\|p-q\|_2^2$	$\sum_i q_i(1-q_i)$

function" is, entry-by-entry, the vector $\big(\mathbb{P}(Y = e_1 \mid P = p), \ldots, \mathbb{P}(Y = e_K \mid P = p)\big)$, i.e., *the true class distribution among inputs that receive prediction p*.

Following the work of DeGroot and Fienberg [31], Bröcker [42], and Berta *et al.* [8], the population risk with a proper loss function ℓ can be decomposed as:

$$\begin{aligned} \mathcal{E}(f) &= \mathbb{E}\big[\ell(f(X), Y)\big] = \mathcal{C}_\ell(f) + \mathcal{R}_\ell(f), \\ \underbrace{\mathcal{C}_\ell(f) := \mathbb{E}\big[d_\ell(P, \gamma(P))\big]}_{\text{Calibration error}}, &\quad \underbrace{\mathcal{R}_\ell(f) := \mathbb{E}\big[e_\ell(\gamma(P))\big]}_{\text{Refinement error}}. \end{aligned} \tag{2.14}$$

The first term, *Calibration error* measures the deviation of predictions from their conditional label distribution and vanishes if and only if f is calibrated. The second term, *Refinement error* measures the intrinsic uncertainty remaining after conditioning on predictions.

Following Bröcker [42], the refinement term can be further decomposed into the uncertainty $\mathcal{U}_\ell(Y)$ (it is independent of f) and a sharpness term $\mathcal{S}_\ell(f)$, yielding the "*calibration-sharpness decomposition*":

$$\mathcal{E}(f) = \mathbb{E}\big[\ell(f(X), Y)\big] = \underbrace{\mathbb{E}\big[d_\ell(P, \gamma(P))\big]}_{\text{Calibration error } (\mathcal{C}_\ell(f))} + \underbrace{\underbrace{e_\ell(\mathbb{E}[Y])}_{\text{Uncertainty}(\mathcal{U}_\ell(Y))} - \underbrace{\mathbb{E}\big[d_\ell(\mathbb{E}[Y], \gamma(P))\big]}_{\text{Resolution / Sharpness } (\mathcal{S}_\ell(f))}}_{\text{Refinement error } (\mathcal{R}_\ell(f))} \tag{2.15}$$

The *sharpness principle* of Gneiting *et al.* [43] restates the goal of probabilistic forecasting as "maximize sharpness subject to calibration." Crucially for the present work, Berta *et al.* [8] give a variational form of this decomposition for cross-entropy and show, theoretically and empirically, that the calibration and refinement terms are *not* minimized at the same point during training. So a model chosen by validation loss is optimal for neither. Their "refine, then calibrate" prescription (optimize refinement during training, fix calibration afterwards) is one response; the converse question of folding calibration *into* training is the subject of this thesis.

It is worth noting explicitly that the expected calibration error (ECE) of section 2.4.1 is itself not a proper loss. This can be seen in two ways. First, structurally: definition 2.1 requires a

proper loss to be a per-example function $\ell(p, y)$ whose population expectation is minimized by truthful reporting; ECE is instead a population-level statistic computed by binning across an entire evaluation set and has no such per-example form. Second, conceptually: a classifier that outputs the constant marginal base rate for every input achieves zero ECE (it is perfectly calibrated in the sense of section 2.2) while discarding all input-dependent information, the failure mode that propriety, via the refinement/sharpness term of Eq. 2.15, is specifically designed to penalize. This distinction matters for the thesis: it is precisely because ECE is not proper, and hence not directly optimizable as a stand-alone training signal without also controlling sharpness, that chapter 3 regularizes with the calibration component of a proper loss Eq. 3.1 rather than with ECE directly.

Total proper losses such as the Brier score or cross-entropy are sometimes reported alongside ECE as an auxiliary, indirect check on calibration. We caution against this practice as a substitute for ECE: since we have already seen in this section that these losses decompose into a calibration term and a refinement term. A raw Brier score or NLL value conflates the two and cannot, by itself, distinguish a well-calibrated-but-unsharp model from a sharp-but-miscalibrated one. The calibration term of the decomposition Eq. 2.14, not the total loss, is the correct like-for-like comparison to ECE.

2.5 Post-hoc Calibration methods

When a trained model is miscalibrated, the most immediate remedy is to correct its outputs without retraining it, this is what we call *recalibrating* a trained model. *Post-hoc calibration* holds the model's parameters fixed, fits a map g on a small held-out calibration set, and applies g to the frozen model's outputs at test time. Different post-hoc calibration methods share a common template: fit a map $g : [0, 1] \to [0, 1]$ (or on the logits) on a held-out set and apply it to the frozen model's outputs. They differ only in the family of maps they fit and in whether they act per-class or on the whole vector. These methods are cheap, model-agnostic, and treat the predictor as a black box. The empirical study of which supervised learners yield well-calibrated probabilities is itself long-standing and dates back to findings by Niculescu-Mizil and Caruana [44].

We describe each principal method in turn, giving its calibration map explicitly, and compare them in Table 2.4. Throughout, $s \in [0, 1]$ denotes a binary score, $\mathbf{z} \in \mathbb{R}^K$ a logit vector, $\mathbf{p} \in \Delta_K$ a predicted probability vector where $\Delta_K := \{p \in [0, 1]^K \ : \ \mathbf{1}^\top p = 1\}$, σ the logistic function, and $\{(s_i, y_i)\}_{i=1}^n$ the held-out calibration data with $y_i \in \{0, 1\}$.

2.5.1 Platt scaling

Platt scaling [5] is the prototypical parametric method which fits a logistic transformation to the scores by passing the model's real-valued score (or logit) z through a logistic sigmoid,

$$g(z) \ = \ \sigma(az + b) \ = \ \frac{1}{1 + \exp(az + b)}, \tag{2.16}$$

and estimates the two parameters a, b by minimizing the negative log-likelihood on the calibration set. The map is smooth and data-efficient, but its two-parameter logistic form assumes the miscalibration is sigmoidal, an assumption that need not hold in practice.

2.5.2 Temperature scaling

Temperature Scaling, the widely popular standard for deep networks [1] is the one-parameter restriction of Platt scaling that divides the logit vector $\mathbf{z}$ by a single scalar temperature $T > 0$ before the softmax,

$$g(\mathbf{z})_k \;=\; \frac{\exp(z_k/T)}{\sum_{j=1}^{K}\exp(z_j/T)}, \qquad T>0, \tag{2.17}$$

where T is fit by minimizing the validation negative log-likelihood. Because the same T rescales every logit, the arg-max and the top-1 accuracy is left unchanged. $T > 1$ softens the distribution (reducing confidence) and $T < 1$ sharpens it. The generalizations *vector* and *matrix scaling* replace $\mathbf{z}/T$ by $\mathbf{W}\mathbf{z} + \mathbf{b}$ with a diagonal or full matrix $\mathbf{W}$; they are more expressive but introduce many parameters and can degrade accuracy [1].

2.5.3 Beta calibration

Beta calibration is a parametric family designed for scores that are already probabilities in $[0, 1]$: the map is derived from the Beta distribution and has three parameters [45]. Its calibrated output has a logit that is linear in $\ln s$ and $\ln(1 - s)$,

$$\ln\frac{g(s)}{1-g(s)} \;=\; c + a\,\ln s - b\,\ln(1-s), \qquad a, b \geq 0, \tag{2.18}$$

which strictly generalizes logistic calibration. Crucially, the family contains the identity map (at $a = b = 1,\ c = 0$, (2.18) reduces to $g(s) = s$), so beta calibration cannot worsen an already well-calibrated classifier in the population limit, a guarantee Platt scaling does not provide.

2.5.4 Histogram binning

This is the simplest non-parametric method that partitions the score range into M bins $B_1, \ldots, B_M$ and maps any score lying in bin B_m to the fraction of positives among held-out points in that bin [46],

$$g(s) \;=\; \frac{1}{|B_m|}\sum_{i:\, s_i \in B_m} \mathbb{1}[\,y_i = 1\,], \qquad s \in B_m. \tag{2.19}$$

It is assumption-free and easy to fit, but its output is piecewise constant, it discards the ordering of scores within a bin, and its quality is sensitive to the number and placement of the bins.

2.5.5 Isotonic regression

This non-parametric method removes the dependence on a fixed grid by fitting a monotonically non-decreasing, piecewise-constant map that minimises squared error subject only to a monotonicity constraint,

$$\hat{g} \;=\; \arg\min_{g \text{ non-decreasing}} \sum_{i=1}^{n} \big(g(s_i) - y_i\big)^2, \tag{2.20}$$

solved efficiently by the pool-adjacent-violators algorithm [6]. It is more flexible than a parametric fit like Platt Scaling and assumes only monotonicity, at a higher risk of overfitting when the calibration set is small.

NOTE: It is important to note that both *histogram binning* and *isotonic regression* map scores to empirical frequencies non-parametrically.

2.5.6 Bayesian Binning into Quantiles (BBQ)

It is a non-parametric refinement of histogram binning that tempers the arbitrariness of a single histogram by performing Bayesian model averaging over a collection of equal-frequency binning models $\{M_1, \ldots, M_R\}$, weighting each by a Bayesian score [7].

$$g(s) \;=\; \sum_{r=1}^{R} P(M_r \mid \mathcal{D})\; P\big(y = 1 \mid s, M_r\big), \tag{2.21}$$

where $P(y = 1 \mid s, M_r)$ is the bin frequency under model M_r and the posterior weight $P(M_r \mid \mathcal{D})$ is proportional to that model's marginal likelihood (a Bayesian score) times its prior. The averaging removes the dependence on a single arbitrary binning and yields smoother estimates than a fixed histogram.

2.5.7 Dirichlet calibration

The natively multiclass generalization of beta calibration [24] applies an affine map to the log-probability vector followed by a softmax,

$$g(\mathbf{p}) \;=\; \mathrm{softmax}\big(\mathbf{W} \ln \mathbf{p} + \mathbf{b}\big), \qquad \mathbf{W} \in \mathbb{R}^{K \times K},\ \mathbf{b} \in \mathbb{R}^{K}, \tag{2.22}$$

equivalently a log-transform, a single linear layer, and a softmax. Unlike temperature scaling, which only rescales confidence, Dirichlet Calibration, with its $K(K+1)$ parameters, can correct the class-wise (marginal) miscalibration of the entire probability vector. Finite-sample guarantees for such recalibration are available through verified calibration [33].

Table 2.4 summarizes these methods along the axes that matter in practice: the map family, parameter count, whether the method is natively multiclass, and whether it provably preserves top-1 accuracy.

Table 2.4: Comparison of common post-hoc calibration methods, ordered as in the text. "Multiclass-native" indicates whether the method calibrates the full vector directly or is applied one-vs-rest; "Preserves acc." indicates whether the map provably leaves the top-1 prediction unchanged.

Method	**Map family**	**#params**	**Multiclass-native**	**Preserves acc.**
Platt scaling [5]	Logistic (sigmoid)	2	×	∼
Temperature scaling [1]	Single-parameter logit	1	✓	✓
Beta calibration [45]	Beta family	3	×	∼
Histogram binning [46]	Non-param. (binning)	M bins	×	∼
Isotonic regression [6]	Non-param. monotone	data-dep.	×	∼
BBQ [7]	Non-param. Bayesian bins	—	×	∼
Dirichlet calibration [24]	Linear-on-log + softmax	$K(K+1)$	✓	×

Legend: ✓ yes × no ∼ depends on setting / not guaranteed. Temperature scaling preserves the arg-max because it scales all logits by the same factor; per-class and binning maps may alter it after renormalisation.

2.5.8 A shared limitation

The strengths of these methods, simplicity and model-agnosticism, come with structural limitations already noted in Chapter 1. They operate on a *frozen* predictor and therefore cannot influence the learned representation of the model or its decision boundary; the correction is confined to the output layer. The simplest variants (temperature scaling) apply only a global, monotone transformation; and because the map is fit on one distribution, the resulting calibration is not guaranteed to survive a distribution shift [3]. These limitations are the direct motivation for studying the alternative: folding calibration into training itself, to which we now turn (Section 2.6).

2.6 Calibration During Training (End-to-End)

To be usable *inside* training, a calibration term must satisfy three requirements. (1) *A well-defined target*: it should estimate a genuine population calibration functional, ideally one that is zero if and only if the model is calibrated, such as the canonical calibration error of Section 2.2. (2) *Statistical soundness*: it should admit a finite-sample estimator, preferably consistent and low-bias, so that driving the term down corresponds to genuinely improving calibration. (3) *Differentiability*: the estimator must be (sub)differentiable with respect to the model parameters θ, so that it can be minimized by the same gradient-based optimization used for the predictive loss.

The standard binned ECE of Section 2.4 fails (3). Assigning each prediction to a bin uses indicator functions, so the estimator is piecewise constant in θ and its gradient is zero almost everywhere; no useful learning signal flows back to the network [10]. Because this prevents backpropagation, the binned ECE cannot be optimized directly by gradient descent. This is precisely why an explicit calibration penalty requires a differentiable *surrogate*. In-training approaches modify the objective so that the learner is shaped by calibration from the outset.

2.6.1 Auxiliary calibration losses

The most direct approach adds a differentiable calibration penalty to the predictive loss, optimizing $\widehat{\mathcal{E}}_\ell(f) + \lambda\,\Omega(f)$ for a penalty Ω and strength λ. Instances include the Difference of Confidence and Accuracy (DCA) term [2], MMCE [9], the soft-binned SB-ECE and S-AvUC objectives [10], and the $\mathrm{ECE}_{\mathrm{KDE}}$-regularised risk [11]. Reported results indicate that such penalties can reduce ECE substantially with little loss of accuracy and can improve robustness under dataset shift relative to cross-entropy plus post-hoc recalibration [10]. (Specific reduction figures are method- and dataset-dependent and should be cited from the corresponding papers rather than generalized.)

In this thesis, we will focus only on "Auxiliary Calibration Loss". In particular, three estimators for the calibration error, of increasing statistical strength, are central to this thesis.

Soft-ECE (kernel-smoothed ECE). The most direct solution is to soften the binning. Hard bin membership is replaced by smooth weights derived from a kernel placed at fixed bin centers, a Gaussian softmax over distances to the centers, or a Laplacian kernel. So every prediction contributes fractionally to every bin and the per-bin confidence and accuracy become differentiable functions of θ [10].

Let $c_1, \ldots, c_M$ be M fixed bin centers uniformly placed on $[0, 1]$. For a prediction p_i (the model's estimated probability for the input i), define the *soft membership weight* to bin j as

$$w_{j,i} = \frac{\exp\left(-\dfrac{(p_i - c_j)^2}{2\sigma^2}\right)}{\displaystyle\sum_{k=1}^{M} \exp\left(-\dfrac{(p_i - c_k)^2}{2\sigma^2}\right)}, \tag{2.23}$$

where $\sigma > 0$ is the kernel bandwidth. This is a Gaussian softmax over distances to bin centers, so every sample contributes fractionally to every bin. The *Differentiable Expected Calibration Error (DECE)* is then

$$\mathcal{L}_{\mathrm{DECE}} = \sum_{j=1}^{M} \frac{\displaystyle\sum_i w_{j,i}}{n} \left(\underbrace{\frac{\displaystyle\sum_i w_{j,i}\, p_i}{\displaystyle\sum_i w_{j,i}}}_{\text{Soft Confidence of bin } j} - \underbrace{\frac{\displaystyle\sum_i w_{j,i}\, y_i}{\displaystyle\sum_i w_{j,i}}}_{\text{Soft Accuracy of bin } j} \right)^2. \tag{2.24}$$

A precision-aware analogue replaces the soft accuracy term with a soft precision term:

$$\text{Soft Precision of bin } j \;=\; \frac{\displaystyle\sum_i w_{j,i}\, \mathbf{1}[\hat{y}_i = 1]\, y_i}{\displaystyle\sum_i w_{j,i}\, \mathbf{1}[\hat{y}_i = 1]}, \tag{2.25}$$

yielding $\mathcal{L}_{\text{DECE-prec}}$ by substituting Eq. 2.25 for the soft accuracy in Eq. 2.24.

The resulting soft-binned calibration error is easy to implement and to add to the loss. Its weaknesses are inherited from binning: it depends on the number and placement of bin centers and on the kernel bandwidth, and it is neither a *consistent* nor a *de-biased* estimator of the underlying calibration error.

Remark 2.2. A Laplacian kernel $\exp(-|p_i - c_k|/\sigma)$ is often preferred over the Gaussian kernel in calibration settings, as it is less sensitive to outliers in the predicted probability distribution and yields sparser effective bin assignments. The bandwidth σ is a hyperparameter selected via cross-validation.

Maximum Mean Calibration Error (MMCE). *Kernel mean embeddings* give the Maximum Mean Calibration Error (MMCE), a trainable RKHS statistic that is zero iff the model is calibrated in top-label sense[9]. It embeds the per-sample calibration residuals in a reproducing-kernel Hilbert space through a differentiable kernel (typically Laplacian) and measures calibration as a maximum mean discrepancy [9]. Since MMCE removes binning entirely and needs no choice of evaluation points, it is mathematically clean for optimization. However, it targets only top-label (scalar-confidence) calibration and has been shown *not* to be a consistent estimator of the L_p canonical calibration error [11].

While sounds similar and abbreviated almost the same, the *Maximum Mean Calibration Error* (MMCE) is not to be confused with the (unrelated) *Maximum Calibration Error* MCE of subsection 2.4.3.

Let $r_i = \mathbf{1}[\hat{y}_i = y_i]$ be the indicator of correct prediction for sample i, and $c_i \in [0, 1]$ its predicted confidence. The *squared MMCE* over a batch of n samples is

$$\mathcal{L}^2_{\mathrm{MMCE}} = \frac{1}{n^2} \sum_{i=1}^{n} \sum_{j=1}^{n} (r_i - c_i)(r_j - c_j)\, \kappa(c_i, c_j), \tag{2.26}$$

where $\kappa(c_i, c_j)$ is a differentiable kernel function, usually the Laplacian kernel $\kappa(c_i, c_j) = \exp(-|c_i - c_j|/\sigma)$.

Intuitively, MMCE tells us *whether the calibration residuals* $(r_i - c_i)$ *of two samples are correlated when their predicted confidences are similar?*

A perfectly calibrated model satisfies $\mathcal{L}_{\mathrm{MMCE}} = 0$ if and only if $\mathrm{CE}_p = 0$ [9], i.e., the model is calibrated. In imbalanced settings such as GRN edge prediction (the specific application case study for this thesis), one may additionally replace r_i with a precision indicator $r_i^{\mathrm{prec}} = \mathbf{1}[\hat{y}_i = y_i = 1]/\max(\mathbf{1}[\hat{y}_i = 1], \epsilon)$, giving a precision-aware MMCE loss that penalizes overconfident false-positive predictions more heavily.

Canonical L_p error via kernel density estimation ($\mathrm{ECE}_{\mathrm{KDE}}$). The two preceding estimators operate at the level of top-label (scalar confidence) calibration. Popordanoska *et al.* [11] provides a stronger alternative which requires the entire probability vector to be calibrated simultaneously. It

is also a consistent, low-bias, and differentiable estimator of the canonical calibration error. It forms a kernel density estimate of the conditional label distribution over the probability simplex using a Dirichlet kernel, and plugs that estimate into the L_p canonical calibration error [11]. Dirichlet Kernel is a natural choice on the simplex, since unlike Gaussian kernel, it places no mass outside it. The estimator is consistent (under mild regularity), differentiable, and admits a geometric-series debiasing scheme that reduces its bias from $\mathcal{O}(n^{-1})$ to $\mathcal{O}(n^{-2})$, at $\mathcal{O}(n^2)$ cost per mini-batch.

Formally, the L_p canonical calibration error is [11]:

$$\mathrm{CE}_p(f) = \mathbb{E}\Big[\big\|\mathbb{E}[Y \mid f(X)] - f(X)\big\|_p^p\Big]^{1/p}. \tag{2.27}$$

A classifier f is perfectly calibrated if and only if $\mathrm{CE}_p(f) = 0$.

Estimating the conditional expectation $\mathbb{E}[Y \mid f(X)]$ requires density estimation over the probability simplex Δ_K. Popordanoska *et al.* [11] uses a *Dirichlet kernel density estimate (KDE)*:

$$\widehat{\mathbb{E}}[Y \mid f(X)]\Big|_{f(x_j)} = \frac{\sum_{i \neq j} \kappa_{\mathrm{Dir}}\big(f(x_j);\, f(x_i)\big)\, y_i}{\sum_{i \neq j} \kappa_{\mathrm{Dir}}\big(f(x_j);\, f(x_i)\big)}, \tag{2.28}$$

where the *Dirichlet kernel* is defined as

$$\kappa_{\mathrm{Dir}}\big(f(x_j),\, f(x_i)\big) = \frac{\Gamma\left(\sum_{k=1}^K \alpha_{ik}\right)}{\prod_{k=1}^K \Gamma(\alpha_{ik})} \prod_{k=1}^K f(x_j)_k^{\alpha_{ik}-1}, \qquad \alpha_i = \frac{f(x_i)}{h} + \mathbf{1}, \tag{2.29}$$

with $h > 0$ a bandwidth parameter satisfying $h \to 0$ as $n \to \infty$, and Γ the Gamma function. The Dirichlet kernel is the natural choice for density estimation over the simplex Δ_K, as it respects the boundary structure of the simplex — unlike Gaussian kernels, which assign positive mass outside Δ_K.

The $\mathrm{ECE}^{\mathrm{KDE}}$ estimator of the L_p canonical calibration error is then

$$\widehat{\mathrm{CE}}_p(f)^p = \frac{1}{n} \sum_{j=1}^n \left\| \widehat{\mathbb{E}}[Y \mid f(X)]\Big|_{f(x_j)} - f(x_j) \right\|_p^p. \tag{2.30}$$

Remark 2.3 (Binary specialization). In a binary ($K = 2$) settings (like the edge prediction between two nodes in a graph), the Dirichlet kernel in Eq. 2.29 specializes to a *Beta kernel*:

$$\kappa_B\big(f(x_j),\, f(x_i)\big) = f(x_j)^{\,\alpha_i - 1} (1 - f(x_j))^{\,\beta_i - 1} \frac{\Gamma(\alpha_i + \beta_i)}{\Gamma(\alpha_i)\,\Gamma(\beta_i)}, \tag{2.31}$$

with $\alpha_i = f(x_i)/h + 1$ and $\beta_i = (1 - f(x_i))/h + 1$. Eq. 2.28— 2.30 then define a consistent, differentiable estimator of the L_p binary calibration error directly applicable to binary settings.

This estimator has the following properties [11]:

Table 2.5: Differentiable calibration estimators considered in this thesis. "Consistent" and "De-biased" refer to estimation of the underlying calibration error; "Canonical" indicates whether the full probability vector (rather than only top-label confidence) is calibrated.

Estimator	Consistent	Differentiable	De-biased	Canonical
Soft-ECE (kernel-smoothed) [10]	×	✓	×	×
MMCE [9]	×	✓	×	×
ECE_{KDE} [11]	✓	✓	✓	✓

- **Consistency:** $\widehat{\text{CE}}_p(f) \xrightarrow{p} \text{CE}_p(f)$ as $n \to \infty$, provided $p_{f(X)}(f(X))$ is Lipschitz continuous over the interior of Δ_K.
- **Convergence rate:** $\mathcal{O}(n^{-1/2})$.
- **Bias:** Converges as $\mathcal{O}(n^{-1})$, reducible to $\mathcal{O}(n^{-2})$ via a geometric-series debiasing scheme.
- **Computational complexity:** $\mathcal{O}(n^2)$ with respect to the mini-batch size n, adding negligible overhead to training.
- **Differentiability:** $\widehat{\text{CE}}_p(f)^p$ is differentiable with respect to f, enabling end-to-end gradient-based training.

To the best of our knowledge, it is the only tractable and consistent estimator of canonical multiclass calibration error.

These three estimators are exactly the instantiations of the calibration term $\widehat{\mathcal{C}}$ that the end-to-end objective of Chapter 3 minimizes. Their differing guarantees, summarized in Table 2.5. The choice of estimator is an empirical question, *which surrogate most reduces miscalibration while preserving predictive accuracy*, that the experimental chapters investigate.

2.6.2 Other methods to improve calibration during training

Implicit regularizers that affect calibration. Several techniques improve calibration as a side effect rather than by an explicit calibration term. *Label smoothing* replaces hard targets with softened ones and is known to improve calibration while changing the geometry of learned representations [47]. *Focal loss* down-weights well-classified examples and has been analysed as an implicit regulariser of predictive confidence, improving calibration [48, 41]. *Deep ensembles* improve both accuracy and uncertainty by averaging independently trained models [49]. These methods are evidence that the training procedure itself—not only post-hoc correction—controls calibration, and that calibration interacts with the inductive bias of the learner.

Reward-based calibration. Beyond supervised classification, the Reinforcement Learning with Calibration Rewards (RLCR) framework trains models with a reward that combines correctness

and a bounded proper scoring rule, yielding provably accurate and calibrated predictions and demonstrating that explicit optimisation for calibration need not cost accuracy [12]. Although developed for reasoning language models, its core argument of a bounded proper scoring rule integrated into the objective shapes the solution toward calibration, is exactly the mechanism this thesis studies.

2.6.3 Calibration, accuracy, and architecture

Empirically, the relationship between calibration and accuracy is neither adversarial nor automatic. Minderer *et al.* [3] find that recent architectures can be both more accurate and better calibrated, and that calibration can improve with model size under distribution shift; large-scale benchmark studies catalog how calibration varies with architecture, training, and metric choice across many thousands of models [4]. Together with the calibration–refinement analysis [8], these results frame end-to-end calibration not as a fixed trade-off but as a problem of *shaping* the learner, which leads naturally to a learning-theoretic treatment.

2.7 Statistical Learning Theory Foundations

This section assembles the theoretical apparatus used in Chapter 1's hypothesis that calibration acts as a structural constraint on the hypothesis space. We review the generalization gap, complexity-based bounds (Rademacher), PAC-Bayesian bounds, and how a calibration constraint can be situated within them.

2.7.1 Empirical risk minimization and the generalization gap

The object of interest is the *generalization gap* $\mathcal{E}_\ell(f) - \widehat{\mathcal{E}}_\ell(f)$ between population and empirical risk. Classical theory bounds the worst-case gap over a hypothesis class $\mathcal{F}$ in terms of a complexity measure of $\mathcal{F}$ [50, 51]. A central modern observation is that the classical capacity measures (e.g. VC dimension) are far too loose for overparameterized deep networks, which can fit random labels yet still generalize [52]; this is what makes *data-* and *algorithm-dependent* notions of complexity, and the regularizing effect of training choices, the right lens for modern models.

2.7.2 Rademacher complexity bounds

The empirical Rademacher complexity of a function class $\mathcal{G}$ on a sample of size n is

$$\widehat{\mathfrak{R}}_m(\mathcal{G}) \;=\; \mathbb{E}_\sigma\left[\sup_{g\in\mathcal{G}} \frac{1}{n}\sum_{i=1}^{n} \sigma_i\, g(z_i)\right], \tag{2.32}$$

where σ_i are independent Rademacher (± 1) variables. It measures the ability of $\mathcal{G}$ to correlate with random noise—i.e. its effective capacity [53]. The standard bound states that, with probability at least $1-\delta$, every $g \in \mathcal{G}$ (taking values in $[0,1]$) satisfies

$$\mathbb{E}[g(Z)] \;\le\; \frac{1}{n}\sum_{i=1}^{n} g(z_i) \;+\; 2\,\widehat{\mathfrak{R}}_m(\mathcal{G}) \;+\; 3\sqrt{\frac{\ln(2/\delta)}{2n}}. \tag{2.33}$$

The relevance to this thesis is structural: if augmenting the objective with a calibration penalty effectively restricts the learner to a subset $\mathcal{F}_{\text{cal}} \subseteq \mathcal{F}$ of better-calibrated hypotheses, then $\widehat{\mathfrak{R}}_m(\mathcal{F}_{\text{cal}}) \le \widehat{\mathfrak{R}}_m(\mathcal{F})$ and the bound tightens—a precise sense in which calibration can behave like capacity control.

2.7.3 PAC-Bayesian bounds

PAC-Bayes bounds are particularly apt for modern, overparameterized models because they bound the risk of *randomized* predictors and have yielded the tightest (sometimes non-vacuous) guarantees for neural networks [54, 55, 56]. For a prior P over hypotheses chosen before seeing the data and any posterior Q (possibly data-dependent), McAllester's bound states that, with probability at least $1-\delta$,

$$\mathbb{E}_{f\sim Q}\big[\mathcal{E}_\ell(f)\big] \;\le\; \mathbb{E}_{f\sim Q}\big[\widehat{\mathcal{E}}_\ell(f)\big] \;+\; \sqrt{\frac{\mathrm{KL}(Q\,\|\,P) + \ln\frac{2\sqrt{n}}{\delta}}{2n}} \tag{2.34}$$

The bound improves when the posterior Q stays close (in KL divergence) to the prior P. This suggests a clean way to formalize calibration as regularization: a calibration objective can be viewed as biasing the posterior Q toward a calibration-respecting region, and if that region can be encoded in (or kept close to) the prior P, the complexity term $\mathrm{KL}(Q\|P)$ shrinks. Establishing such a connection precisely is a theoretical aim of Chapter 1's RQ2.

2.7.4 Calibration as a constraint on the hypothesis space

Two threads make the "structural constraint" framing concrete. First, the calibration–refinement decomposition (Section 2.4.5) shows that a proper-loss risk already *contains* a calibration term; penalizing calibration explicitly re-weights this existing component rather than adding an unrelated objective [8, 42]. Second, impossibility and trade-off results show that calibration genuinely *constrains* the attainable set of predictors: Pleiss *et al.* [57] prove that calibration is compatible with only a restricted set of group error-rate constraints, an explicit example of calibration acting as a hard structural restriction on the hypothesis space rather than a free improvement. Taken together, these results support treating end-to-end calibration as an inductive bias whose effect on generalization is analyzable with the tools above.

2.8 Gene Regulatory Network Inference

2.8.1 Biological background

A gene regulatory network is a directed graph in which an edge $a \to b$ encodes that the product of gene a (typically a transcription factor) influences the expression of gene b. Reconstructing GRNs from expression measurements is foundational for understanding differentiation, cellular response, and disease [13, 14]. Two features make the problem a stringent testbed for probabilistic prediction. The edges of interest are *directed* and ideally *causal*, so a method that merely captures statistical association can score well on association-based metrics while misrepresenting the regulatory logic; and the practical output is a *ranked, uncertain* set of candidate edges, whose usefulness depends on whether the attached confidences are trustworthy.

2.8.2 Data and its pathologies

GRN inference uses two broad data modalities. *Bulk* expression data, such as the microarray compendia of the DREAM5 challenge, aggregate signal over many cells across many conditions [22]. *Single-cell* RNA-sequencing (scRNA-seq) resolves individual cells but introduces severe pathologies: high technical noise, extreme sparsity, and "dropout" events in which expressed transcripts are recorded as zero. Simulation studies make the consequences explicit—technical noise can dominate the accuracy of network inference [15]—and many single-cell methods further rely on *pseudotime*, an inferred ordering of cells along a trajectory, whose errors propagate into the inferred network [14, 58]. These properties violate clean i.i.d. assumptions and elevated noise levels make calibrated uncertainty both harder to achieve and more valuable.

2.8.3 Classical and statistical approaches

Early and still-competitive methods are unsupervised. *Correlation- and information-theoretic* methods score edges by pairwise dependence: ARACNE uses mutual information with a data-processing-inequality pruning step [59], CLR uses a context-likelihood correction [60], and PIDC uses partial information decomposition for single-cell data [19]. *Probabilistic graphical models* provide the most rigorous paradigm: Bayesian networks model the joint distribution of expression and support reasoning about conditional independence and, under assumptions, causal structure and pseudo-temporal dynamics [13, 61]. The formal machinery of causal discovery—the do-calculus and constraint-based search—sits in this tradition [62, 63]. The decisive obstacle is computational: learning Bayesian-network structure is NP-hard in general [16], and exact structure learning does not scale to the thousands of genes assayed at genome scale, which is the gap that motivates faster, approximate, learning-based methods. *ODE- and dynamics-based* methods such as SCODE [58] and time-stamped approaches such as SINCERITIES [64] target temporal data specifically.

2.8.4 Machine-learning and deep-learning approaches

Scalability is supplied by feature-selection and regression formulations. GENIE3 decomposes network inference into p independent regression problems, predicting each gene from the others with tree ensembles and ranking edges by variable importance [17]; it was a top performer in the DREAM challenges. GRNBoost2 retains this scheme with gradient-boosted trees for far greater speed [18], and both serve as the regression backbone of the SCENIC pipeline [65]. More recently, *deep* models have been applied: CNNC casts gene-pair inference as image classification over co-expression histograms [20]; GENELink uses a graph attention network to predict regulatory links from single-cell data [21]; and DeepSEM embeds a structural equation model within a β-VAE to jointly infer the network and a denoised representation [66]. These supervised and generative deep methods supply the scalability that classical structure learning lacks. However, they are trained for, and judged by, predictive ranking, and often do not produce calibrated edge probabilities or guarantees of causal validity, which is the gap the thesis targets.

Table 2.6 summarizes the principal method families along with the axes most relevant to this thesis—edge directionality, causal semantics, genome-scale scalability, output type, and whether calibrated uncertainty is an explicit training objective.

Table 2.6: Families of gene regulatory network inference methods and the properties relevant to this thesis. The final column is the key observation: across all families, including deep methods that do emit probability-like scores, calibrated uncertainty is not an explicit objective, and edge confidences are neither trained for nor evaluated as calibrated quantities.

Family	**Representative methods**	**Dir.**	**Causal**	**Scale**	**Output**	**Calib.**
Correlation & mutual information	ARACNE [59], CLR [60], PIDC [19]	×	×	✓	Association score (ranking)	×
Bayesian networks / graphical models	Score-and-search BNs [13, 61]	✓	∼	×[a]	Structure / edge posterior	×
ODE & time-stamped dynamics	SCODE [58], SINCERITIES [64]	✓	∼	∼	Regression coeff. (ranking)	×
Tree-ensemble regression	GENIE3 [17], GRNBoost2 [18] (SCENIC [65])	✓	×	✓	Importance score (ranking)	×
Deep learning (supervised / generative)	CNNC [20], GENELink [21], DeepSEM [66]	✓	×[b]	✓	Edge probability / weighted adjacency	×

Legend: ✓ yes × no ∼ partial / only under modelling assumptions. *Dir.* = directed edges; *Causal* = explicit causal semantics; *Scale* = applicability at genome scale; *Calib.* = calibrated uncertainty is an explicit objective.
[a] Exact Bayesian-network structure learning is NP-hard [16], limiting genome-scale use. [b] DeepSEM's structural-equation formulation carries a partial causal interpretation under its modelling assumptions.

2.8.5 Benchmarking, simulation, and evaluation

Progress in the field is organised around shared benchmarks. The DREAM5 challenge provides standardised datasets with gold-standard networks and established that aggregating predictions across methods is robust [22]; GeneNetWeaver supplies in-silico benchmark generation and profiling [67]; BEELINE systematises single-cell GRN evaluation against curated and synthetic ground truth [14]; and SERGIO simulates single-cell expression from a user-specified GRN with realistic stochasticity and noise [15]. Evaluation almost universally uses *ranking* metrics—area under the ROC and precision–recall curves and early precision [14, 22]. These quantify whether true edges are ranked above false ones but are *silent on calibration*: they do not test whether a predicted probability of 0.8 corresponds to an 80% empirical edge frequency, nor whether the recovered structure is causal. The combination of (i) controlled simulators that expose the true network and (ii) an evaluation culture that ignores calibration is exactly what makes GRN inference a suitable and underexplored testbed for end-to-end calibrated learning.

2.9 Synthesis and Positioning

Three observations bring this chapter to a close and motivate the central focus of the thesis. *First*, calibration is a well-defined and hierarchically structured property (Section 2.2), but its assessment is far from straightforward. In particular, training objectives and evaluation estimators must be carefully distinguished (Section 2.4), and both are rooted in the theory of proper scoring rules and the calibration–refinement decomposition (Section 2.4.5). *Second*, although post-hoc recalibration methods are well established (Section 2.5), they remain fundamentally separate from the learning process itself. By contrast, in-training approaches (Section 2.6), including differentiable calibration penalties, implicit regularization strategies, and reward-based formulations, modify the learner itself during optimization. The learning-theoretic framework introduced in Section 2.7 provides the tools to examine whether such interventions can be understood as a form of capacity control. *Third*, gene regulatory network (GRN) inference (Section 2.8) presents an especially compelling application domain. While scalable deep learning models have achieved strong predictive performance, they are typically designed and evaluated with ranking accuracy as the primary objective. As a result, calibrated and causally interpretable probabilistic inference remains largely unexplored. At the same time, the availability of simulation frameworks with controlled ground truth makes GRN inference uniquely suited for rigorous studies of calibration and generalization. The gap addressed by this thesis lies at the intersection of these three strands: an end-to-end treatment of calibration as an explicit learning constraint, analysed through the lens of generalisation theory and evaluated in the setting of simulation-supervised GRN inference.

3 End-to-End Calibration as a Structural Constraint

Chapter 1 advanced the hypothesis that integrating a calibration objective into empirical risk minimization (ERM) behaves like *implicit regularization*: it restricts the set of admissible predictors and thereby reshapes the bias–variance trade-off (RQ1, RQ2). This chapter makes that hypothesis precise. We (i) formalize the composite, calibration-regularized objective and relate its special cases to proper losses (Section 3.1); (ii) recall the exact calibration–refinement decomposition that the objective re-weights (Section 3.2); (iii) express penalized training as *constrained* ERM over a calibration-restricted hypothesis class (Section 3.3); and (iv) analyze the generalization of that restricted class through Rademacher complexity (Section 3.4) and a PAC-Bayesian lens (Section 3.5). Section 3.6 interprets the results as capacity control, while Section 3.8 states the assumptions and the questions that remain open.

Throughout the chapter, we adopt the notation of Section 2.1: data $(X, Y) \sim \mathcal{D}$ where $X \in \mathcal{X}, Y \in \mathcal{Y} \subset \Delta_K$, a hypothesis class $\mathcal{F}$ of probabilistic predictors $f : \mathcal{X} \to \Delta_K$, a sample $S = \{(x_i, y_i)\}_{i=1}^n$, empirical and population risks $\widehat{\mathcal{E}}_\ell, \mathcal{E}_\ell$, and the Bayes predictor $f^\star(x) = \eta(x) = \pi(\cdot \mid x)$.

3.1 The Composite Calibration-Regularized Objective

Motivated by the calibration–refinement decomposition, briefly introduced in Section 2.4.5 and the observation that optimization dynamics may yield poorly calibrated predictors, we consider an *empirical, differentiable surrogate* $\widehat{\mathcal{C}}(f) : \mathcal{F} \to \mathbb{R}_{\geq 0}$ of the calibration error, for instance the kernel-density canonical estimator $\mathrm{ECE}_{\mathrm{KDE}}$ [11], the maximum mean calibration error [9], or a soft-binned error [10]. We also consider $\widehat{\mathcal{E}} : \mathcal{F} \to \mathbb{R}_{\geq 0}$ to denote the total empirical proper loss (e.g. the Brier score).

Our objective is to solve the optimization problem

$$\min_{\theta \in \Theta} \mathcal{L}_\lambda(f_\theta) = \min_{\theta \in \Theta} \widehat{\mathcal{C}}(f_\theta) + \lambda \widehat{\mathcal{R}}(f_\theta) = \min_{\theta \in \Theta} (1 - \lambda)\widehat{\mathcal{C}}(f_\theta) + \lambda \widehat{\mathcal{E}}(f_\theta), \tag{3.1}$$

with refinement error, $\widehat{\mathcal{R}}(f_\theta) := \widehat{\mathcal{E}}(f_\theta) - \widehat{\mathcal{C}}(f_\theta)$ and with $\lambda \in (0, 1]$ being a regularization parameter. Following the calibration–refinement decomposition, we do not specify the refinement term independently; we *define* it as the residual. The goal is to shape the training trajectory $\{f_{\theta_t}\}$ such that the final predictor is both accurate and calibrated.

Definition 3.1 (Calibration-regularised ERM)**.** For weight λ, the calibration-regularised estimator is $\hat{f}_\lambda \in \arg\min_{\theta\in\Theta} \mathcal{L}_\lambda(f_\theta)$.

Two features are central. First, by Eq. 3.1 the proper-loss ERM is recovered *exactly* at the *proper-score point* $\lambda = 1$, where the calibration penalty vanishes and $\mathcal{L}_1 = \widehat{\mathcal{E}}$; decreasing λ toward 0 places increasing weight on calibration. Second, the surrogate $\widehat{C}$ should be *bounded* and built from a bounded proper scoring rule such as the Brier score [40, 12], rather than the unbounded log-loss, so that the objective is not dominated by a few confidently wrong examples; this is the same boundedness argument used by the RLCR framework [12] and is one reason this thesis prefers Brier-based surrogates.

Choosing the weights. The weight λ may be set in several ways, which the experimental chapters compare: *(i)* a fixed grid sweep over λ with selection by a validation criterion; *(ii) homoscedastic uncertainty weighting*, in which the weights are learned jointly with f by treating each term as a Gaussian log-likelihood with a learned observation-noise scale, after Kendall *et al.* [68]; and *(iii)* validation-based selection of a single trade-off parameter. Each induces a different effective constraint (Section 3.3); the uncertainty-weighting scheme in particular tends to drive the calibration weight to large values, which must be anticipated when interpreting its solutions. A core empirical comparison task of this thesis is to determine which estimator most effectively reduces miscalibration while preserving predictive accuracy.

3.2 The Calibration–Refinement Decomposition, Made Precise

The decomposition that Eq. 3.1 re-weights is classical. We have already seen in subsection 2.4.5 that for a proper loss (e.g., Brier Score) ℓ, the expected score admits the population decomposition [31, 42, 38]

$$\mathcal{E}(f) = \mathbb{E}\big[\ell(f(X), Y)\big] = \underbrace{\mathbb{E}\big[d_\ell\big(f(X), \gamma(f(X))\big)\big]}_{\mathcal{C}_\ell(f)\ \geq 0} + \underbrace{\mathbb{E}\big[e_\ell\big(\bar{\gamma}(f(X))\big)\big]}_{\mathcal{R}_\ell(f)} \tag{3.2}$$

where $\gamma(\mathbf{p}) = \mathbb{P}(Y \mid f(X) = \mathbf{p})$ is the true class distribution among inputs that receive prediction $\mathbf{p}$, d_ℓ is the Bregman divergence (reduces to KL divergence when we use log-loss) associated with ℓ, and $\mathcal{R}_\ell$ collects the resolution (sharpness) and the irreducible uncertainty $H(\gamma(Y))$. The calibration term $\mathcal{C}_\ell(f)$ is zero exactly when f is canonically calibrated (Section 2.2).

Berta *et al.* [8] give a variational form of Eq. 3.2 for cross-entropy and show that its two components are not minimized at the same iterate during training. In our composite objective, we estimate the calibration component by the surrogate $\widehat{C} \approx \mathcal{C}_\ell$ and obtain the refinement component as the residual $\widehat{\mathcal{R}}_\ell = \widehat{\mathcal{E}} - \widehat{\mathcal{C}}_\ell$ (Eq. 2.14), the empirical counterpart of $\mathcal{R}_\ell$, so the two estimated parts sum to the measured proper loss by construction. At $\boldsymbol{\lambda} = 1$, the proper loss, and hence the consistent estimation of $f^\star$, is recovered; as λ decreases toward 0, the learner is pushed increasingly toward calibration at a controlled cost in refinement.

Remark 3.2 (Surrogate gap). $\widehat{\mathcal{C}}$ is a finite-sample surrogate for $\mathcal{C}_\ell$ and is in general *biased* (section 2.4.2) [25, 33]. Because refinement is defined as the residual $\widehat{\mathcal{E}} - \widehat{\mathcal{C}}_\ell$, any bias in $\widehat{\mathcal{C}}_\ell$ enters $\widehat{\mathcal{R}}_\ell$ with the opposite sign; only the total $\widehat{\mathcal{E}}$ is estimator-independent. The theory below treats $\widehat{\mathcal{C}}_\ell$ as the quantity actually optimized; relating the optimized surrogate to the population calibration $\mathcal{C}_\ell$ is itself part of the analysis and is revisited in section 3.8.

3.3 From Penalized to Constrained ERM

The claim that calibration "constrains the hypothesis space" is made literal by the standard correspondence between penalized and constrained optimization.

Proposition 3.3 (Penalized–constrained correspondence)**.** *Let $\lambda \in (0, 1]$. By Eq. 3.1, minimizing the composite objective is equivalent to the penalized proper-loss problem $\min_{f\in\mathcal{F}} \widehat{\mathcal{E}}(f) + \mu\,\widehat{\mathcal{C}}_\ell(f)$ with $\mu = (1-\lambda)/\lambda \geq 0$. For every $\mu \geq 0$ there exists a level $c = c(\mu) \geq 0$ such that any minimizer $\hat{f}_\mu$ of the penalized problem is also a minimizer of the constrained problem*

$$\min_{f\in\mathcal{F}_c} \widehat{\mathcal{E}}(f), \qquad \mathcal{F}_c := \{\, f \in \mathcal{F} : \widehat{\mathcal{C}}_\ell(f) \leq c\}, \tag{3.3}$$

and $c(\mu)$ is non-increasing in μ (equivalently, non-decreasingly in λ).

Proof. The equivalence to the penalized proper-loss problem is the algebraic identity in Eq. 3.1 divided by λ. For the constrained form, take $c = \widehat{\mathcal{C}}(\hat{f}_\mu)$. Any f feasible for Eq. 3.3 satisfies $\widehat{\mathcal{C}}_\ell(f) \leq c$, and by optimality of $\hat{f}_\mu$, $\widehat{\mathcal{E}}(\hat{f}_\mu) + \mu c \leq \widehat{\mathcal{E}}(f) + \mu\widehat{\mathcal{C}}_\ell(f) \leq \widehat{\mathcal{E}}(f) + \mu c$, whence $\widehat{\mathcal{E}}(\hat{f}_\mu) \leq \widehat{\mathcal{E}}(f)$. Monotonicity of $c(\cdot)$ follows from the standard envelope argument for the Lagrangian. A formal statement requires mild regularity (e.g. lower semicontinuity of $\widehat{\mathcal{C}}$); see [51] for the convex case. □

Proposition 3.3 licenses the central object of the analysis: the *calibration-restricted class* $\mathcal{F}_c \subseteq \mathcal{F}$. Increasing the calibration weight shrinks c and hence $\mathcal{F}_c$. The remaining question is whether this shrinkage improves generalization.

3.4 Generalization via Rademacher Complexity

We measure the capacity of a class by its empirical Rademacher complexity $\widehat{\mathfrak{R}}_m(\cdot)$ (Section 2.7) [53]. Let $\ell \circ \mathcal{F} = \{(x, y) \mapsto \ell(f(x), y) : f \in \mathcal{F}\}$ denote the induced loss class for a $[0, 1]$-bounded loss ℓ.

Proposition 3.4 (Capacity is monotone in the calibration constraint)**.** *For any $c \geq 0$, the calibration-restricted class $\mathcal{F}_c \subseteq \mathcal{F}$ of Eq. 3.3 satisfies*

$$\widehat{\mathfrak{R}}_m(\ell \circ \mathcal{F}_c) \;\leq\; \widehat{\mathfrak{R}}_m(\ell \circ \mathcal{F}). \tag{3.4}$$

Consequently, with probability at least $1-\delta$*, every* $f \in \mathcal{F}_c$ *satisfies*

$$\mathcal{E}_\ell(f) \;\le\; \widehat{\mathcal{E}}_\ell(f) \;+\; 2\,\widehat{\mathfrak{R}}_m(\ell \circ \mathcal{F}_c) \;+\; 3\sqrt{\tfrac{\ln(2/\delta)}{2n}}, \tag{3.5}$$

so the uniform generalisation bound for the calibration-restricted class is no looser than that for $\mathcal{F}$*, and is strictly tighter whenever the constraint removes hypotheses that attain the supremum in* $\widehat{\mathfrak{R}}_m(\ell \circ \mathcal{F})$.

Proof sketch. $\mathcal{F}_c \subseteq \mathcal{F}$ implies $\ell \circ \mathcal{F}_c \subseteq \ell \circ \mathcal{F}$, and the Rademacher complexity is a supremum over the class, hence monotone under set inclusion; the bound Eq. 3.5 is the standard Rademacher generalisation bound [53, 51] applied to $\mathcal{F}_c$. □

Proposition 3.4 is deliberately modest: monotonicity is an immediate consequence of inclusion. Its value is conceptual—it identifies the precise mechanism by which calibration can act as capacity control—and it isolates the genuinely substantive question, namely *how much* the constraint reduces complexity. Quantifying $\widehat{\mathfrak{R}}_m(\ell \circ \mathcal{F}) - \widehat{\mathfrak{R}}_m(\ell \circ \mathcal{F}_c)$ for concrete classes (e.g. norm-bounded networks) is the non-trivial step and is taken up empirically in Chapter 4; obtaining a closed-form lower bound on the reduction is left as an open problem (Section 3.8).

3.5 A PAC-Bayesian View

PAC-Bayes bounds are well suited to the overparameterised models used for GRN inference, since they bound the risk of randomised predictors and have given the tightest known guarantees for neural networks [54, 55, 56]. McAllester's bound states that for a data-independent prior P and any posterior Q, with probability at least $1-\delta$,

$$\mathbb{E}_{f\sim Q}\big[\mathcal{E}_\ell(f)\big] \;\le\; \mathbb{E}_{f\sim Q}\big[\widehat{\mathcal{E}}_\ell(f)\big] \;+\; \sqrt{\frac{\mathrm{KL}(Q\,\|\,P) + \ln\frac{2\sqrt{n}}{\delta}}{2n}}. \tag{3.6}$$

We propose to read calibration-regularized training through Eq. 3.6 as follows. The composite objective Eq. 3.1 biases the learned posterior $Q_{\boldsymbol{\lambda}}$ toward the calibration-respecting region $\{f : \widehat{C}(f) \le c\}$. If the prior P encodes a preference for the same region—for example a prior concentrated on well-calibrated predictors, or a data-dependent prior in the sense of [55]—then the posterior need not travel far from the prior to fit the data, so the complexity term $\mathrm{KL}(Q_{\boldsymbol{\lambda}}\|P)$ is small and the bound tightens. Informally, calibration and a calibration-aware prior are "aligned," and alignment is what PAC-Bayes rewards.

Remark 3.5 (Status of this argument). The reading above is a *framework*, not a theorem. Turning it into a guarantee requires (a) constructing a valid prior P that provably charges the calibrated region without peeking at the test data, and (b) bounding $\mathrm{KL}(Q_{\boldsymbol{\lambda}}\|P)$ in terms of $\boldsymbol{\lambda}$. Both are non-trivial and are stated here as objectives of the thesis rather than as established results.

3.6 Interpretation: Calibration as Capacity Control

Sections 3.3–3.5 support a single picture. The calibration penalty does not add an objective unrelated to generalization; by Eq. 3.2 it re-weights a component that the proper risk already contains, and by Proposition 3.3 it is equivalent to optimizing within a shrunken class $\mathcal{F}_c$. Reducing the class, lowers variance (Proposition 3.4) at the possible cost of increased bias if $f^\star \notin \mathcal{F}_c$, which is precisely the behavior of a regularizer. The proper-score point $\boldsymbol{\lambda} = 1$ is the bias-free setting (it targets $f^\star$); decreasing λ towards 0 trades a controlled amount of bias for reduced variance and better calibration. This is the formal content of the thesis's hypothesis that end-to-end calibration shapes the bias–variance trade-off, and it yields a concrete, testable prediction: as λ decreases, calibration error and the generalization gap should fall together until bias from an over-restricted $\mathcal{F}_c$ dominates. We will verify this hypothesis in controlled experiments with synthetically generated data in later chapter.

3.7 Robustness Under Covariate Shift

The generalization guarantees of sections 3.4-3.5 are stated for the in-distribution setting, $\mathcal{D}_{\text{test}} = \mathcal{D}_{\text{train}}$. Under covariate shift (section 2.3), a natural question is whether the calibration-restricted class $\mathcal{F}_c$ (section 3.3) that improves in-distribution generalization also yields predictors whose calibration is more stable as $\mathbb{P}(X)$ shifts or whether the restriction only helps in absence of the shift. We distinguish three notions of robustness relevant to this question, following the empirical framework of Ovadia *et al.* [27], who benchmark both accuracy and calibration degradation under dataset shift and find that post-hoc recalibration in particular does not transfer reliably.

1. *Predictive Robustness*: degradation in a ranking-based metric (e.g. AUROC, AUPR) from $\mathcal{D}_{\text{train}}$ to $\mathcal{D}_{\text{test}}$, the standard notion of robustness in the broader shift literature.

2. *Calibration Robustness*: degradation in a calibration-specific metric (e.g. ECE, RMSCE, KSCE, or the calibration component $\mathcal{C}_\ell(f)$ of the decomposition Eq. 2.14) from $\mathcal{D}_{\text{train}}$ to $\mathcal{D}_{\text{test}}$, isolated from predictive performance. This is the notion most directly relevant to whether calibration-regularized training (Eq. 3.1) confers a benefit specific to calibration, rather than accuracy in general.

3. *Recalibration-map Transfer Robustness*: whether a post-hoc map g (section 2.5), fit on $\mathcal{D}_{\text{train}}$-distributed validation data, remains effective on $\mathcal{D}_{\text{test}}$ without refitting. Ovadia *et al.* [27] find that this typically fails to hold for standard post-hoc methods, which is the shift-specific instance of the general limitation already noted in section 2.5.8.

For a metric M, we define the *robustness gap* as below:

$$\text{RobustnessGap}_{\text{M}}(\text{method}) = \text{M}(\text{method}; \mathcal{D}_{\text{test}}) - \text{M}(\text{method}; \mathcal{D}_{\text{train}}) \tag{3.7}$$

For metrics where lower is better (e.g., ECE, RMSCE, KSCE etc.), a positive gap indicates

degradation under shift; for metrics where higher is better (AUROC, AUPR), a negative gap indicates the same. Comparing |`RobustnessGap`| across methods puts both metric orientations on the same footing. Under this framework, end-to-end calibration is considered more robust than a predictive-only baseline if |`RobustnessGap`| is smaller for the calibration-specific metrics of notion (2), independent of whether the predictive-robustness gap of notion (1) is comparable or worse. It is precisely this separation, calibration robustness without necessarily improved predictive robustness, that constitutes evidence for the mechanism this thesis studies (RQ1–RQ2), rather than a generic accuracy-under-shift effect already documented elsewhere.

This framework is deliberately dataset-agnostic. The exact setup depends on the experimental design, including which distributions are used as $\mathcal{D}_{\text{train}}/\mathcal{D}_{\text{test}}$ pairs, how many pairs are available, and how much detail the sample size allows in the analysis. These aspects are discussed in more detail in later chapter.

3.8 Assumptions, Scope, and Open Problems

For transparency we collect the assumptions on which the analysis rests and the questions it leaves open.

Assumption 3.6 (Standing assumptions)**.** *(i)* The sample S is drawn i.i.d. from $\mathcal{D}$; *(ii)* the loss ℓ is bounded (e.g. Brier), so the Rademacher bound applies; *(iii)* the calibration surrogate $\widehat{C}$ is lower semicontinuous, so Proposition 3.3 holds.

The following points qualify the results and mark the work that remains.

1. **Surrogate-to-population gap.** The analysis constrains the *empirical surrogate* $\widehat{C}$, whereas the quantity of interest is the population calibration Cal_S. Because $\widehat{C}$ is biased (Remark 3.2), a complete argument must bound $|\widehat{C}(f) - \mathrm{Cal}_S(f)|$ uniformly over $\mathcal{F}$ [33, 11].

2. **Magnitude of the complexity reduction.** Proposition 3.4 gives only monotonicity. A quantitative theory needs a lower bound on $\widehat{\mathfrak{R}}_m(\ell \circ \mathcal{F}) - \widehat{\mathfrak{R}}_m(\ell \circ \mathcal{F}_c)$ for a concrete class.

3. **PAC-Bayes prior.** The guarantee sketched in Section 3.5 depends on constructing an admissible calibration-aware prior (Remark 3.5).

4. **The i.i.d. assumption in the application.** Assumption 3.6(i) is benign in the controlled experiments but is *structurally violated* in the GRN setting, where candidate edges share genes and samples and the adjacency is fixed; no splitting scheme fully restores independence. The application chapter therefore frames its GRN study as an empirical case study rather than a validation of these bounds, and is explicit about this gap.

5. **Robustness under covariate shift.** The theory above targets in-distribution generalization. Whether calibration-restricted training (Section 3.3) improves calibration specifically, as opposed to predictive performance generally, under covariate shift is a distinct empirical

question, addressed in the GRN case study via a per-network robustness comparison across DREAM5's organism networks.

In summary, this chapter has reduced the thesis's central hypothesis to a small number of precise, checkable statements: a decomposition that the objective re-weights, an equivalence between penalised training and a calibration-restricted class, a monotone reduction in that class's complexity, and a PAC-Bayesian alignment argument. The established components are cited as such; the quantitative refinements and the PAC-Bayes construction are the theoretical contributions the remainder of the thesis develops and tests.

4 Simulation-Supervised Experiments

4.1 Overview and Objectives

The central question of this thesis is whether calibration can be improved by embedding it into the training objective as a structural component, that is, by solving

$$\min_{\theta\in\Theta} \widetilde{C}(f_\theta) + \lambda\,\widetilde{R}(f_\theta) \;=\; \min_{\theta\in\Theta} (1-\lambda)\,\widetilde{C}(f_\theta) + \lambda\,\widetilde{E}(f_\theta), \tag{4.1}$$

while statistical accuracy is preserved through an appropriate choice of $\lambda > 0$. Here $\widetilde{C}$ is an empirical calibration term, $\widetilde{E}$ the empirical total proper loss and $\widetilde{R} = \widetilde{E} - \widetilde{C}$ the refinement term, as introduced in Chapter 3.

This chapter examines the question in the one setting where every quantity involved can be compared with the truth. We use a one-dimensional binary classification problem whose conditional class probability $\eta(x) = \mathbb{P}(Y = 1 \mid X = x)$ is known in closed form. On real data, calibration can only be estimated from observed labels, typically through binned statistics whose own bias and variance blur the comparison. Here, in addition, we can measure how far a trained model is from η itself. That distance, MSE_η, is the reference quantity of the chapter: it separates a model that is genuinely closer to the true posterior from one that merely produces a flatter reliability diagram.

The chapter is primarily an answer to RQ3, which asks how end-to-end calibrated models compare with baselines on exact measures of calibration, refinement and generalisation across model capacity and regularisation strength. It also supplies the empirical evidence for RQ1 (does the calibration term act as an implicit regulariser, and how does end-to-end training compare with predictive-only training followed by post-hoc calibration?) and it constrains how RQ2 can be answered, since the theoretical arguments of Chapter 3 make predictions that can be checked here. RQ4, which concerns gene regulatory network inference, is addressed in the following chapters.

A controlled setting is necessary for four reasons. First, MSE_η can be computed only when η is known. Second, capacity can be varied in a way whose approximation error is known exactly (Section 4.2.2), so that any remaining miscalibration can be attributed to estimation rather than to a hypothesis class that cannot represent the truth. Third, training sets can be redrawn from the same distribution as often as needed, which makes variance across training samples a measurable quantity rather than a thought experiment. Fourth, samples are independent and identically distributed by construction, so the assumptions under which the results of Chapter 3 are derived hold exactly. This last point is the main difference to the real-data case study later in the thesis, where i.i.d. sampling cannot be assumed.

Table 4.1: Experiment questions of Track 1, the experiment that answers each of them, and where the result is reported. "1 seed" refers to data seed 1 and initialisation seed 42; "30 seeds" to data seeds $s = 1, \ldots, 30$ with initialisation seed $10000 + s$.

	Question	Seeds	Reported in
Q1.1	Does the calibration–refinement decomposition hold during training, and do the two components move in opposite directions?	1	Sec. 4.3.1
Q1.2	Is there a phase transition during training that the penalty can prevent?	1	Sec. 4.3.1
Q1.3	How do post-hoc methods compare with each other and with end-to-end training across capacity?	1	Sec. 4.3.5
Q1.4	Bias and variance of TS-based and isotonic refinement estimators	30	Sec. 4.3.5
Q1.5	Does post-hoc calibration degrade for overparameterised models?	1	Sec. 4.3.5
Q1.6	Brier score or cross-entropy as base loss?	1	Sec. 4.3.2
Q1.7	Does focal loss match explicit end-to-end calibration?	30	Sec. 4.3.6
Q1.8	Does overparameterisation help under end-to-end training?	1 and 30	Sec. 4.3.4
Q1.9	Does the penalty help more when $m \gg n$?	1 and 30	Sec. 4.3.4
Q1.10	Does the penalty reduce bias and variance simultaneously?	30	Sec. 4.3.3
Q1.11	Does the penalty move the model towards η, or only improve the reliability diagram?	1	Sec. 4.3.2
Q1.12	Is there an optimal λ, and does it depend on m?	1	Sec. 4.3.2
A	Is the gain caused by the penalty or by the reduced step size on $\widetilde{E}$?	30	Sec. 4.3.7
B	Additive penalty at a fixed learning rate	30	Sec. 4.3.7
C	Choosing λ on validation data versus validation early stopping	30	Sec. 4.3.7
D	Learned loss weights (uncertainty weighting)	30	Sec. 4.3.7

Table 4.1 lists the experiment questions of this track, the experiment that answers each of them and where the answer is reported. The table also records how many seeds each experiment uses. The distinction matters: the single-seed experiments are used to inspect mechanisms, and only the 30-seed experiments carry significance tests.

4.2 Experimental Setup

4.2.1 Data-Generating Process

Inputs and labels are generated as

$$X \sim \mathrm{Uniform}[-1,1], \qquad Y \mid X = x \sim \mathrm{Bernoulli}\big(\eta(x)\big), \qquad \eta(x) = 0.1 + 0.8\,\sigma\big(4\sin(\pi x)\big), \tag{4.2}$$

with $\sigma(t) = 1/(1+e^{-t})$. The Bayes-optimal predictor is $f^* = \eta$. Because $\eta(x) \in (0.1, 0.9)$ for every x, every label is genuinely random: even f^* misclassifies with probability $\min\{\eta(x), 1-\eta(x)\} \geq 0.1$. This Bernoulli draw is the only source of noise; no additional noise term is introduced. Two consequences are used repeatedly below.

- The irreducible Brier risk is $\mathbb{E}[\eta(X)(1-\eta(X))] \approx 0.147$ (estimated on the test set). For any predictor f that does not depend on the test labels, $\mathbb{E}[(f(X)-Y)^2] = \mathrm{MSE}_\eta(f) + \mathbb{E}[\eta(X)(1-\eta(X))]$. The Brier score and MSE_η therefore rank models almost identically, and they differ mainly by this constant.
- The Bayes classifier is correct with probability $\max\{\eta(x), 1-\eta(x)\} \leq 0.9$. A top-label confidence above 0.9 is therefore always overconfident in this problem, whichever model produces it.

Each training set has $n_{\mathrm{train}} = 100$ points and is redrawn for every data seed. The test set has $n_{\mathrm{test}} = 50{,}000$ points, is drawn once with a fixed seed and is shared by all runs, so that differences between runs are not confounded by test-set noise. The training set is deliberately small so that the larger models can interpolate it.

4.2.2 Model and Hypothesis Class

The learner is a logistic model on a Legendre polynomial basis,

$$p_\theta(x) = \sigma\Big(\sum_{k=0}^{m} \theta_k\, \phi_k(x)\Big), \tag{4.3}$$

where $\phi_0, \ldots, \phi_m$ are the Legendre polynomials on $[-1,1]$ and the degree m controls capacity (the model has $m+1$ parameters). Following the supervisor's recommendation, we use this model instead of a neural network because it isolates the effect of capacity from confounders such as depth, normalisation layers and optimiser heuristics. The Legendre basis spans the same space as the monomials but is numerically well conditioned.

The capacity grid is $m \in \{5, 10, 100, 120, 200, 300, 1000\}$. Using the rule of thumb $m \approx n^r$, it covers an underparameterised regime ($m \in \{5, 10\}$, $r \approx 0.35$–0.5), the interpolation threshold ($m = 100 = n_{\mathrm{train}}$, $r = 1$), a near-threshold overparameterised regime ($m \in \{120, 200, 300\}$, $r \approx 1.04$–1.24) and a strongly overparameterised regime ($m = 1000$, $r = 1.5$). The three primary

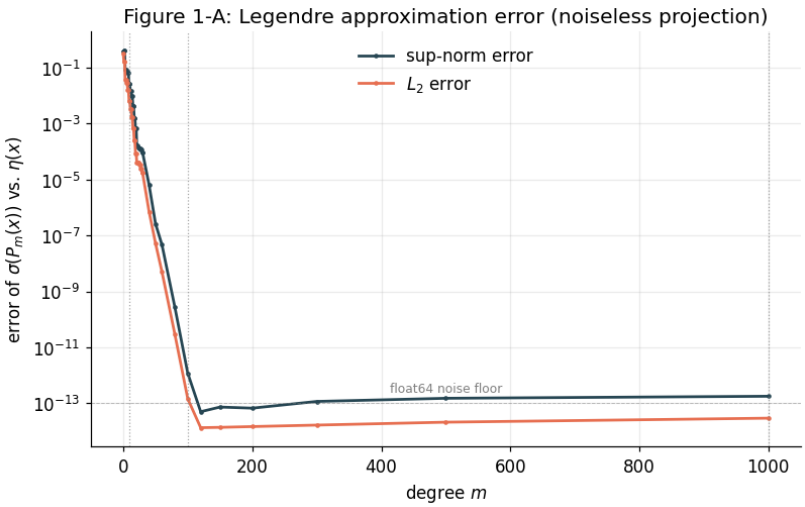


Figure 4.1: Approximation error of $\sigma(P_m)$ relative to η as a function of the degree m, for the noiseless Legendre projection of $g = \sigma^{-1} \circ \eta$ (no training data involved). Dotted vertical lines mark the primary capacities $m \in \{10, 100, 1000\}$.

capacities are $m \in \{10, 100, 1000\}$; the full grid is used where the question concerns the shape of a curve in m. Parameters are initialised as $\theta \sim \mathcal{N}(0, 0.01\, I)$, so all models start with predictions close to 0.5. The initialisation seed is set independently of the data seed.

Can the hypothesis class represent η? Since η is bounded away from 0 and 1, the function $g = \sigma^{-1} \circ \eta$ is continuous and bounded on $[-1, 1]$. By the Weierstrass approximation theorem it can be approximated uniformly by polynomials, and because σ is uniformly continuous, $\sigma(P_m) \to \eta$ uniformly as $m \to \infty$. Figure 4.1 shows how fast this happens. The figure uses no data at all: g is projected onto the first $m + 1$ Legendre polynomials by Gauss–Legendre quadrature, and σ of the projection is compared with η.

The convergence is spectral, which is expected because g is real-analytic. At $m = 10$ the approximation error is small but not zero (sup-norm 2.6×10^{-2}, L_2 error 6.6×10^{-3}). At $m = 100$ it has fallen to about 10^{-12}, which is at the level of floating-point noise, and it stays there up to $m = 1000$. Both $m = 100$ and $m = 1000$ can therefore represent η essentially exactly, and every deviation from η observed at these capacities is estimation error: it comes from the finite sample, the optimisation and overfitting, not from the function class.

The error at $m = 10$ is also small in absolute terms. Its squared L_2 error, about 4×10^{-5}, is a lower bound on the MSE_η any model of degree 10 can reach. The trained models in this chapter have $\mathrm{MSE}_\eta \approx 10^{-2}$ at best, so approximation bias accounts for well under one percent of their error even at $m = 10$.

4.2.3 Training Objective and Weight Schemes

End-to-end objective. All end-to-end models minimise the convex combination

$$\mathcal{L}_\lambda(\theta) = (1-\lambda)\,\widetilde{C}_{\text{soft}}(\theta) + \lambda\,\widetilde{E}(\theta), \tag{4.4}$$

where $\widetilde{E}$ is either the cross-entropy (CE) or the Brier score on the training set. The calibration term is the soft-binned estimator

$$\widetilde{C}_{\text{soft}}(\theta) = \sum_{b=1}^{B}\Big(\frac{1}{n}\sum_{i=1}^{n} w_{ib}\Big)\big(\bar{y}_b - \bar{p}_b\big)^2, \qquad w_{ib} = \text{softmax}_b\Big(-\frac{(p_i - c_b)^2}{2h^2}\Big), \tag{4.5}$$

with bin centres $c_b = (b - \frac{1}{2})/B$, $B = 15$ bins and bandwidth $h = 0.08$. Here $\bar{p}_b$ and $\bar{y}_b$ are the $w_{\cdot b}$-weighted means of predictions and labels in bin b. The term is differentiable, and it is a smoothed version of a binned squared calibration error on the training predictions.

In computing its gradient, the bin weights w_{ib} are treated as constants (a straight-through approximation). The gradient therefore flows into the penalty through the predictions inside $\bar{p}_b$, but not through the assignment of predictions to bins. This is an approximation of the true gradient of (4.5), and all results below refer to it.

Role of λ and the limiting cases. At $\lambda = 1$, (4.4) reduces to the plain proper loss, which is the predictive-only baseline. The objective is formally defined for $\lambda \in (0, 1]$. We nevertheless include $\lambda = 0$ as a limiting case, because it shows what the calibration term does on its own. Unlike $\widetilde{E}$, the calibration term is not a proper scoring rule: any calibrated predictor minimises it, including the constant predictor that outputs the base rate everywhere. *Decreasing* λ increases the weight on calibration. This is the opposite of the direction suggested by the wording of Q1.10 in the experiment plan; we follow the direction implied by (4.4) throughout.

An identity that matters for interpretation. All models are trained with full-batch gradient descent at a fixed learning rate η_{lr}. One step on (4.4) can be rewritten as

$$\theta \leftarrow \theta - \eta_{\text{lr}}\big[(1-\lambda)\nabla\widetilde{C}_{\text{soft}} + \lambda\nabla\widetilde{E}\big] = \theta - (\eta_{\text{lr}}\lambda)\Big[\nabla\widetilde{E} + \tfrac{1-\lambda}{\lambda}\nabla\widetilde{C}_{\text{soft}}\Big]. \tag{4.6}$$

Training on the convex objective with learning rate η_{lr} is therefore *exactly* the same as training on the additive objective $\widetilde{E} + \beta\,\widetilde{C}_{\text{soft}}$ with $\beta = (1-\lambda)/\lambda$ and learning rate $\eta_{\text{lr}}\lambda$. We confirmed this numerically: after 1,200 epochs the two parameter vectors agree to within 3×10^{-16}.

The consequence is that lowering λ changes two things at once. It strengthens the calibration term relative to the proper loss, and it shrinks the effective step size on the proper loss by a factor of λ. With a fixed budget of epochs and no early stopping, the second effect alone behaves like stopping plain training early. Berta et al. [8] study exactly that mechanism. Separating the two effects is the purpose of ablation A in Section 4.3.7.

Weight schemes. Four ways of setting the relative weight of the two terms are studied.

1. *Fixed grid.* The main experiments use $\lambda \in \{0, 0.1, 0.5, 1\}$. The single-seed sweep behind Figure 4.5 uses a finer 21-point grid $\lambda \in \{0, 0.05, \ldots, 1\}$; the finer grid changes where the optimum is located (Section 4.3.2).

2. *Additive penalty.* The objective $\widetilde{E} + \beta\,\widetilde{C}_{\text{soft}}$ is trained at the unchanged learning rate 0.5, with $\beta \in \{0, 0.5, 1, 2\}$ from the experiment plan. We add $\beta \in \{9, 19\}$, the values that correspond to $\lambda = 0.1$ and $\lambda = 0.05$ via (4.6). Because the step size on $\widetilde{E}$ is the same in every run, this form isolates the effect of the penalty.

3. *Validation-based selection.* λ is chosen from $\{0.05, 0.1, 0.2, 0.3, 0.5, 0.7, 1\}$ by the Brier score on a held-out validation set of 20 points (an 80/20 split of the training data). This is the realistic counterpart of the fixed-grid experiments, whose "best" λ is read off the ground-truth MSE_η on the test set. That quantity is not available in practice, so selecting on it is an oracle choice.

4. *Learned weights (uncertainty weighting).* Following Kendall et al. [**?**], the loss weights are learned jointly with θ by minimising

$$\mathcal{L}_{\text{UW}}(\theta, s) = \tfrac{1}{2} e^{-s_C}\,\widetilde{C}_{\text{soft}}(\theta) + \tfrac{1}{2} e^{-s_E}\,\widetilde{E}(\theta) + \tfrac{1}{2}(s_C + s_E), \tag{4.7}$$

   with log-variances $s = (s_C, s_E)$ initialised at zero, updated with the same learning rate as θ, and clipped to $[-8, 8]$. For fixed θ the minimiser is $s_i^* = \log L_i$, which gives each loss the effective weight $1/(2L_i)$. The alternative parameterisation $e^{-s}L + \frac{1}{2}s$ leads to the same effective weight. The exact form of the scheme was not fixed in the experiment plan, so this experiment is exploratory.

4.2.4 Baselines

Predictive-only training. The model is trained on the plain proper loss ($\lambda = 1$) with CE or Brier as the base loss. The CE baseline is the reference in all significance tests.

Predictive-only training followed by post-hoc calibration. The 100 training points are split 80/20. The baseline model is trained on the 80 points, and four recalibration maps are fitted on the 20 held-out points:

- temperature scaling (one parameter, fitted by NLL) [1];
- Platt scaling (two parameters) [5];
- isotonic regression [6];
- histogram binning with 15 equal-width bins [6].

The end-to-end model it is compared with is trained on all 100 points. This is deliberate: not having to give up training data for a calibration hold-out is one of the practical arguments for end-to-end calibration, and equalising the data would remove that trade-off from the comparison.

Focal loss. Focal loss [41, 48] is included as an implicit calibration method, with $\gamma \in \{0, 1, 2, 5\}$. We use the standard asymmetric form

$$\mathcal{L}_{\text{Focal}}(\theta) = -\frac{1}{n}\sum_{i=1}^{n}\Big[y_i(1-p_i)^{\gamma}\log p_i + (1-y_i)\,p_i^{\gamma}\log(1-p_i)\Big], \tag{4.8}$$

which down-weights easy examples of both classes. The experiment plan wrote a single modulating factor $(1-p_i)^{\gamma}$ for both terms. That version does not down-weight confidently correct negative examples (for $y = 0$ and $p = 0.05$ its factor is 0.90 instead of 0.0025 at $\gamma = 2$), so it does not implement the mechanism of the cited works. Both forms coincide at $\gamma = 0$, where they reduce to CE; we verified that $\gamma = 0$ reproduces the CE training trajectory exactly.

Controls introduced for the ablations. Two further baselines are used in Section 4.3.7. The *step-size-matched control* is plain CE trained with learning rate 0.5λ. By (4.6), it differs from end-to-end training at λ only in the absence of the penalty. *Validation early stopping* trains plain CE on 80 points and keeps the checkpoint with the lowest Brier score on the 20 validation points among all 1,200 epochs. This is checkpoint selection rather than stopping with a patience rule, and it is the natural competitor to validation-based selection of λ.

4.2.5 Evaluation Metrics

All metrics are computed on the fixed test set.

Metrics exact up to Monte Carlo error. These are computed against η rather than the labels:

- $\text{MSE}_{\eta} = \frac{1}{N}\sum_j \big(p_\theta(x_j) - \eta(x_j)\big)^2$, the primary metric of this chapter.
- The *ground-truth refinement* used for Q1.4: $\sum_b \frac{n_b}{N}\bar{\eta}_b(1-\bar{\eta}_b)$, where $\bar{\eta}_b$ is the mean of the true $\eta(x)$ over test points in bin b of p_θ (30 bins). It approximates the refinement term $\mathbb{E}[\,C(1-C)\,]$ with $C = \mathbb{E}[Y \mid f(X)]$, following Bröcker [42] and Berta et al. [8], but uses the true η instead of noisy labels.

Metrics estimated from test labels.

- Test CE and Brier score.

- The hard-binned Brier decomposition with 10 equal-width bins in p: $\widehat{C}_{\text{Brier}} = \sum_b \frac{n_b}{N}(\bar{p}_b - \bar{y}_b)^2$, with refinement defined as the residual so that total = calibration + refinement holds exactly.
- For the CE training curves in Figure 4.2 only, the binned analogue $\sum_b \frac{n_b}{N} \mathrm{KL}(\bar{y}_b \,\|\, \bar{p}_b)$. It follows the correspondence between log-loss and KL divergence in [8].
- Top-label ECE with 10 bins of confidence $\max(p, 1-p) \in [0.5, 1]$, compared with the accuracy of the thresholded prediction.
- Its worst-bin (MCE) and root-mean-square (RMSCE) [26] variants.
- The binning-free KS calibration error (KSCE) [36].

The two calibration notions differ in an important way. The Brier calibration term compares the predicted probability with the empirical frequency of $Y = 1$, which is a canonical (distribution-level) notion. Top-label ECE compares confidence with the accuracy of the decision and ignores which class the probability refers to. As Section 4.3.6 shows, the two can rank methods differently.

4.2.6 Implementation and Reproducibility

Optimisation. All models are trained by full-batch gradient descent with learning rate 0.5 for 1,200 epochs, without early stopping unless stated otherwise, in double precision. The soft-ECE penalty uses $B = 15$ and $h = 0.08$; the evaluation decomposition uses 10 bins. Experiments Q1.1–Q1.12 were implemented in PyTorch with automatic differentiation. The ablations A–D were implemented in NumPy with the analytic gradients of the same objectives. The NumPy implementation reproduces a value from the PyTorch runs (test CE 0.5854648820627176 for one fixed configuration) to within 10^{-15}, and it reproduces the corresponding rows of Table 4.3 to all reported decimal places.

Seeds. Mechanism-level experiments (Figures 4.2–4.8) use one data seed (1) and one initialisation seed (42). Multi-seed experiments use data seeds $s = 1, \ldots, 30$ with initialisation seed $10000 + s$ and, where a split is needed, split seed $20000 + s$. The single-seed post-hoc experiment uses split seed 123.

Statistical testing. Paired comparisons over the 30 seeds use the Wilcoxon signed-rank test with a Bonferroni correction over the number of comparisons stated with each table. Markers are * for $p_{\text{Bonf}} < 0.05$ and ** for $p_{\text{Bonf}} < 0.01$. Variances are reported descriptively; no test on variances was specified in the protocol.

Compute and code. The largest sweep (360 training runs) took about eight minutes on Google Colab; the ablations took of the order of ten minutes in total. All notebooks and the full-resolution

figures are available at the project repository.

4.3 Results

The results are presented in the order in which the experiments build on each other.

1. How calibration and refinement evolve during a single training run (Section 4.3.1).
2. What happens at the end of training as λ varies (Section 4.3.2).
3. Generalisation and variance across 30 seeds (Section 4.3.3).
4. The role of capacity (Section 4.3.4).
5. The comparison with post-hoc calibration and focal loss (Sections 4.3.5 and 4.3.6).
6. The ablations of Section 4.3.7.

The ablations change the interpretation of several earlier results, so the first sections describe what was observed and point forward where the ablations qualify it. The interpretation itself is collected in Section 4.4.

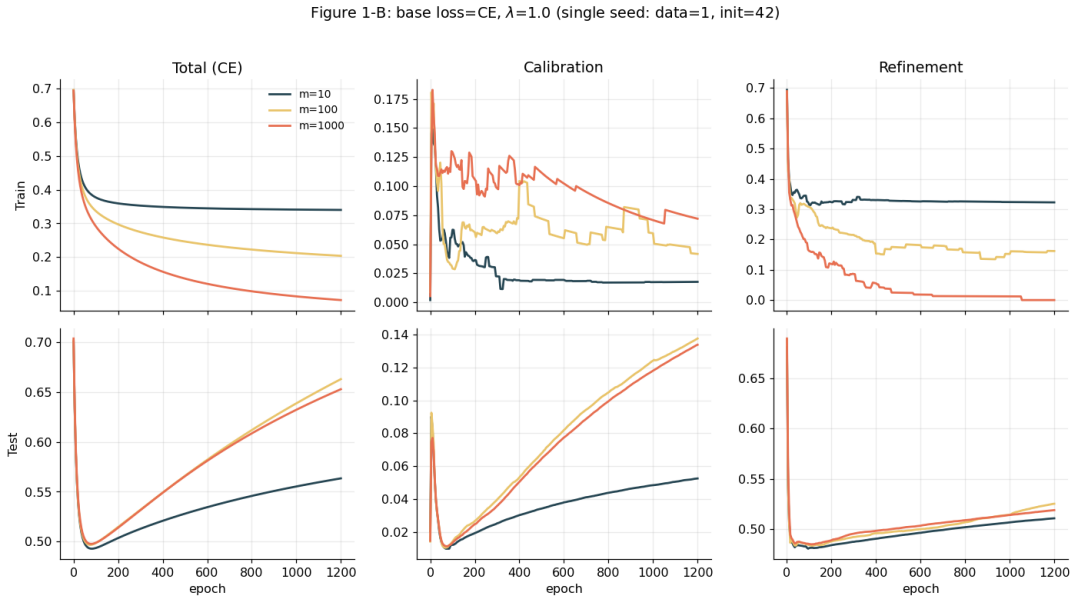
4.3.1 Training Dynamics of the Calibration–Refinement Decomposition

Figure 4.2 decomposes the cross-entropy into its calibration and refinement components over the course of training, on the training set (upper rows) and the test set (lower rows). It shows two configurations: the predictive-only baseline ($\lambda = 1$) and end-to-end training with $\lambda = 0.1$.

Q1.1: do calibration and refinement move in opposite directions? In (4.4) the decomposition holds by construction, because refinement is defined as the residual of the total loss. The substantive question is how the two components behave during training.

Under the baseline, their behaviour on the test set diverges clearly once the model is large enough to fit the training labels. Between initialisation and the final epoch, test refinement at $m = 100$ falls from 0.688 to 0.525, while test calibration rises from 0.014 to 0.138. The values at $m = 1000$ are nearly identical ($0.689 \to 0.519$ and $0.015 \to 0.134$). At $m = 10$ the calibration term also increases, but much less ($0.020 \to 0.053$).

The curves show the structure behind these endpoints. During roughly the first 80 to 100 epochs, both components decrease together. After that, test calibration increases almost linearly, while test refinement stays nearly flat and even rises slightly. On the training set the picture is different: refinement keeps falling and reaches almost zero at $m = 1000$, which reflects the model memorising the training labels.



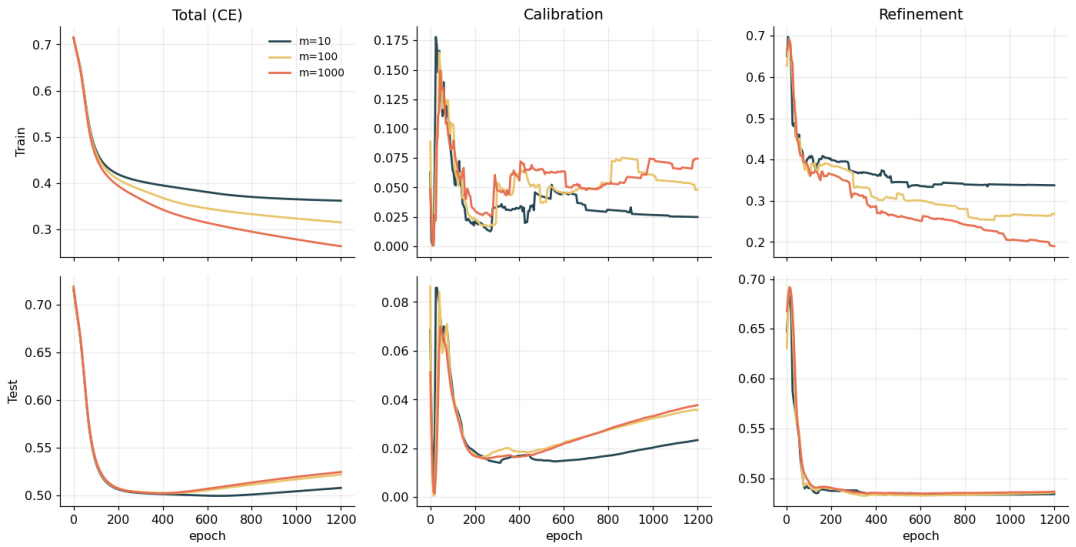


Figure 4.2: Calibration–refinement decomposition of the cross-entropy over 1,200 epochs, for $m \in \{10, 100, 1000\}$ (single seed). Top block: predictive-only baseline ($\lambda = 1$). Bottom block: end-to-end training with $\lambda = 0.1$. Within each block, the upper row is the training set and the lower row the test set. The figure is cropped from the full grid, which covers $\lambda \in \{0, 0.1, 0.5, 1\}$ for both the CE and the Brier base loss and is available on GitHub.

The two components are therefore not minimised at the same time, which is the pattern Berta et al. [8] describe. A precise summary of what we observe is this: after an initial phase of joint improvement, test calibration degrades while test refinement has essentially saturated. The late increase in test loss is almost entirely a calibration effect.

Q1.2: is there a phase transition, and does the penalty prevent it? For $m \geq 100$ the transition between the two phases is sharp and occurs early, at roughly epoch 80. At $m = 10$ it is present but mild.

With $\lambda = 0.1$ the transition is delayed and much weaker. Test calibration reaches its minimum at around epoch 250 to 400 and ends at 0.036 ($m = 100$) and 0.038 ($m = 1000$) instead of 0.138 and 0.134. Test refinement ends at 0.486–0.487, slightly lower than under the baseline. The penalty therefore postpones and damps the calibration-degradation phase; it does not remove it.

The identity (4.6) gives a second reading of the same curves. At $\lambda = 0.1$ the proper-loss part of the objective moves ten times more slowly than under the baseline. After 1,200 epochs it has progressed about as far as 120 epochs of plain CE, which is close to where the baseline's test loss is lowest. Whether the damping comes from the penalty or from this slower time scale cannot be decided from Figure 4.2; ablation A (Section 4.3.7) addresses exactly this.

Reliability diagrams. Figure 4.3 shows the same two configurations as snapshots of the confidence histogram and the reliability diagram.

Under the baseline, the models at $m \geq 100$ become increasingly confident. By epoch 1,200 roughly 60% of the test points have a top-label confidence above 0.95, and the reliability curves bend clearly below the diagonal at high confidence (ECE 0.114 and 0.127, against 0.071 at $m = 10$). Confidence above 0.9 is always overconfident in this problem (Section 4.2.1), so this mass is miscalibrated by construction.

With $\lambda = 0.1$, the confidence mass concentrates between 0.90 and 0.95, the curves stay close to the diagonal over most of the range, and the final ECE is 0.061, 0.082 and 0.082 for $m = 10$, 100 and 1000. Early snapshots (epochs 1 and 50) are dominated by a few sparsely populated bins and should not be over-interpreted.

4.3.2 Calibration–Refinement Trade-off Across Weights

This section looks at the final epoch as a function of λ. Figure 4.4 shows the learned functions. Figure 4.5 shows MSE_η, ECE and the Brier score on the 21-point grid, and Figure 4.6 splits the Brier score into its calibration and refinement terms. All three figures use a single seed.

Q1.11: does the penalty move the model towards η? The limiting case $\lambda = 0$ answers the question for the calibration term on its own, and the answer is no. Without the proper loss, the

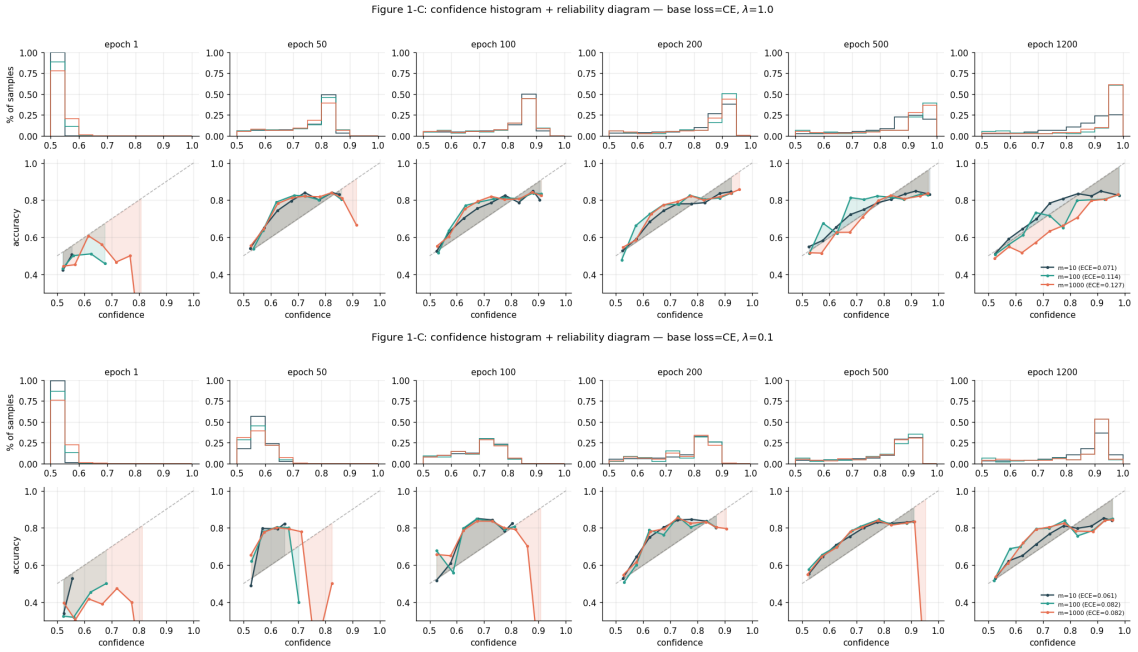


Figure 4.3: Confidence histograms (upper rows) and reliability diagrams (lower rows) at epochs 1, 50, 100, 200, 500 and 1,200 for the CE base loss (single seed). Top block: $\lambda = 1$. Bottom block: $\lambda = 0.1$. Shaded areas show the gap to the diagonal. The legend in the last panel of each block gives the final ECE. The snapshot at epoch 500 replaces the epoch 400 listed in the experiment plan. Cropped from the full figure with $\lambda \in \{0, 0.1, 0.5, 1\}$ and both base losses, available on GitHub.

model collapses to a nearly constant prediction around 0.5. At $m = 1000$ the test predictions have mean 0.4995 and standard deviation 0.032, and the CE and Brier families coincide exactly because the base loss no longer enters the objective.

Measured against η, this solution is useless: $\mathrm{MSE}_\eta \approx 0.105$ at every capacity, which is essentially the 0.103 of the constant predictor 0.5. It is nevertheless well calibrated in the canonical sense. Its Brier calibration term is the lowest of all runs at $m = 1000$ (0.0036), and all of its loss sits in refinement (0.243–0.250, close to the 0.25 of a coin flip; Figure 4.6). This is the calibration–refinement trade-off in its purest form.

Its top-label ECE is not particularly low (0.087, 0.078 and 0.059 for the three capacities). In the 30-seed experiment of Section 4.3.3, however, its ECE at $m \geq 100$ is lower than at $\lambda = 0.1$, although its MSE_η is roughly four to five times higher. A calibration metric alone would not have identified this model as degenerate.

For $\lambda > 0$ the picture changes. At every capacity and for both base losses, MSE_η decreases monotonically as λ is lowered from 1 towards the optimum. Figure 4.4 shows what this looks like:

- At $\lambda = 1$, the models with $m \geq 100$ produce narrow spikes towards 0 and 1 (for example near $x \approx -0.7$ and $x \approx -0.6$), where they fit individual noisy training labels. They also

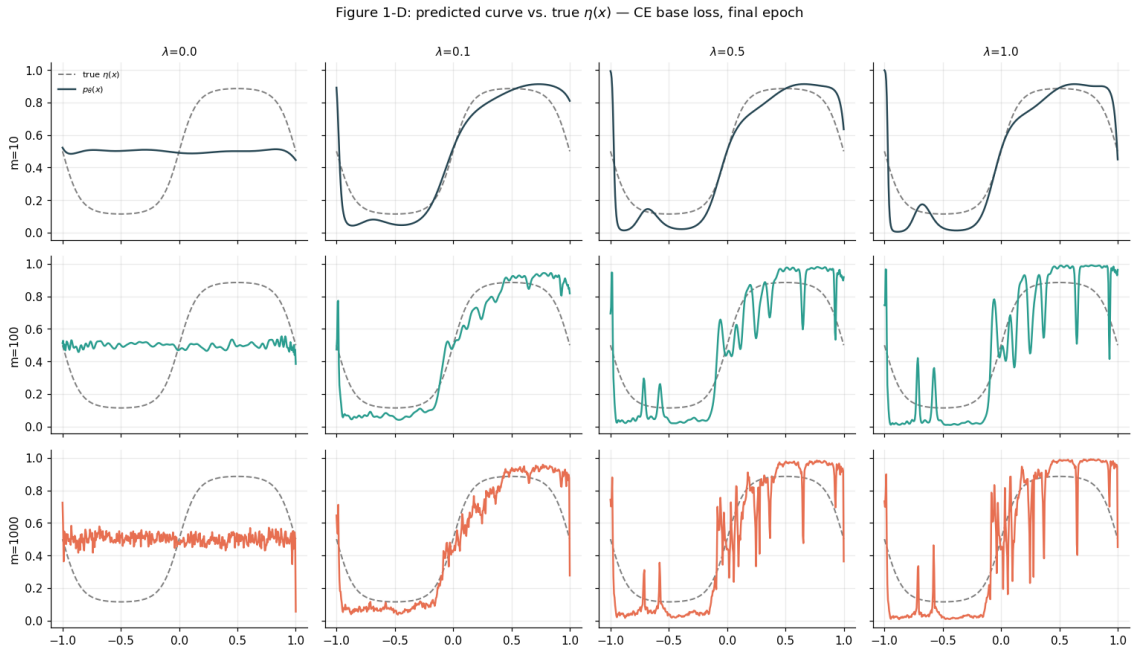


Figure 4.4: Learned function $p_\theta(x)$ (solid) against the true posterior $\eta(x)$ (dashed) at the final epoch, CE base loss, for $m \in \{10, 100, 1000\}$ (rows) and $\lambda \in \{0, 0.1, 0.5, 1\}$ (columns), single seed. The corresponding figure for the Brier base loss is available on GitHub.

produce a plateau at 0.95–1 on the interval where $\eta \approx 0.88$.

- At $\lambda = 0.1$, most spikes disappear and the function follows η over most of the domain.
- Three kinds of deviation remain at $\lambda = 0.1$: a moderate overshoot on the upper plateau, high-frequency ripple at $m = 1000$, and boundary effects at $x = \pm 1$. The boundary effects are most visible for $m = 10$ at $x = -1$, where the prediction approaches 0.9 although $\eta(-1) = 0.5$.

Within the interior of the λ range, then, the penalised objective does move the model towards η and not only towards a better-looking reliability diagram.

Q1.12: is there an optimal λ, and does it depend on m? On the fine grid, the λ that minimises MSE_η is:

- 0.05 for the CE base loss at all three capacities, with $\mathrm{MSE}_\eta = 0.0097$, 0.0118 and 0.0123;
- 0.15, 0.10 and 0.10 for the Brier base loss, with $\mathrm{MSE}_\eta = 0.0091$, 0.0112 and 0.0112.

The optimum is thus interior in both families and nearly independent of capacity, but it depends on the base loss.

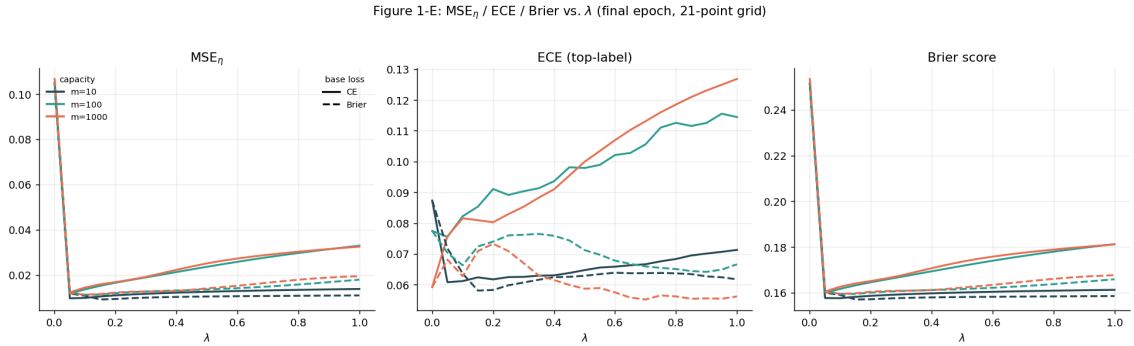


Figure 4.5: MSE_η, top-label ECE and Brier score at the final epoch as functions of λ (21-point grid, single seed). Colour encodes capacity and line style the base loss.

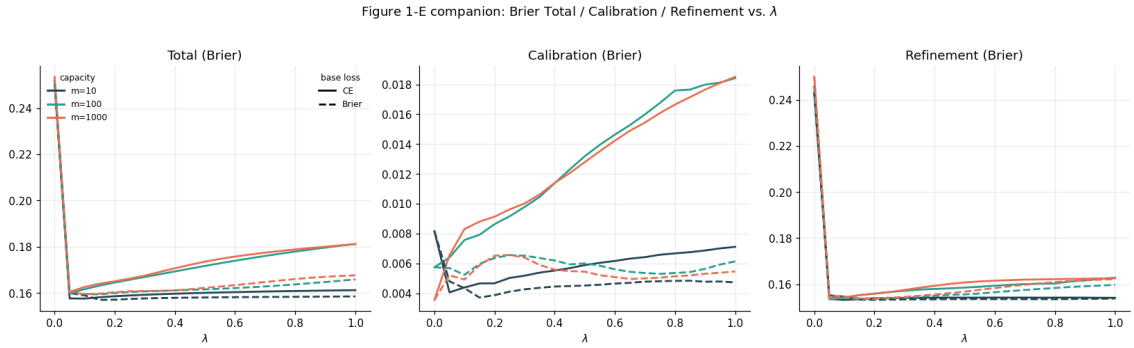


Figure 4.6: Hard-binned Brier decomposition (total = calibration + refinement) at the final epoch as a function of λ, for the same runs as Figure 4.5. Every model is scored with the Brier decomposition, whichever loss it was trained with.

The CE optimum lies at the second-smallest grid point. We cannot exclude that a grid refined near zero would place it lower still; the only thing established is that it lies between 0 and 0.1. The coarse grid $\{0, 0.1, 0.5, 1\}$ used in the multi-seed experiments places the same optimum at 0.1, simply because 0.05 is not in that grid. This is why the experiments below use $\lambda = 0.1$ in some places and $\lambda = 0.05$ in others: the difference reflects grid resolution, not a different optimum.

At the optimum, sharpness is not sacrificed. The Brier refinement term in Figure 4.6 is lowest for $\lambda \in [0.05, 0.1]$ (for example 0.154 against 0.163 at $\lambda = 1$ for CE and $m = 1000$), so both components improve together.

The ECE-optimal and MSE_η-optimal values of λ do not always agree. For the Brier base loss at $m \geq 100$, ECE is lowest at $\lambda = 0.7$–0.9, while MSE_η is lowest at 0.10. A λ chosen to minimise ECE would therefore not select the model closest to η.

All values in this paragraph come from a single seed and are selected with MSE_η on the test set, which is an oracle criterion. The realistic counterpart is ablation C.

Q1.6: Brier score or cross-entropy as base loss? At $\lambda = 1$, the Brier base loss is better than CE on both metrics at every capacity, and the gap grows with capacity:

- at $m = 1000$: MSE_η 0.0195 against 0.0325, ECE 0.056 against 0.127;
- at $m = 10$: MSE_η 0.0109 against 0.0138, ECE 0.062 against 0.071.

The Brier family is also much less sensitive to λ. Its calibration term stays around 0.004–0.007 over nearly the whole range, while the CE calibration term at $m \geq 100$ grows steadily with λ, from about 0.006 to 0.018 (Figure 4.6). At the respective optima, however, the difference largely vanishes (at most 0.0011 in MSE_η).

This agrees with the boundedness argument for the Brier score made in Chapter 3. We add a hypothesis that we have not tested separately. The gradient of the Brier score with respect to the logit, $2(p - y)\,p(1 - p)$, is at most half that of CE and vanishes for confident predictions. It therefore slows the fitting of individual noisy labels, which is a step-size effect of the same kind that ablation A isolates for λ.

These comparisons rest on a single seed. The 30-seed experiments were run for the CE base loss only.

4.3.3 Effect on Generalisation and on Variance Across Seeds

Table 4.3 and Figure 4.7 report the 30-seed results for the CE base loss on the coarse grid.

End-to-end ($\lambda = 0.1$) against the baseline ($\lambda = 1$). With $\lambda = 0.1$ the model is better than the baseline on every metric at every capacity, and the difference is significant in 25 of 27 metric–capacity combinations (paired Wilcoxon test, Bonferroni over three capacities). The exceptions are MCE at $m = 100$ and $m = 1000$. The average calibration error (ECE, RMSCE, KSCE and the Brier calibration term) improves reliably, but the gap in the single worst bin does not.

The size of the effect depends strongly on capacity:

- MSE_η falls from 0.0189 to 0.0101 at $m = 10$, from 0.0482 to 0.0127 at $m = 100$ and from 0.0378 to 0.0129 at $m = 1000$;
- ECE falls from 0.069 to 0.046, from 0.122 to 0.055 and from 0.114 to 0.055;
- the Brier refinement term also falls significantly (for example $0.177 \to 0.155$ at $m = 100$), so the gain in calibration is not paid for with sharpness.

Generalisation gap. In the single-seed runs of Figure 4.2, the gap between training and test cross-entropy at the final epoch shrinks under the penalty. At $m = 1000$ it goes from approximately 0.58 (training loss about 0.07, test loss 0.65) to approximately 0.26 (about 0.26 against 0.52). At

Table 4.2: Mean and across-seed variance of MSE_η at $m = 1000$ (CE base loss, 30 seeds). Decreasing λ means increasing weight on the calibration term.

λ	mean MSE_η	std	variance
1.0	0.0378	0.0083	7.0×10^{-5}
0.5	0.0289	0.0075	5.6×10^{-5}
0.1	0.0129	0.0052	2.7×10^{-5}
0.0	0.0580	0.0456	2.1×10^{-3}

$m = 10$ it goes from approximately 0.22 to approximately 0.15. These values are read from the curves and are approximate.

Q1.10: are bias and variance reduced simultaneously? Table 4.2 shows the mean and the across-seed variance of MSE_η at $m = 1000$ as the weight on calibration increases, that is, as λ decreases.

From $\lambda = 1$ over 0.5 to 0.1, both quantities fall together: the mean by about two thirds, the variance from 7.0×10^{-5} to 2.7×10^{-5}. The same pattern holds at $m = 100$ (standard deviation $0.0128 \to 0.0090 \to 0.0054$) and at $m = 10$ ($0.0078 \to 0.0072 \to 0.0054$).

At $\lambda = 0$ the relationship breaks down completely. The mean rises to 0.058, worse than the baseline, and the variance to 2.1×10^{-3}, about eighty times its value at $\lambda = 0.1$. Once the proper loss no longer anchors the model to the data, which degenerate solution it reaches depends heavily on the seed.

The answer to Q1.10 is therefore yes, over the interior range $\lambda \in [0.1, 1]$ that was tested. The effect is not monotone down to $\lambda = 0$.

Two qualifications apply. First, "bias" and "variance" here are the mean and the across-seed variance of MSE_η. They are not the squared-bias and variance terms of the bias–variance decomposition of the predictor itself, which would require the pointwise mean and variance of $p_\theta(x)$ across seeds; we did not compute these. Second, ablation A shows that the step-size control reproduces this pattern as well (Section 4.3.7).

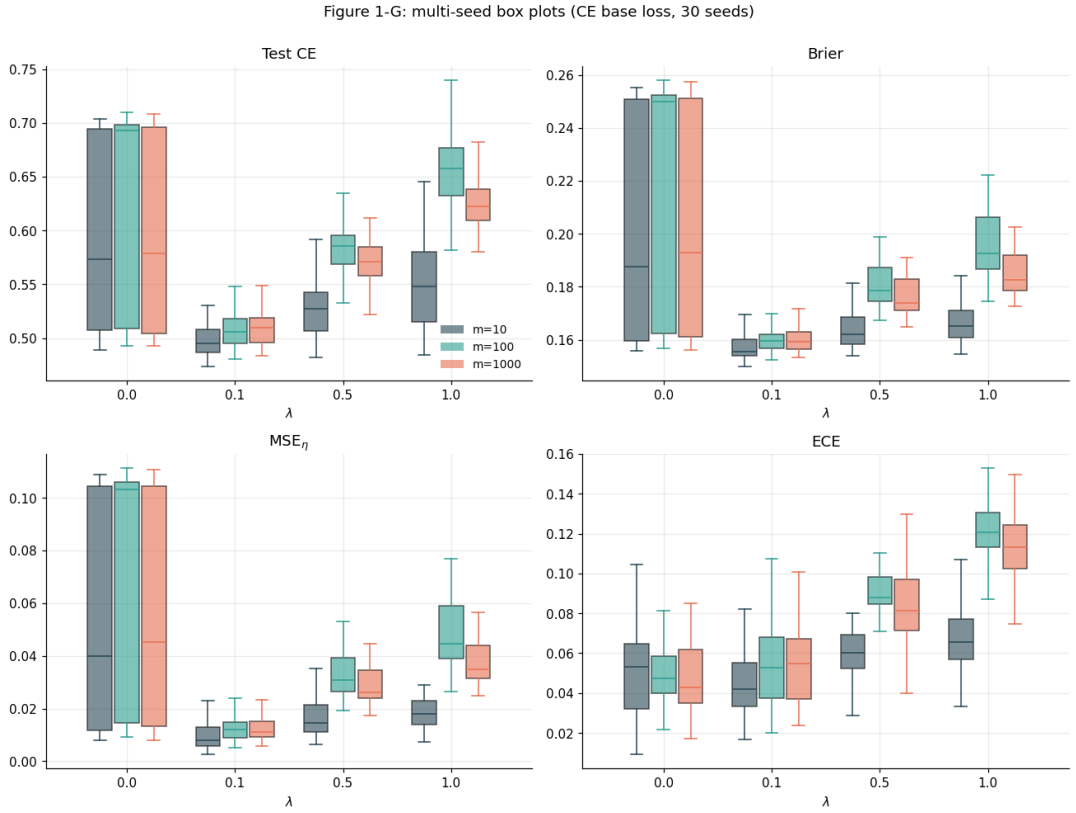


Figure 4.7: Distribution over 30 seeds of test CE, Brier score, MSE_η and ECE for the CE base loss, grouped by λ, with one box per capacity (median, interquartile range, whiskers without outliers).

Table 4.3: Final-epoch test metrics, CE base loss, mean $\pm$ standard deviation over 30 seeds. Markers on the $\lambda = 0.1$ rows give the result of a paired Wilcoxon test against the baseline $\lambda = 1$, Bonferroni-corrected over the three capacities ($^{*}\, p < 0.05$, $^{**}\, p < 0.01$, n.s. not significant). All marked differences favour $\lambda = 0.1$.

m	λ	Test CE	Brier	ECE	MSE_η	Cal (Brier)	Ref (Brier)	MCE	RMSCE	KSCE
10	0.0	0.5969 ± 0.0905	0.2032 ± 0.0436	0.0528 ± 0.0269	0.0559 ± 0.0443	0.0060 ± 0.0052	0.1973 ± 0.0447	0.1628 ± 0.1489	0.0621 ± 0.0307	0.0478 ± 0.0236
	0.1	$0.4981 \pm 0.0170^{**}$	$0.1577 \pm 0.0054^{**}$	$0.0459 \pm 0.0179^{**}$	$0.0101 \pm 0.0054^{**}$	$0.0051 \pm 0.0038^{**}$	$0.1525 \pm 0.0029^{**}$	$0.1594 \pm 0.1486^{**}$	$0.0552 \pm 0.0181^{**}$	$0.0337 \pm 0.0160^{*}$
	0.5	0.5293 ± 0.0292	0.1637 ± 0.0071	0.0630 ± 0.0184	0.0163 ± 0.0072	0.0075 ± 0.0036	0.1562 ± 0.0050	0.1991 ± 0.1324	0.0775 ± 0.0193	0.0368 ± 0.0167
	1.0	0.5512 ± 0.0424	0.1663 ± 0.0077	0.0694 ± 0.0195	0.0189 ± 0.0078	0.0090 ± 0.0039	0.1573 ± 0.0053	0.2080 ± 0.1265	0.0870 ± 0.0204	0.0388 ± 0.0164
100	0.0	0.6233 ± 0.0938	0.2163 ± 0.0450	0.0497 ± 0.0177	0.0692 ± 0.0458	0.0048 ± 0.0041	0.2115 ± 0.0471	0.3999 ± 0.2429	0.0624 ± 0.0206	0.0399 ± 0.0201
	0.1	$0.5077 \pm 0.0181^{**}$	$0.1604 \pm 0.0055^{**}$	$0.0545 \pm 0.0212^{**}$	$0.0127 \pm 0.0054^{**}$	$0.0060 \pm 0.0042^{**}$	$0.1545 \pm 0.0026^{**}$	0.1821 ± 0.1359 n.s.	$0.0633 \pm 0.0219^{**}$	$0.0347 \pm 0.0166^{**}$
	0.5	0.5841 ± 0.0268	0.1805 ± 0.0088	0.0917 ± 0.0175	0.0330 ± 0.0090	0.0123 ± 0.0046	0.1682 ± 0.0072	0.1533 ± 0.0433	0.1004 ± 0.0183	0.0496 ± 0.0180
	1.0	0.6573 ± 0.0363	0.1958 ± 0.0125	0.1217 ± 0.0178	0.0482 ± 0.0128	0.0190 ± 0.0054	0.1768 ± 0.0103	0.1930 ± 0.0513	0.1316 ± 0.0187	0.0632 ± 0.0189
1000	0.0	0.6001 ± 0.0939	0.2054 ± 0.0449	0.0506 ± 0.0302	0.0580 ± 0.0456	0.0058 ± 0.0070	0.1995 ± 0.0468	0.5076 ± 0.2773	0.0628 ± 0.0298	0.0367 ± 0.0222
	0.1	$0.5099 \pm 0.0176^{**}$	$0.1607 \pm 0.0053^{**}$	$0.0546 \pm 0.0202^{**}$	$0.0129 \pm 0.0052^{**}$	$0.0061 \pm 0.0042^{**}$	$0.1546 \pm 0.0026^{**}$	0.1590 ± 0.1398 n.s.	$0.0620 \pm 0.0201^{**}$	$0.0355 \pm 0.0178^{**}$
	0.5	0.5724 ± 0.0274	0.1766 ± 0.0074	0.0854 ± 0.0212	0.0289 ± 0.0075	0.0105 ± 0.0047	0.1661 ± 0.0069	0.1361 ± 0.0362	0.0925 ± 0.0207	0.0479 ± 0.0168
	1.0	0.6252 ± 0.0368	0.1855 ± 0.0082	0.1143 ± 0.0184	0.0378 ± 0.0083	0.0162 ± 0.0048	0.1694 ± 0.0076	0.1580 ± 0.0330	0.1198 ± 0.0177	0.0594 ± 0.0173

4.3.4 Effect of Model Capacity

Capacity under predictive-only training. Under the baseline, the error is not monotone in capacity. Table 4.3 gives $\mathrm{MSE}_\eta = 0.0189$, 0.0482 and 0.0378 for $m = 10$, 100 and 1000. ECE $(0.069, 0.122, 0.114)$ and test CE $(0.551, 0.657, 0.625)$ show the same ordering. The worst model is the one at the interpolation threshold $m = n_{\text{train}}$, and the strongly overparameterised model partly recovers.

This is consistent with a double-descent-type curve, but three capacities are not enough to establish one. The single-seed post-hoc sweep over the full grid points in the same direction (Figure 4.8, trained on 80 points). There, the raw ECE is 0.03–0.06 for $m \leq 10$, 0.136–0.146 for $m \in [100, 300]$ and 0.131 at $m = 1000$.

Either way, the degradation starts at the threshold itself, not only deep in the overparameterised regime. Since $m = 100$ and $m = 1000$ can both represent η essentially exactly (Figure 4.1), the degradation is estimation error.

Q1.8: does overparameterisation help under end-to-end training? No. At $\lambda = 0.1$ the three capacities reach $\mathrm{MSE}_\eta = 0.0101$, 0.0127 and 0.0129, and at the fine-grid optimum of the single-seed sweep 0.0097, 0.0118 and 0.0123 (CE). In both cases the smallest model is numerically the best. The larger models are only slightly worse, and the differences between capacities are smaller than one standard deviation across seeds.

The calibration penalty thus removes most of the damage done by overparameterisation, but it does not turn capacity into an advantage. In this problem there is little for extra capacity to gain, because the approximation error at $m = 10$ is already negligible (Section 4.2.2). A target function that the small model cannot represent would be needed to test whether overparameterisation can help under end-to-end training.

Q1.9: does the penalty help more when $m \gg n$? In absolute terms, yes. Moving from $\lambda = 1$ to $\lambda = 0.1$ reduces MSE_η by 0.0088 at $m = 10$, by 0.0355 at $m = 100$ and by 0.0249 at $m = 1000$. These are relative reductions of 47%, 74% and 66%. The benefit is largest at the interpolation threshold and remains large in the strongly overparameterised regime, because that is where the baseline does most damage. Ablation A shows that the same capacity dependence appears for the step-size control.

4.3.5 Comparison with Post-hoc Calibration

Q1.3 and Q1.5: post-hoc methods across capacity. Figure 4.8 and Table 4.4 compare four post-hoc methods, fitted on 20 held-out points, with end-to-end training at $\lambda = 0.1$ over the full capacity grid (single seed). The ranking depends mainly on how miscalibrated the raw model already is, and that in turn depends on capacity.

For $m \leq 10$ the raw CE model is already reasonably calibrated (ECE 0.027 at $m = 5$, 0.059 at $m = 10$), and *every* post-hoc method makes it worse. Temperature scaling, for example, doubles the ECE at $m = 5$ ($0.027 \to 0.057$). With little real miscalibration to correct and only 20 noisy calibration points, the fitted maps mostly fit noise.

From $m = 100$ onwards, the raw model is clearly miscalibrated (ECE 0.131–0.146), and the methods separate:

- *Temperature scaling* is the best method at every capacity in this range, including end-to-end training. It reaches 0.035 at $m = 100$, but its ECE grows steadily with capacity to 0.075 at $m = 1000$.
- *Platt scaling* is somewhat worse (0.075–0.090).
- *Isotonic regression* leaves the ECE essentially unchanged (0.129–0.136).
- *Histogram binning* is often worse than no calibration at all (up to 0.214 at $m = 300$).
- *End-to-end training* stays in a narrow band across the whole grid: 0.049–0.083 for the CE family and 0.061–0.069 for the Brier family.

For the Brier base loss, the raw models are already fairly well calibrated at every capacity (ECE 0.040–0.070). Temperature scaling helps only for $m \in \{100, 120, 200, 300\}$, and at $m = 1000$ the raw Brier model (0.056) is better than both temperature scaling (0.076) and end-to-end training (0.063).

These single-split numbers have to be read against their variability. Repeating the 80/20 split ten times at $m = 1000$ (CE) gives the following mean $\pm$ standard deviation of ECE:

- raw model: 0.130 ± 0.006;
- temperature scaling: 0.073 ± 0.041 (range 0.032–0.134);
- Platt scaling: 0.099 ± 0.036;
- isotonic regression: 0.133 ± 0.029;
- histogram binning: 0.205 ± 0.048.

The advantage of temperature scaling over end-to-end training at a single split (0.075 against 0.082) is therefore well within the split-to-split variation of temperature scaling itself. The failure of isotonic regression and histogram binning, on the other hand, is systematic at this calibration-set size.

In summary, post-hoc calibration does not degrade in a simple monotone way. Two effects combine as capacity grows. The raw model becomes badly miscalibrated from the interpolation threshold onwards. The best global correction (temperature scaling) becomes less effective beyond the threshold and much less stable. That is consistent with the hypothesis that a single global map

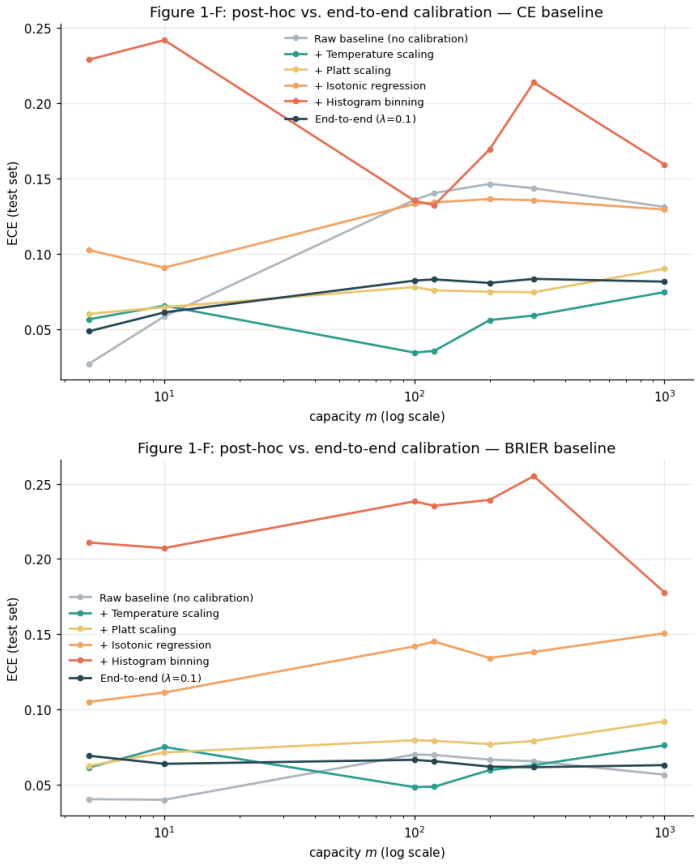


Figure 4.8: Test ECE of the raw baseline, the four post-hoc methods and end-to-end training ($\lambda = 0.1$) across the full capacity grid, for the CE (top) and Brier (bottom) base losses (single seed, single 80/20 split). The baseline is trained on 80 points and the maps are fitted on the remaining 20; the end-to-end model uses all 100 points.

cannot fully correct miscalibration that varies across the input space. It is not evidence that end-to-end training is better calibrated than temperature scaling: in this comparison it is not.

Q1.4: estimating refinement with a recalibration map. A post-hoc map can also be used to estimate the refinement of a model, since the loss after optimal recalibration equals the refinement term [8]. Figure 4.9 and Table 4.5 compare two such estimators against the ground-truth refinement of Section 4.2.5:

- a *TS-refinement* estimator, which uses temperature scaling as the recalibration class, as proposed by Berta et al.;
- an *isotonic-refinement* estimator.

Both are fitted and evaluated with the Brier score, on 20 held-out points, for CE baseline models

Table 4.4: Test ECE before and after post-hoc calibration, and for end-to-end training ($\lambda = 0.1$), by base loss and capacity (single seed, single 80/20 split).

Loss	m	Raw	Temp. scaling	Platt	Isotonic	Histogram	End-to-end
CE	5	0.0271	0.0566	0.0602	0.1025	0.2288	0.0487
	10	0.0585	0.0656	0.0646	0.0908	0.2417	0.0612
	100	0.1358	0.0345	0.0779	0.1331	0.1353	0.0822
	120	0.1403	0.0356	0.0758	0.1341	0.1322	0.0830
	200	0.1464	0.0561	0.0749	0.1363	0.1694	0.0807
	300	0.1435	0.0591	0.0745	0.1355	0.2137	0.0834
	1000	0.1311	0.0746	0.0901	0.1294	0.1592	0.0816
Brier	5	0.0401	0.0611	0.0624	0.1049	0.2109	0.0690
	10	0.0398	0.0749	0.0712	0.1111	0.2072	0.0636
	100	0.0699	0.0481	0.0794	0.1418	0.2382	0.0663
	120	0.0695	0.0484	0.0790	0.1450	0.2353	0.0654
	200	0.0665	0.0595	0.0768	0.1340	0.2392	0.0618
	300	0.0654	0.0627	0.0789	0.1381	0.2551	0.0614
	1000	0.0564	0.0759	0.0919	0.1506	0.1777	0.0628

over 30 seeds.

Both estimators are positively biased at every capacity; they overestimate refinement. This is expected for two reasons. A restricted class such as temperature scaling cannot reach the ideal recalibration map, and a map fitted on 20 noisy points is further from it still.

TS-refinement has the smaller mean bias everywhere. The difference is significant at $m = 100$ (0.016 against 0.045, $p_{\text{Bonf}} < 0.01$) and at $m = 1000$ (0.020 against 0.036, $p_{\text{Bonf}} = 0.017$), but not at $m = 10$ (0.028 against 0.035).

The variance ranking does change with capacity. Isotonic regression is the more stable estimator at $m = 10$ (3.3×10^{-4} against 4.1×10^{-4}). TS-refinement is more than twice as stable at $m = 100$. The two are equal at $m = 1000$.

With small calibration sets, TS-refinement is therefore the safer default once the model is at or beyond the interpolation threshold. This supports its use in the refine-then-calibrate procedure of Berta et al. The variance comparison is descriptive only.

4.3.6 Comparison with Focal Loss

Figure 4.10 and Table 4.6 compare focal loss at $\gamma \in \{0, 1, 2, 5\}$ with end-to-end training at $\lambda = 0.05$, the MSE_η optimum of the fine grid for CE (30 seeds). As a consistency check, focal loss at $\gamma = 0$ reproduces the $\lambda = 1$ rows of Table 4.3 exactly.

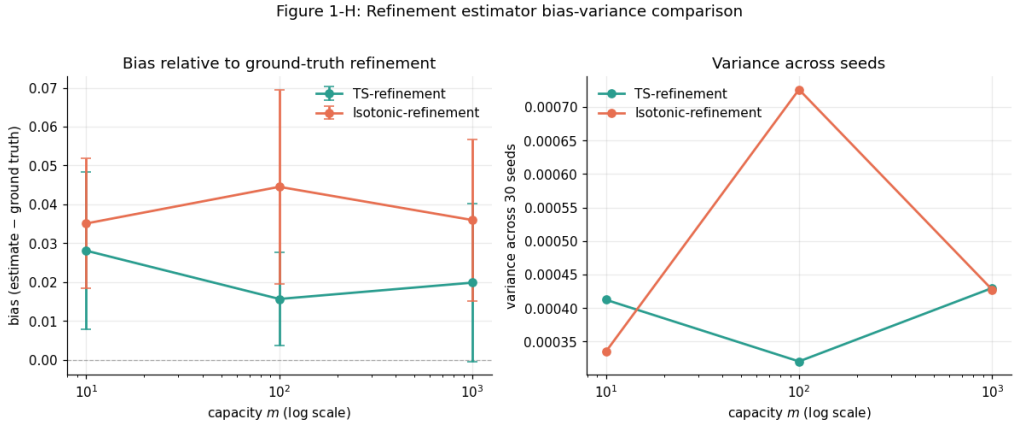


Figure 4.9: Bias (left; mean ± standard deviation over 30 seeds) and across-seed variance (right) of TS-based and isotonic refinement estimators relative to the ground-truth refinement, for CE baseline models.

Table 4.5: Refinement estimators relative to the ground-truth refinement (30 seeds, CE baseline, Brier-based estimators, 20 calibration points). Bias is mean ± standard deviation of (estimate − ground truth); variance is the across-seed variance of the estimate. The last column tests $|\text{bias}_{\text{TS}}|$ against $|\text{bias}_{\text{iso}}|$ (paired Wilcoxon, Bonferroni over three capacities).

	TS-refinement		Isotonic refinement		
m	bias	variance	bias	variance	$\lvert\text{bias}\rvert$ test
10	$+0.0281 \pm 0.0203$	4.1×10^{-4}	$+0.0351 \pm 0.0167$	3.3×10^{-4}	n.s. ($p = 0.40$)
100	$+0.0156 \pm 0.0120$	3.2×10^{-4}	$+0.0445 \pm 0.0251$	7.3×10^{-4}	** ($p < 0.001$)
1000	$+0.0198 \pm 0.0203$	4.3×10^{-4}	$+0.0359 \pm 0.0208$	4.3×10^{-4}	* ($p = 0.017$)

Q1.7: does focal loss match explicit end-to-end calibration? The answer differs between the two metrics.

On MSE_η, end-to-end training is better than every γ at every capacity, and every difference is significant. Compared with the best γ, it is about 2.0, 3.3 and 2.3 times closer to η at $m = 10$, 100 and 1000 (0.0097 against 0.0189, 0.0111 against 0.0365, 0.0117 against 0.0268). It is also best on test CE and the Brier score at every capacity.

On top-label ECE, the ranking depends on capacity. At $m = 10$, end-to-end training is clearly best (0.045 against 0.069 for the best focal setting, $\gamma = 0$). At $m = 100$ and $m = 1000$, focal loss with $\gamma = 2$ is significantly better than end-to-end training (0.035 against 0.051, and 0.032 against 0.052).

The best γ also shifts with capacity. At $m = 10$, any $\gamma > 0$ makes both metrics worse, monotonically in γ. At $m \geq 100$, $\gamma = 2$ is best on both metrics and $\gamma = 5$ overshoots, ending worse than plain CE on ECE.

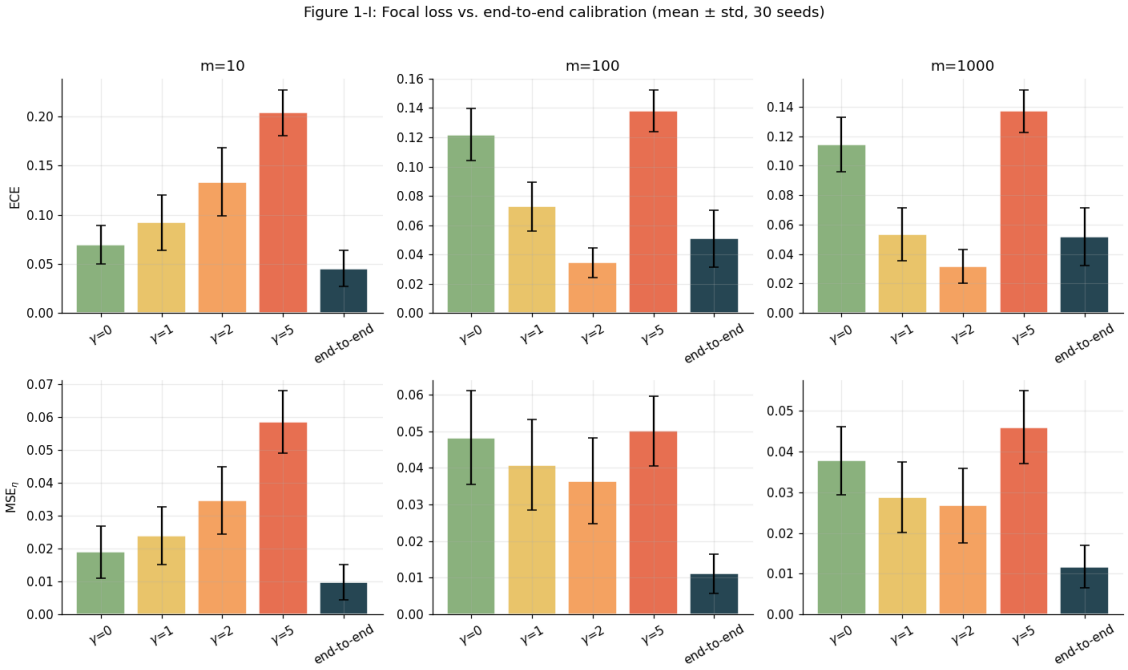


Figure 4.10: ECE (top) and MSE_η (bottom) for focal loss with $\gamma \in \{0, 1, 2, 5\}$ and for end-to-end training ($\lambda = 0.05$, CE base loss), mean $\pm$ standard deviation over 30 seeds.

The gap on MSE_η therefore does not narrow with capacity; it is largest at the interpolation threshold. The ranking on ECE reverses. This is the clearest case in the chapter of the two metrics disagreeing: focal loss with $\gamma = 2$ aligns confidence with accuracy on average better than any other method, yet its predicted probabilities are two to three times further from η than those of end-to-end training.

Top-label ECE only asks whether confidence matches the accuracy of the decision. It is blind to how the probability mass is distributed within the confidence range, and to whether the probability of the class is correct. Two cautions apply to this comparison. The λ of the end-to-end model was selected for MSE_η and not for ECE. The best γ was selected after the fact for each capacity, so both methods are compared at an oracle choice of their hyperparameter.

4.3.7 Ablations and Sensitivity Analysis

The identity (4.6) shows that every result obtained so far confounds two effects of lowering λ: a stronger calibration term and a smaller step size on the proper loss. The four ablations below separate these effects and test weight schemes that do not rely on an oracle. All use the CE base loss and 30 seeds. Significance markers compare the stated pair of methods. Because an end-to-end model can be significantly *worse* than its control, we use $^*/^{**}$ for a significant improvement and $^\dagger/^{\dagger\dagger}$ for a significant deterioration, at $p_{\mathrm{Bonf}} < 0.05$ and < 0.01 respectively.

Table 4.6: Focal loss and end-to-end training ($\lambda = 0.05$), mean $\pm$ standard deviation over 30 seeds. The best mean per capacity and column is in bold. Paired Wilcoxon tests of end-to-end training against each γ (Bonferroni over 12 comparisons per metric) are significant at $p < 0.01$ for ECE and MSE_η in all cases except ECE at $m = 1000$, $\gamma = 1$ (n.s.).

m	Method	Test CE	Brier	ECE	MSE_η
10	focal, $\gamma = 0$	0.5512 ± 0.0424	0.1663 ± 0.0077	0.0694 ± 0.0195	0.0189 ± 0.0078
	focal, $\gamma = 1$	0.5341 ± 0.0247	0.1711 ± 0.0086	0.0919 ± 0.0280	0.0240 ± 0.0087
	focal, $\gamma = 2$	0.5524 ± 0.0238	0.1816 ± 0.0101	0.1334 ± 0.0343	0.0346 ± 0.0102
	focal, $\gamma = 5$	0.6019 ± 0.0199	0.2053 ± 0.0094	0.2036 ± 0.0234	0.0585 ± 0.0094
	end-to-end	$\mathbf{0.4963} \pm 0.0173$	$\mathbf{0.1574} \pm 0.0054$	$\mathbf{0.0453} \pm 0.0184$	$\mathbf{0.0097} \pm 0.0054$
100	focal, $\gamma = 0$	0.6573 ± 0.0363	0.1958 ± 0.0125	0.1217 ± 0.0178	0.0482 ± 0.0128
	focal, $\gamma = 1$	0.5734 ± 0.0262	0.1882 ± 0.0121	0.0727 ± 0.0167	0.0408 ± 0.0124
	focal, $\gamma = 2$	0.5513 ± 0.0258	0.1837 ± 0.0116	$\mathbf{0.0345} \pm 0.0101$	0.0365 ± 0.0118
	focal, $\gamma = 5$	0.5837 ± 0.0201	0.1970 ± 0.0094	0.1381 ± 0.0144	0.0500 ± 0.0095
	end-to-end	$\mathbf{0.5014} \pm 0.0172$	$\mathbf{0.1589} \pm 0.0054$	0.0508 ± 0.0194	$\mathbf{0.0111} \pm 0.0054$
1000	focal, $\gamma = 0$	0.6252 ± 0.0368	0.1855 ± 0.0082	0.1143 ± 0.0184	0.0378 ± 0.0083
	focal, $\gamma = 1$	0.5430 ± 0.0196	0.1763 ± 0.0085	0.0535 ± 0.0180	0.0288 ± 0.0086
	focal, $\gamma = 2$	0.5299 ± 0.0206	0.1741 ± 0.0090	$\mathbf{0.0317} \pm 0.0115$	0.0268 ± 0.0091
	focal, $\gamma = 5$	0.5754 ± 0.0189	0.1930 ± 0.0088	0.1370 ± 0.0145	0.0460 ± 0.0089
	end-to-end	$\mathbf{0.5039} \pm 0.0169$	$\mathbf{0.1595} \pm 0.0053$	0.0521 ± 0.0196	$\mathbf{0.0117} \pm 0.0052$

A: step-size-matched control. Figure 4.11 and Table 4.7 compare end-to-end training at $\lambda \in \{0.05, 0.1, 0.5\}$ with plain CE trained at the learning rate 0.5λ. By (4.6), the two differ only in the presence of the penalty. The results fall into three regimes.

- **At $\lambda = 0.1$, the penalty adds nothing measurable to MSE_η or calibration.** End-to-end training and its control reach the same MSE_η at every capacity (0.0101 against 0.0100, 0.0127 against 0.0127, 0.0129 against 0.0129). ECE and the Brier calibration term do not differ significantly; at $m \geq 100$ the control is even numerically better on ECE (0.048 against 0.055). The only significant effect of the penalty is a very small reduction of the refinement term at $m \geq 100$ (about 0.001). The step-size effect alone accounts for essentially all of the improvement over the baseline that Section 4.3.3 attributed to $\lambda = 0.1$ (between 99.8% and 101% of the MSE_η gain). The reduction in across-seed variance is reproduced as well: the control has standard deviation 0.0049 against 0.0052 at $m = 1000$.

- **At $\lambda = 0.5$, the penalty is harmful.** At $m \geq 100$ it makes MSE_η, ECE and the calibration term significantly *worse* than the control (for example 0.0289 against 0.0277 in MSE_η and 0.085 against 0.080 in ECE at $m = 1000$).

- **At $\lambda = 0.05$, the penalty contributes a genuine improvement.** End-to-end training is significantly better than its control on MSE_η at all three capacities (0.0097 against 0.0131, 0.0111 against 0.0146, 0.0117 against 0.0155) and on the Brier calibration term. ECE improves as well, significantly only at $m = 10$ after correction; at $m \geq 100$ the improvement is numerical (0.051 against 0.069). The step-size effect still explains most of the gain over

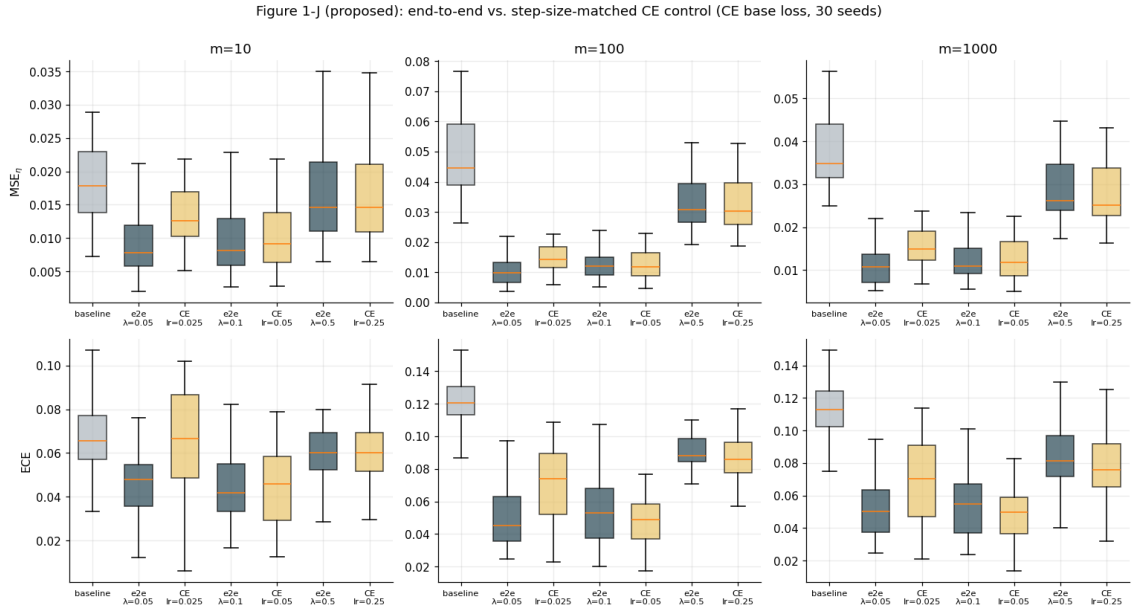


Figure 4.11: Ablation A: MSE_η (top) and ECE (bottom) for the baseline, end-to-end training at $\lambda \in \{0.05, 0.1, 0.5\}$ (dark) and plain CE at the matched learning rate 0.5λ (light), over 30 seeds.

the baseline (63%, 91% and 86% for MSE_η at $m = 10$, 100 and 1000), so the penalty's own share is roughly 10–40%.

The control at learning rate 0.025 underfits: it is worse than the control at 0.05. The penalty appears to compensate for this. As a result, end-to-end training at $\lambda = 0.05$ is numerically better at $m \geq 100$ than the best of the three controls (0.0111 against 0.0127 and 0.0117 against 0.0129). This last comparison is across different pairs and was not tested.

B: additive penalty at a fixed learning rate. Figure 4.12 and Table 4.8 add the penalty to plain CE without changing the learning rate. The result is unambiguous: at the full step size, the penalty never helps.

- At $m = 10$, no value of β changes any metric significantly.
- At $m = 100$ and $m = 1000$, MSE_η, ECE and the calibration term get worse monotonically as β increases, and significantly so for every $\beta \geq 0.5$. At $m = 1000$, $\beta = 19$ raises MSE_η from 0.0378 to 0.0474, ECE from 0.114 to 0.142 and the calibration term from 0.0162 to 0.0232.

The same $\beta = 19$ at learning rate 0.025, which is end-to-end training at $\lambda = 0.05$, gave the best results of ablation A. Whether the penalty helps or hurts therefore depends on how fast the proper loss is being fitted, not on its weight alone.

Table 4.7: Ablation A: end-to-end training against plain CE at the matched learning rate 0.5λ (mean $\pm$ standard deviation, 30 seeds). Markers on end-to-end rows compare with the control directly below (paired Wilcoxon, Bonferroni over nine comparisons per metric). $^{*}/^{**}$: significantly better; $^{\dagger}/^{\dagger\dagger}$: significantly worse; unmarked: not significant.

m	Condition	MSE_η	ECE	Cal (Brier)	Ref (Brier)
10	baseline (lr 0.5)	0.0189 ± 0.0078	0.0694 ± 0.0195	0.0090 ± 0.0039	0.1573 ± 0.0053
	end-to-end, $\lambda = 0.05$	$0.0097 \pm 0.0054^{*}$	$0.0453 \pm 0.0184^{*}$	$0.0050 \pm 0.0037^{*}$	0.1524 ± 0.0031
	CE, lr 0.025	0.0131 ± 0.0045	0.0644 ± 0.0263	0.0083 ± 0.0040	0.1523 ± 0.0019
	end-to-end, $\lambda = 0.1$	0.0101 ± 0.0054	0.0459 ± 0.0179	0.0051 ± 0.0038	0.1525 ± 0.0029
	CE, lr 0.05	0.0100 ± 0.0047	0.0447 ± 0.0171	0.0052 ± 0.0035	0.1523 ± 0.0025
	end-to-end, $\lambda = 0.5$	0.0163 ± 0.0072	0.0630 ± 0.0184	0.0075 ± 0.0036	0.1562 ± 0.0050
	CE, lr 0.25	0.0161 ± 0.0071	0.0623 ± 0.0178	0.0074 ± 0.0036	0.1561 ± 0.0049
100	baseline (lr 0.5)	0.0482 ± 0.0128	0.1217 ± 0.0178	0.0190 ± 0.0054	0.1768 ± 0.0103
	end-to-end, $\lambda = 0.05$	$0.0111 \pm 0.0054^{**}$	0.0508 ± 0.0194	$0.0057 \pm 0.0041^{*}$	$0.1532 \pm 0.0021^{**}$
	CE, lr 0.025	0.0146 ± 0.0048	0.0685 ± 0.0242	0.0083 ± 0.0040	0.1539 ± 0.0020
	end-to-end, $\lambda = 0.1$	0.0127 ± 0.0054	0.0545 ± 0.0212	0.0060 ± 0.0042	$0.1545 \pm 0.0026^{**}$
	CE, lr 0.05	0.0127 ± 0.0048	0.0484 ± 0.0158	0.0051 ± 0.0033	0.1551 ± 0.0027
	end-to-end, $\lambda = 0.5$	$0.0330 \pm 0.0090^{\dagger\dagger}$	$0.0917 \pm 0.0175^{\dagger\dagger}$	$0.0123 \pm 0.0046^{\dagger\dagger}$	$0.1682 \pm 0.0072^{**}$
	CE, lr 0.25	0.0326 ± 0.0090	0.0877 ± 0.0178	0.0115 ± 0.0046	0.1686 ± 0.0075
1000	baseline (lr 0.5)	0.0378 ± 0.0083	0.1143 ± 0.0184	0.0162 ± 0.0048	0.1694 ± 0.0076
	end-to-end, $\lambda = 0.05$	$0.0117 \pm 0.0052^{**}$	0.0521 ± 0.0196	$0.0058 \pm 0.0043^{*}$	$0.1538 \pm 0.0018^{**}$
	CE, lr 0.025	0.0155 ± 0.0049	0.0683 ± 0.0245	0.0084 ± 0.0040	0.1546 ± 0.0020
	end-to-end, $\lambda = 0.1$	0.0129 ± 0.0052	0.0546 ± 0.0202	0.0061 ± 0.0042	$0.1546 \pm 0.0026^{**}$
	CE, lr 0.05	0.0129 ± 0.0049	0.0473 ± 0.0161	0.0050 ± 0.0033	0.1556 ± 0.0028
	end-to-end, $\lambda = 0.5$	$0.0289 \pm 0.0075^{\dagger\dagger}$	$0.0854 \pm 0.0212^{\dagger\dagger}$	$0.0105 \pm 0.0047^{\dagger\dagger}$	$0.1661 \pm 0.0069^{\dagger\dagger}$
	CE, lr 0.25	0.0277 ± 0.0072	0.0795 ± 0.0223	0.0097 ± 0.0047	0.1657 ± 0.0066

We offer a hypothesis for this dependence; it is consistent with ablations A and B but was not tested directly. The penalty is computed on the *training* predictions. When the proper loss is fitted quickly, those predictions soon approach the training labels. In the bins close to 0 and 1, the label mean $\bar{y}_b$ then lies further from 0.5 than the prediction mean $\bar{p}_b$, so the gradient of (4.5) pushes predictions further towards the extremes. The penalty then rewards training-set calibration of a model that is overfitting, and this increases its overconfidence on the test set. When the proper loss is fitted slowly, the training-set bin gaps still resemble the population gaps, and the penalty corrects genuine miscalibration.

C: validation-based selection of λ against validation early stopping. All "best" values of λ so far were read off the ground-truth MSE_η. Ablation C replaces this oracle with a 20-point validation set and compares the result with plain CE stopped early on the same validation set (Figure 4.13, Table 4.9).

Selecting λ by validation Brier score works reasonably well. The selected value is $\lambda = 0.05$ in 15, 19 and 19 of the 30 seeds for $m = 10$, 100 and 1000, and $\lambda = 1$ in 6, 1 and 3 seeds. Compared with the oracle choice $\lambda = 0.1$ on the same 80 training points, the selection costs about 0.002–0.003 in MSE_η.

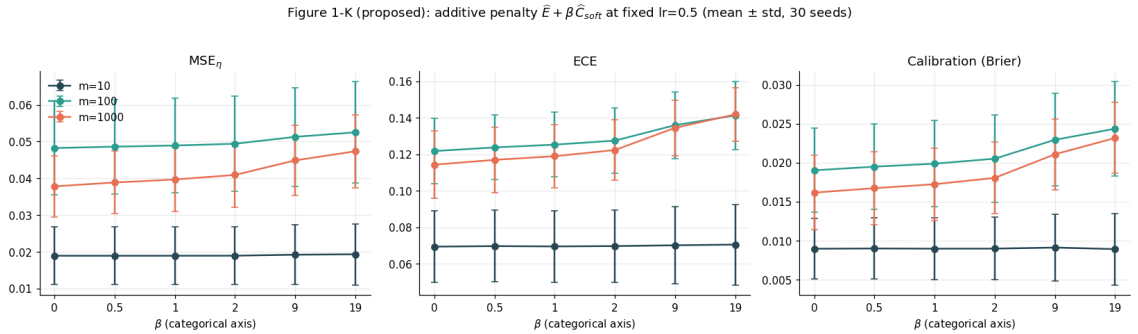


Figure 4.12: Ablation B: MSE_η, ECE and Brier calibration term for the additive objective $\widetilde{E} + \beta\,\widetilde{C}_{\mathrm{soft}}$ at the fixed learning rate 0.5 (mean $\pm$ standard deviation, 30 seeds). $\beta = 9$ and $\beta = 19$ correspond to $\lambda = 0.1$ and $\lambda = 0.05$ at a correspondingly reduced learning rate.

Against validation early stopping, end-to-end training with a selected λ reaches a lower MSE_η at all capacities:

- 0.0142 against 0.0210 at $m = 10$;
- 0.0169 against 0.0232 at $m = 100$;
- 0.0185 against 0.0250 at $m = 1000$.

The difference is significant at $m = 10$ and $m = 100$ ($p_{\mathrm{Bonf}} < 0.05$) but not at $m = 1000$ ($p_{\mathrm{Bonf}} = 0.07$). The same pattern holds when both pipelines are refitted on all 100 points. The ECE differences (0.059 against 0.076, and similar values at the other capacities) are not significant at any capacity.

This advantage has at least two possible sources. Ablation A offers one: the selected λ is mostly 0.05, the one value at which the penalty contributes on its own. The second concerns how noisy each selection rule is. Early stopping selects one of 1,201 checkpoints on 20 points, and the chosen epoch varies enormously between seeds (median 102–137 epochs, standard deviation 226–439, range from a few epochs to 1,200). Selecting one of seven values of λ may simply be less noisy. The experiment cannot separate the two explanations; selecting the learning rate of plain CE from the same seven-point grid would.

D: learned loss weights. Uncertainty weighting (4.7) fails in this setting (Table 4.10, Figure 4.14). It is significantly worse than both the baseline and end-to-end training at $\lambda = 0.1$, on MSE_η and on ECE, at every capacity. At $m = 100$, for example, it reaches $\mathrm{MSE}_\eta = 0.103$ against 0.048 for the baseline, and a test CE of 1.69. The learned effective weight $\lambda_{\mathrm{eff}} = w_E/(w_C + w_E)$ fluctuates between roughly 0.01 and 0.2 throughout training rather than settling, and the clipping bound on the log-variances is reached in 70 of 90 runs.

The effective weights $1/(2L_i)$ explain the failure. As the model fits the training data, the training

Table 4.8: Ablation B: additive penalty at learning rate 0.5 (mean ± standard deviation, 30 seeds). $\beta = 0$ is the CE baseline. Markers compare each $\beta > 0$ with $\beta = 0$ (paired Wilcoxon, Bonferroni over 15 comparisons per metric); ††: significantly worse at $p < 0.01$; unmarked: not significant.

m	β	MSE_η	ECE	Cal (Brier)	Ref (Brier)
10	0	0.0189 ± 0.0078	0.0694 ± 0.0195	0.0090 ± 0.0039	0.1573 ± 0.0053
	0.5	0.0189 ± 0.0078	0.0697 ± 0.0196	0.0090 ± 0.0039	0.1573 ± 0.0053
	1	0.0189 ± 0.0079	0.0695 ± 0.0196	0.0090 ± 0.0040	0.1573 ± 0.0053
	2	0.0189 ± 0.0079	0.0697 ± 0.0198	0.0090 ± 0.0040	0.1573 ± 0.0054
	9	0.0192 ± 0.0081	0.0701 ± 0.0213	0.0091 ± 0.0043	0.1575 ± 0.0055
	19	0.0193 ± 0.0083	0.0705 ± 0.0221	0.0089 ± 0.0046	0.1578 ± 0.0055
100	0	0.0482 ± 0.0128	0.1217 ± 0.0178	0.0190 ± 0.0054	0.1768 ± 0.0103
	0.5	$0.0486 \pm 0.0129^{\dagger\dagger}$	$0.1237 \pm 0.0177^{\dagger\dagger}$	$0.0195 \pm 0.0055^{\dagger\dagger}$	0.1767 ± 0.0103
	1	$0.0489 \pm 0.0129^{\dagger\dagger}$	$0.1253 \pm 0.0177^{\dagger\dagger}$	$0.0199 \pm 0.0055^{\dagger\dagger}$	0.1766 ± 0.0102
	2	$0.0494 \pm 0.0130^{\dagger\dagger}$	$0.1275 \pm 0.0179^{\dagger\dagger}$	$0.0205 \pm 0.0056^{\dagger\dagger}$	0.1764 ± 0.0101
	9	$0.0513 \pm 0.0134^{\dagger\dagger}$	$0.1359 \pm 0.0183^{\dagger\dagger}$	$0.0229 \pm 0.0059^{\dagger\dagger}$	0.1759 ± 0.0097
	19	$0.0525 \pm 0.0139^{\dagger\dagger}$	$0.1413 \pm 0.0186^{\dagger\dagger}$	$0.0243 \pm 0.0061^{\dagger\dagger}$	0.1757 ± 0.0096
1000	0	0.0378 ± 0.0083	0.1143 ± 0.0184	0.0162 ± 0.0048	0.1694 ± 0.0076
	0.5	$0.0389 \pm 0.0085^{\dagger\dagger}$	$0.1169 \pm 0.0178^{\dagger\dagger}$	$0.0167 \pm 0.0047^{\dagger\dagger}$	$0.1699 \pm 0.0077^{\dagger\dagger}$
	1	$0.0397 \pm 0.0087^{\dagger\dagger}$	$0.1189 \pm 0.0173^{\dagger\dagger}$	$0.0172 \pm 0.0047^{\dagger\dagger}$	$0.1702 \pm 0.0078^{\dagger\dagger}$
	2	$0.0409 \pm 0.0089^{\dagger\dagger}$	$0.1223 \pm 0.0166^{\dagger\dagger}$	$0.0180 \pm 0.0046^{\dagger\dagger}$	$0.1706 \pm 0.0079^{\dagger\dagger}$
	9	$0.0449 \pm 0.0095^{\dagger\dagger}$	$0.1345 \pm 0.0152^{\dagger\dagger}$	$0.0211 \pm 0.0046^{\dagger\dagger}$	$0.1715 \pm 0.0082^{\dagger\dagger}$
	19	$0.0474 \pm 0.0099^{\dagger\dagger}$	$0.1418 \pm 0.0147^{\dagger\dagger}$	$0.0232 \pm 0.0045^{\dagger\dagger}$	$0.1719 \pm 0.0083^{\dagger\dagger}$

CE falls, so the weight on the proper loss *increases*. The effective step size on $\widetilde{E}$ therefore grows exactly when the model begins to overfit, which is the opposite of the mechanism that helped in ablation A. At the same time, the small training calibration error gives the penalty a large weight in a fast-fitting regime, which is where ablation B shows it to be harmful.

The experiment is exploratory. The learning rate of the log-variances, the clipping range and the choice to compute the weights on training rather than validation losses were fixed without tuning, and a different configuration may behave differently.

4.4 Discussion

4.4.1 The Central Question in the Light of the Controlled Experiments

The central question asks whether calibration can be improved by solving (4.1) while statistical accuracy is maintained through a suitable λ. Taken at face value, the experiments of this chapter answer it positively. Compared with predictive-only training under the same budget of 1,200 epochs, the convex objective with $\lambda \approx 0.05$–0.1 improves every calibration measure except the worst-bin gap. It also improves the distance to the true posterior, the test cross-entropy and the

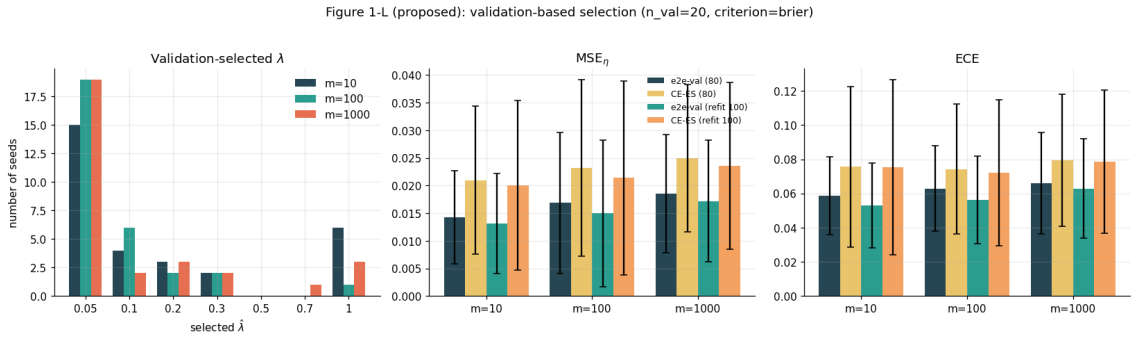


Figure 4.13: Ablation C: distribution of the validation-selected λ over 30 seeds (left), and test MSE_η (middle) and ECE (right) for end-to-end training with validation-selected λ and for CE with validation early stopping, trained on 80 points or refitted on all 100 (mean $\pm$ standard deviation).

refinement term, significantly and at every capacity (Table 4.3). Nothing is traded away: at a well-chosen λ, calibration and statistical accuracy improve together, and an interior optimum exists for both base losses (Figure 4.5).

The ablations, however, change what this result means. The convex combination couples the weight on calibration to the step size on the proper loss (4.6), and in our setting most of the benefit comes from the second channel:

- At $\lambda = 0.1$, plain cross-entropy trained with a ten times smaller learning rate reproduces the improvement completely: in MSE_η, in the calibration measures and in the across-seed variance.
- At $\lambda = 0.5$ the penalty makes things measurably worse than this control.
- At the full step size, adding the penalty never helps and, for $m \geq n$, always hurts.
- The calibration term contributes on its own in only one regime: at the smallest weight tested ($\lambda = 0.05$), where it accounts for roughly 10–40% of the gain in MSE_η and significantly lowers the calibration error.

A more precise answer to the central question is therefore this. The objective (4.1), optimised with a fixed learning rate and budget, does improve calibration without sacrificing accuracy. It does so mainly because it slows the fitting of the proper loss, which acts as an implicit form of early stopping, and only secondarily because the calibration term steers the model. In the realistic setting where λ has to be chosen on held-out data, end-to-end training still reaches a lower MSE_η than validation-based early stopping of plain cross-entropy, significantly so for $m \leq n$.

Table 4.9: Ablation C: validation-based pipelines (20 validation points, criterion: Brier score; mean ± standard deviation, 30 seeds). Markers on end-to-end rows compare with the early-stopping row of the same version (80 points or refit; paired Wilcoxon, Bonferroni over six comparisons per metric). The last two rows per capacity are references: the oracle $\lambda = 0.1$, and plain CE trained for all 1,200 epochs, both on 80 points.

m	Pipeline	MSE_η	ECE	Brier
10	end-to-end, selected λ (80)	$0.0142 \pm 0.0084^*$	0.0587 ± 0.0227	$0.1618 \pm 0.0082^*$
	CE, early stopping (80)	0.0210 ± 0.0134	0.0757 ± 0.0469	0.1683 ± 0.0132
	end-to-end, selected λ (refit)	$0.0131 \pm 0.0091^*$	0.0529 ± 0.0248	0.1607 ± 0.0089
	CE, early stopping (refit)	0.0200 ± 0.0154	0.0755 ± 0.0512	0.1674 ± 0.0152
	end-to-end, $\lambda = 0.1$ (80)	0.0122 ± 0.0058	0.0535 ± 0.0198	0.1598 ± 0.0058
	CE, 1,200 epochs (80)	0.0240 ± 0.0097	0.0794 ± 0.0213	0.1714 ± 0.0095
100	end-to-end, selected λ (80)	$0.0169 \pm 0.0128^*$	0.0629 ± 0.0251	$0.1646 \pm 0.0125^*$
	CE, early stopping (80)	0.0232 ± 0.0160	0.0743 ± 0.0379	0.1706 ± 0.0157
	end-to-end, selected λ (refit)	$0.0150 \pm 0.0133^*$	0.0564 ± 0.0257	$0.1627 \pm 0.0130^*$
	CE, early stopping (refit)	0.0214 ± 0.0176	0.0720 ± 0.0427	0.1688 ± 0.0173
	end-to-end, $\lambda = 0.1$ (80)	0.0153 ± 0.0059	0.0622 ± 0.0226	0.1630 ± 0.0060
	CE, 1,200 epochs (80)	0.0528 ± 0.0140	0.1366 ± 0.0169	0.2003 ± 0.0137
1000	end-to-end, selected λ (80)	0.0185 ± 0.0106	0.0661 ± 0.0297	0.1663 ± 0.0105
	CE, early stopping (80)	0.0250 ± 0.0133	0.0794 ± 0.0387	0.1725 ± 0.0132
	end-to-end, selected λ (refit)	0.0172 ± 0.0110	0.0629 ± 0.0290	0.1650 ± 0.0109
	CE, early stopping (refit)	0.0235 ± 0.0151	0.0786 ± 0.0419	0.1711 ± 0.0149
	end-to-end, $\lambda = 0.1$ (80)	0.0151 ± 0.0053	0.0605 ± 0.0203	0.1629 ± 0.0054
	CE, 1,200 epochs (80)	0.0396 ± 0.0081	0.1200 ± 0.0183	0.1873 ± 0.0082

4.4.2 RQ1: Calibration as an Implicit Regulariser

Does the calibration objective act as an implicit regulariser? Descriptively, yes. Lowering λ from 1 to 0.1 produces every signature of regularisation one would look for:

- the mean of MSE_η falls, and so does its variance across training samples;
- the gap between training and test loss shrinks;
- the spikes with which the overparameterised baseline fits individual noisy labels disappear (Figure 4.4);
- the damage done at the interpolation threshold is largely undone.

What ablation A shows is where this regularisation comes from. For $\lambda \geq 0.1$ it is an effect of the optimisation trajectory: a smaller effective step on the proper loss within a fixed number of epochs, which is algorithmic regularisation of the kind studied by Berta et al. [8]. It is not an effect of the calibration term. The term itself regularises only when fitting is slow, and it works against generalisation when fitting is fast (ablation B).



Figure 4.14: Ablation D: implied convex weight $\lambda_{\text{eff}} = w_E/(w_C + w_E)$ (left, logarithmic scale) and log-variance s_C of the calibration term (right) during training under uncertainty weighting (mean and 10–90% band over 30 seeds). The dashed line marks the lower clipping bound.

Table 4.10: Ablation D: uncertainty weighting compared with the baseline and end-to-end training at $\lambda = 0.1$ (mean $\pm$ standard deviation, 30 seeds). ††: significantly worse than both references at $p < 0.01$ (paired Wilcoxon, Bonferroni over six comparisons per metric).

m	Method	MSE_η	ECE	Test CE	final λ_{eff}
10	uncertainty weighting	$0.0670 \pm 0.0852^{\dagger\dagger}$	$0.1576 \pm 0.1197^{\dagger\dagger}$	1.0119 ± 0.7193	0.038 ± 0.041
	baseline ($\lambda = 1$)	0.0189 ± 0.0078	0.0694 ± 0.0195	0.5512 ± 0.0424	–
	end-to-end, $\lambda = 0.1$	0.0101 ± 0.0054	0.0459 ± 0.0179	0.4981 ± 0.0170	–
100	uncertainty weighting	$0.1028 \pm 0.0853^{\dagger\dagger}$	$0.2257 \pm 0.0963^{\dagger\dagger}$	1.6949 ± 1.1359	0.165 ± 0.306
	baseline ($\lambda = 1$)	0.0482 ± 0.0128	0.1217 ± 0.0178	0.6573 ± 0.0363	–
	end-to-end, $\lambda = 0.1$	0.0127 ± 0.0054	0.0545 ± 0.0212	0.5077 ± 0.0181	–
1000	uncertainty weighting	$0.0697 \pm 0.0135^{\dagger\dagger}$	$0.1946 \pm 0.0170^{\dagger\dagger}$	1.1628 ± 0.2438	0.148 ± 0.156
	baseline ($\lambda = 1$)	0.0378 ± 0.0083	0.1143 ± 0.0184	0.6252 ± 0.0368	–
	end-to-end, $\lambda = 0.1$	0.0129 ± 0.0052	0.0546 ± 0.0202	0.5099 ± 0.0176	–

Our hypothesis for this dependence is that the penalty is estimated on the training predictions. Once a model has begun to fit individual labels, low training-set calibration error is no longer evidence of calibration on new data, and minimising it pushes predictions further towards the labels. If this hypothesis holds, a calibration penalty estimated on the same data as the proper loss can only act as a regulariser while the model is still far from interpolating. That would limit how far the idea of calibration as a structural constraint can be taken in the overparameterised regime.

How does it reshape the bias–variance trade-off relative to post-hoc calibration? Post-hoc calibration and training-time weighting act on different components of the loss. A recalibration map that is injective, such as temperature or Platt scaling, does not change which inputs receive which ranks, so it cannot reduce the refinement term [42, 8]; it can only remove calibration error. In the overparameterised baseline, however, the excess loss is partly a refinement problem. The

Brier refinement term at $m = 100$ is 0.177 under the baseline and 0.155 at $\lambda = 0.1$ (Table 4.3), a gap larger than the whole calibration term. End-to-end training (or, equivalently here, a smaller step size) reduces both components. Post-hoc calibration can at best reduce one.

On the calibration measure alone, temperature scaling on a 20-point hold-out is competitive with end-to-end training and is better on the one split we examined (Table 4.4), but its result varies strongly with the choice of that split. Isotonic regression and histogram binning do not work at this calibration-set size. End-to-end training is the only method whose calibration is stable across the whole capacity range, and it does not give up training data for calibration.

These comparisons are single-split and single-seed, and they measure ECE only. They should be read as showing that the two approaches solve different problems, not as a ranking.

4.4.3 RQ2: What the Experiments Imply for the Theory

Chapter 3 approaches generalisation under end-to-end calibration through the complexity of a calibration-constrained hypothesis class and through PAC-Bayesian bounds. The experiments cannot confirm or refute a bound, but they do indicate which kind of argument fits the observed behaviour. Three observations are relevant.

First, the improvements in generalisation that we observe are algorithm-dependent. They depend on the product of learning rate and number of epochs, and they disappear or reverse when the penalty is added at the full step size. An argument that bounds generalisation uniformly over a hypothesis class, such as one based on Rademacher complexity, does not see the optimisation trajectory. It therefore cannot account for most of the effect measured here. Algorithm-dependent analyses, for instance of early-stopped gradient descent, or PAC-Bayesian bounds with a posterior that depends on the training procedure, are closer to the mechanism we observe.

Second, a soft penalty is not a hard constraint. The set of models with small empirical calibration error does not shrink the hypothesis class in a useful way when the model can interpolate: an interpolating model can be well calibrated on its own training data. This is consistent with ablation B.

Third, where the penalty does contribute on its own (ablation A at $\lambda = 0.05$), the effect is modest but significant. A theoretical account of it would need to explain why it appears only when the proper loss is fitted slowly.

These are interpretations of the experiments and are offered as directions for the analysis, not as results.

4.4.4 RQ3: Summary of the Controlled Comparison

Table 4.11 collects the answers to the experiment questions. Measured against the true posterior, three statements hold across capacities.

- End-to-end calibrated models at a small λ are closer to η than predictive-only models, better calibrated on average and no less sharp.
- The benefit is largest at and beyond the interpolation threshold, where predictive-only training does the most damage.
- Overparameterisation brings no advantage under end-to-end training in this problem, but it no longer brings a disadvantage either.

Two findings concern the evaluation itself rather than the methods. The first is that label-based calibration metrics can be misleading. The degenerate model at $\lambda = 0$ has a top-label ECE comparable to the best end-to-end model while being four to five times further from η. Focal loss with $\gamma = 2$ has the best ECE at $m \geq 100$ while being two to three times further from η than end-to-end training. A model should therefore never be judged by a calibration metric alone; a proper score or its refinement term has to be reported alongside it. The second finding is that selecting λ on the test-set ground truth is optimistic. Selection on 20 validation points costs about 0.002–0.003 in MSE_η.

4.4.5 Limitations

Scope of the simulation. All results come from a single one-dimensional problem with a very smooth η, a fixed training-set size of $n = 100$ and one model family. Because even $m = 10$ represents η almost exactly, the setting cannot show a benefit of capacity. The results should not be extrapolated to problems whose target cannot be represented by small models.

Optimisation. Every experiment uses full-batch gradient descent with one learning rate and one budget of epochs. The central mechanism identified here, the coupling between λ and the effective step size, is specific to optimisers whose step scales with the gradient. Adaptive optimisers such as Adam are approximately invariant to a rescaling of the whole loss. For them the step-size channel should be much weaker, and the relative weight of the two terms should matter more. The soft-ECE gradient uses a straight-through approximation, and its hyperparameters ($B = 15$, $h = 0.08$) were not varied.

Statistical basis. The following rest on a single seed:

- the training dynamics, the reliability diagrams and the curves over the fine λ grid;
- the comparison of base losses;
- the post-hoc comparison, which in addition uses a single split whose variability is large.

The 30-seed experiments use the CE base loss only. The "best" λ in all experiments except ablation C is an oracle choice based on the test-set MSE_η, and the optimum for CE lies at the

Table 4.11: Summary of the answers to the Track 1 experiment questions.

Q	Answer
Q1.1	Yes, the two components are not minimised at the same time. After an early phase of joint improvement, test calibration degrades while test refinement has saturated, clearly so for $m \geq n$.
Q1.2	Yes, with a transition at about epoch 80 for $m \geq n$. The penalty delays and damps it but does not prevent it; at $\lambda \geq 0.1$ this is mostly the slower effective time scale.
Q1.3	Temperature scaling is the strongest post-hoc method and competitive with end-to-end training on ECE for $m \geq n$, but unstable across calibration splits. Isotonic regression and histogram binning fail with 20 calibration points. All post-hoc methods hurt for $m \leq 10$.
Q1.4	TS-refinement has lower bias at every capacity (significant for $m \geq n$). The variance ranking flips: isotonic is lower at $m = 10$, TS at $m = 100$, equal at $m = 1000$.
Q1.5	The raw model degrades from the interpolation threshold on. Temperature scaling becomes less effective and less stable beyond it, but does not collapse.
Q1.6	Brier is better than CE at $\lambda = 1$ at every capacity (single seed), with the gap growing in m; at the respective optimal λ the difference is small.
Q1.7	Not on MSE_η: end-to-end training is two to three times closer to η. On ECE, focal loss with $\gamma = 2$ is better for $m \geq n$.
Q1.8	No: $m = 10$ is numerically best under end-to-end training, but capacity matters little.
Q1.9	Yes in absolute terms (largest reduction at $m = n$), but the step-size control shows the same dependence.
Q1.10	Yes for $\lambda \in [0.1, 1]$ (mean and across-seed variance of MSE_η fall together); not at $\lambda = 0$. At $\lambda = 0.1$ the step-size control reproduces it.
Q1.11	Towards η for $\lambda \in (0, 1)$; the calibration term alone ($\lambda = 0$) produces a useless, near-constant predictor.
Q1.12	Yes: interior optimum near 0.05 (CE) and 0.10–0.15 (Brier), almost independent of m.
A–D	The gain at $\lambda \geq 0.1$ is explained by the reduced step size. The penalty contributes on its own only at $\lambda = 0.05$ and hurts at full step size. Validation-selected λ beats validation early stopping on MSE_η for $m \leq n$. Uncertainty weighting fails in the configuration tested.

edge of the grid. Variances were compared descriptively. The words bias and variance refer to the mean and the across-seed variance of MSE_η, not to a pointwise bias–variance decomposition of the predictor.

Metrics and definitions.

- The top-label ECE and MCE are binned estimators with their own bias; MCE in particular is dominated by sparsely populated bins.
- The calibration term of the cross-entropy used in Figure 4.2 is a binned analogue of the Brier formula.
- The focal loss follows the standard asymmetric form rather than the formula written in the experiment plan.

Comparisons and explanations.

- The post-hoc comparison deliberately gives end-to-end training 100 training points and the post-hoc pipeline 80.
- The uncertainty-weighting experiment uses a single, untuned configuration.
- The explanations offered for ablation B and for the advantage in ablation C are hypotheses that the experiments are consistent with but do not isolate.

4.4.6 Implications for the GRN Experiments

The controlled setting leads to five concrete requirements for the experiments on simulated and real gene regulatory networks.

1. **Baselines.** End-to-end calibrated models have to be compared not only with predictive-only training but also with early stopping and, where the optimiser is not scale-invariant, with a learning-rate-matched control. Otherwise an improvement cannot be attributed to the calibration term.
2. **Selection of λ.** λ has to be chosen on validation data, and the cost of that selection has to be part of the comparison.
3. **Reporting.** Calibration metrics have to be reported together with a proper score and its calibration–refinement decomposition, because ECE alone can rank a degenerate or distorted model as well calibrated. This matters even more for the imbalanced edge-prediction task, where a near-constant prediction of the base rate is a tempting degenerate solution.
4. **Refinement estimation.** Where refinement must be estimated from a small calibration set, TS-refinement is preferable to isotonic refinement.
5. **Held-out calibration term (suggestion).** If the hypothesis behind ablation B is correct, estimating the calibration term on data not used for the proper loss could preserve its regularising effect for large models. This is not tested here.

Together with the rest of the chapter, these points narrow the claim that the thesis can make. Calibration-aware training helps. In the setting where its effect can be measured exactly, however, most of that help comes from how the calibration weight changes the optimisation, and only a smaller part from the calibration term itself.

Bibliography

[1] Chuan Guo, Geoff Pleiss, Yu Sun, and Kilian Q. Weinberger. On calibration of modern neural networks. In *Proceedings of the 34th International Conference on Machine Learning (ICML)*, volume 70 of *PMLR*, pages 1321–1330, 2017. arXiv:1706.04599.

[2] Gongbo Liang, Yu Zhang, Xiaoqin Wang, and Nathan Jacobs. Improved trainable calibration method for neural networks on medical imaging classification. In *Proceedings of the British Machine Vision Conference (BMVC)*, 2020. arXiv:2009.04057.

[3] Matthias Minderer, Josip Djolonga, Rob Romijnders, Frances Hubis, Xiaohua Zhai, Neil Houlsby, Dustin Tran, and Mario Lucic. Revisiting the calibration of modern neural networks. In *Proceedings of the 35th International Conference on Neural Information Processing Systems (NeurIPS)*, volume 34, pages 15682–15694, 2021. arXiv:2106.07998.

[4] Linwei Tao, Younan Zhu, Haolan Guo, Minjing Dong, and Chang Xu. A benchmark study on calibration. In *Proceedings of the International Conference on Learning Representations (ICLR)*, 2024. arXiv:2308.11838.

[5] John C. Platt. Probabilistic outputs for support vector machines and comparisons to regularized likelihood methods. In *Advances in Large Margin Classifiers*, pages 61–74. MIT Press, 1999.

[6] Bianca Zadrozny and Charles Elkan. Transforming classifier scores into accurate multiclass probability estimates. In *Proceedings of the 8th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (KDD)*, pages 694–699, 2002.

[7] Mahdi Pakdaman Naeini, Gregory F. Cooper, and Milos Hauskrecht. Obtaining well calibrated probabilities using bayesian binning. In *Proceedings of the 29th AAAI Conference on Artificial Intelligence (AAAI)*, volume 29, pages 2901–2907, 2015.

[8] Eugène Berta, David Holzmüller, Michael I. Jordan, and Francis Bach. Rethinking early stopping: Refine, then calibrate. *arXiv preprint arXiv:2501.19195*, 2025.

[9] Aviral Kumar, Sunita Sarawagi, and Ujjwal Jain. Trainable calibration measures for neural networks from kernel mean embeddings. In *Proceedings of the 35th International Conference on Machine Learning (ICML)*, volume 80 of *PMLR*, pages 2805–2814, 2018.

[10] Archit Karandikar, Nicholas Cain, Dustin Tran, Balaji Lakshminarayanan, Jonathon Shlens, Michael C. Mozer, and Becca Roelofs. Soft calibration objectives for neural networks. In *Proceedings of the 35th International Conference on Neural Information Processing Systems*

(NeurIPS), volume 34, 2021. arXiv:2108.00106.

[11] Teodora Popordanoska, Raphael Sayer, and Matthew B. Blaschko. A consistent and differentiable L_p canonical calibration error estimator. In *Proceedings of the 36th International Conference on Neural Information Processing Systems (NeurIPS)*, volume 35, pages 7933–7946, 2022. arXiv:2210.07810.

[12] Mehul Damani, Isha Puri, Stewart Slocum, Idan Shenfeld, Leshem Choshen, Yoon Kim, and Jacob Andreas. Beyond binary rewards: Training lms to reason about their uncertainty. In *Proceedings of the International Conference on Learning Representations (ICLR)*, 2026. arXiv:2507.16806.

[13] Nir Friedman, Michal Linial, Iftach Nachman, and Dana Pe'er. Using bayesian networks to analyze expression data. *Journal of Computational Biology*, 7(3-4):601–620, 2000.

[14] Aditya Pratapa, Amogh P. Jalihal, Jeffrey N. Law, Aditya Bharadwaj, and T. M. Murali. Benchmarking algorithms for gene regulatory network inference from single-cell transcriptomic data. *Nature Methods*, 17(2):147–154, 2020.

[15] Payam Dibaeinia and Saurabh Sinha. Sergio: A single-cell expression simulator guided by gene regulatory networks. *Cell Systems*, 11(3):252–271.e11, 2020.

[16] David Maxwell Chickering, David Heckerman, and Christopher Meek. Large-sample learning of bayesian networks is np-hard. *Journal of Machine Learning Research*, 5:1287–1330, 2004. arXiv:1212.2468.

[17] Vân Anh Huynh-Thu, Alexandre Irrthum, Louis Wehenkel, and Pierre Geurts. Inferring regulatory networks from expression data using tree-based methods. *PLoS ONE*, 5(9):e12776, 2010.

[18] Thomas Moerman, Sara Aibar Santos, Carmen Bravo González-Blas, Jaak Simm, Yves Moreau, Jan Aerts, and Stein Aerts. Grnboost2 and arboreto: Efficient and scalable inference of gene regulatory networks. *Bioinformatics*, 35(12):2159–2161, 2019.

[19] Thalia E. Chan, Michael P. H. Stumpf, and Ann C. Babtie. Gene regulatory network inference from single-cell data using multivariate information measures. *Cell Systems*, 5(3):251–267, 2017.

[20] Ye Yuan and Ziv Bar-Joseph. Deep learning for inferring gene relationships from single-cell expression data. *Proceedings of the National Academy of Sciences (PNAS)*, 116(52):27151–27158, 2019.

[21] Guo Chen and Zhi-Ping Liu. Graph attention network for link prediction of gene regulations from single-cell RNA-sequencing data. *Bioinformatics*, 38(19):4522–4529, 2022.

[22] Daniel Marbach, James C. Costello, Robert Küffner, Nicole M. Vega, Robert J. Prill, Diogo M. Camacho, Kyle R. Allison, Manolis Kellis, James J. Collins, and Gustavo Stolovitzky. Wisdom

of crowds for robust gene network inference. *Nature Methods*, 9(8):796–804, 2012.

[23] A. Philip Dawid. The well-calibrated bayesian. *Journal of the American Statistical Association*, 77(379):605–610, 1982.

[24] Meelis Kull, Miquel Perelló Nieto, Markus Kängsepp, Telmo Silva Filho, Hao Song, and Peter Flach. Beyond temperature scaling: Obtaining well-calibrated multi-class probabilities with dirichlet calibration. In *Proceedings of the 33rd International Conference on Neural Information Processing Systems (NeurIPS)*, volume 32, pages 12316–12326, 2019. arXiv:1910.12656.

[25] Juozas Vaicenavicius, David Widmann, Carl Andersson, Fredrik Lindsten, Jacob Roll, and Thomas B. Schön. Evaluating model calibration in classification. In *Proceedings of the 22nd International Conference on Artificial Intelligence and Statistics (AISTATS)*, volume 89 of *PMLR*, 2019. arXiv:1902.06977.

[26] Jeremy Nixon, Michael W. Dusenberry, Linchuan Zhang, Ghassen Jerfel, and Dustin Tran. Measuring calibration in deep learning. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) Workshops*, pages 38–41, 2019. arXiv:1904.01685.

[27] Yaniv Ovadia, Emily Fertig, Jie Ren, Zachary Nado, D. Sculley, Sebastian Nowozin, Joshua V. Dillon, Balaji Lakshminarayanan, and Jasper Snoek. Can you trust your model's uncertainty? evaluating predictive uncertainty under dataset shift. In *Proceedings of the 33rd International Conference on Neural Information Processing Systems (NeurIPS)*, number 1254, pages 14003–140014, 2019.

[28] Hidetoshi Shimodaira. Improving predictive inference under covariate shift by weighting the log-likelihood function. *Journal of Statistical Planning and Inference*, 90(2):227–244, 2000.

[29] Masashi Sugiyama. *Learning under Non-Stationarity: Covariate Shift Adaptation*. Springer Nature Link, 2012.

[30] Masashi Sugiyama and Motoaki Kawanabe. *Machine Learning in Non-Stationary Environments: Introduction to Covariate Shift Adaptation*. MIT Press, 2012.

[31] Morris H. DeGroot and Stephen E. Fienberg. The comparison and evaluation of forecasters. *Journal of the Royal Statistical Society: Series D (The Statistician)*, 32(1):12–22, 1983.

[32] Bianca Zadrozny and Charles Elkan. Learning and making decisions when costs and probabilities are both unknown. In *Proceedings of the Seventh ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pages 204–213. Association for Computing Machinery, 2001.

[33] Ananya Kumar, Percy Liang, and Tengyu Ma. Verified uncertainty calibration. In *Proceedings of the 33rd International Conference on Neural Information Processing Systems (NeurIPS)*, volume 32, pages 3792–3803, 2019. arXiv:1909.10155.

[34] Arsenii Ashukha, Alexander Lyzhov, Dmitry Molchanov, and Dmitry Vetrov. Pitfalls of in-domain uncertainty estimation and ensembling in deep learning. *International Conference on Learning Representations (ICLR)*, 2020. arXiv:2002.06470.

[35] Dan Hendrycks, Mantas Mazeika, and Thomas G. Dietterich. Deep anomaly detection with outlier exposure. In *International Conference on Learning Representations (ICLR)*, 2019. arXiv:1812.04606.

[36] Kartik Gupta, Amir Rahimi, Thalaiyasingam Ajanthan, Thomas Mensink, Cristian Sminchisescu, and Richard Hartley. Calibration of neural networks using splines. In *International Conference on Learning Representations (ICLR)*, 2021. arXiv:2006.12800.

[37] Andreas Kirsch and Yarin Gal. A note on "assessing generalization of sgd via disagreement". *Transactions on Machine Learning Research*, 2022. arXiv:2202.01851.

[38] Tilmann Gneiting and Adrian E. Raftery. Strictly proper scoring rules, prediction, and estimation. *Journal of the American Statistical Association*, 102(477):359–378, 2007.

[39] Claude E. Shannon. A mathematical theory of communication. *The Bell System Technical Journal*, 27(3–4):379–423, 623–656, 1948.

[40] Glenn W. Brier. Verification of forecasts expressed in terms of probability. *Monthly Weather Review*, 78(1):1–3, 1950.

[41] Tsung-Yi Lin, Priya Goyal, Ross Girshick, Kaiming He, and Piotr Dollár. Focal loss for dense object detection. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 42(2):318–327, 2020.

[42] Jochen Bröcker. Reliability, sufficiency, and the decomposition of proper scores. *Quarterly Journal of the Royal Meteorological Society*, 135(643):1512–1519, 2009.

[43] Tilmann Gneiting, Fadoua Balabdaoui, and Adrian E. Raftery. Probabilistic forecasts, calibration and sharpness. *Journal of the Royal Statistical Society: Series B*, 69(2):243–268, 2007.

[44] Alexandru Niculescu-Mizil and Rich Caruana. Predicting good probabilities with supervised learning. In *Proceedings of the 22nd International Conference on Machine Learning (ICML)*, pages 625–632, 2005.

[45] Meelis Kull, Telmo M. Silva Filho, and Peter Flach. Beyond sigmoids: How to obtain well-calibrated probabilities from binary classifiers with beta calibration. *Electronic Journal of Statistics*, 11(2):5052–5080, 2017.

[46] Bianca Zadrozny and Charles Elkan. Obtaining calibrated probability estimates from decision trees and naive bayesian classifiers. In *Proceedings of the 18th International Conference on Machine Learning (ICML)*, pages 609–616, 2001.

[47] Rafael Müller, Simon Kornblith, and Geoffrey Hinton. When does label smoothing help? In

Proceedings of the 33rd International Conference on Neural Information Processing Systems (NeurIPS), volume 32, pages 4694–4702, 2019. arXiv:1906.02629.

[48] Jishnu Mukhoti, Viveka Kulharia, Amartya Sanyal, Stuart Golodetz, Philip H. S. Torr, and Puneet K. Dokania. Calibrating deep neural networks using focal loss. In *Proceedings of the 34th International Conference on Neural Information Processing Systems (NeurIPS)*, volume 33, pages 15288–15299, 2020. arXiv:2002.09437.

[49] Balaji Lakshminarayanan, Alexander Pritzel, and Charles Blundell. Simple and scalable predictive uncertainty estimation using deep ensembles. In *Proceedings of the 31st International Conference on Neural Information Processing Systems (NeurIPS)*, volume 30, pages 6405–6416, 2017. arXiv:1612.01474.

[50] Vladimir N. Vapnik. *Statistical Learning Theory*. Wiley, 1998.

[51] Shai Shalev-Shwartz and Shai Ben-David. *Understanding Machine Learning: From Theory to Algorithms*. Cambridge University Press, 2014.

[52] Chiyuan Zhang, Samy Bengio, Moritz Hardt, Benjamin Recht, and Oriol Vinyals. Understanding deep learning (still) requires rethinking generalization. *Communications of the ACM*, 64(3):107–115, 2021.

[53] Peter L. Bartlett and Shahar Mendelson. Rademacher and gaussian complexities: Risk bounds and structural results. *Journal of Machine Learning Research*, 3:463–482, 2002.

[54] David A. McAllester. Pac-bayesian model averaging. In *Proceedings of the 12th Annual Conference on Computational Learning Theory (COLT)*, pages 164–170, 1999.

[55] Gintare Karolina Dziugaite and Daniel M. Roy. Computing nonvacuous generalization bounds for deep (stochastic) neural networks with many more parameters than training data. In *Proceedings of the 33rd Conference on Uncertainty in Artificial Intelligence (UAI)*, 2017. arXiv:1703.11008.

[56] Behnam Neyshabur, Srinadh Bhojanapalli, and Nathan Srebro. A pac-bayesian approach to spectrally-normalized margin bounds for neural networks. In *Proceedings of the International Conference on Learning Representations (ICLR)*, 2018. arXiv:1707.09564.

[57] Geoff Pleiss, Manish Raghavan, Felix Wu, Jon Kleinberg, and Kilian Q. Weinberger. On fairness and calibration. In *Proceedings of the 31st International Conference on Neural Information Processing Systems (NeurIPS)*, pages 5684–5693, 2017. arXiv:1709.02012.

[58] Hirotaka Matsumoto, Hisanori Kiryu, Chikara Furusawa, Minoru S. H. Ko, Shigeru B. H. Ko, Norio Gouda, Tetsutaro Hayashi, and Itoshi Nikaido. Scode: An efficient regulatory network inference algorithm from single-cell RNA-Seq during differentiation. *Bioinformatics*, 33(15):2314–2321, 2017.

[59] Adam A. Margolin, Ilya Nemenman, Katia Basso, Chris Wiggins, Gustavo Stolovitzky,

Riccardo Dalla Favera, and Andrea Califano. Aracne: An algorithm for the reconstruction of gene regulatory networks in a mammalian cellular context. *BMC Bioinformatics*, 7(S7), 2006.

[60] Jeremiah J. Faith, Boris Hayete, Joshua T. Thaden, Ilaria Mogno, Jamey Wierzbowski, Guillaume Cottarel, Simon Kasif, James J. Collins, and Timothy S. Gardner. Large-scale mapping and validation of Escherichia coli transcriptional regulation from a compendium of expression profiles. *PLoS Biology*, 5(1), 2007.

[61] Daphne Koller and Nir Friedman. *Probabilistic Graphical Models: Principles and Techniques*. MIT Press, 2009.

[62] Judea Pearl. *Causality: Models, Reasoning, and Inference*. Cambridge University Press, 2nd edition, 2009.

[63] Peter Spirtes, Clark Glymour, and Richard Scheines. *Causation, Prediction, and Search*. MIT Press, 2nd edition, 2000.

[64] Nan Papili Gao, S. M. Minhaz Ud-Dean, Olivier Gandrillon, and Rudiyanto Gunawan. Sincerities: Inferring gene regulatory networks from time-stamped single cell transcriptional expression profiles. *Bioinformatics*, 34(2):258–266, 2018.

[65] Sara Aibar, Carmen Bravo González-Blas, Thomas Moerman, Vân Anh Huynh-Thu, Hana Imrichova, Gert Hulselmans, Florian Rambow, Jean-Christophe Marine, Pierre Geurts, Jan Aerts, Joost van den Oord, Zeynep Kalender Atak, Jasper Wouters, and Stein Aerts. Scenic: Single-cell regulatory network inference and clustering. *Nature Methods*, 14(11):1083–1086, 2017.

[66] Hantao Shu, Jingtian Zhou, Qiuyu Lian, Han Li, Dan Zhao, Jianyang Zeng, and Jianzhu Ma. Modeling gene regulatory networks using neural network architectures. *Nature Computational Science*, 1(7):491–501, 2021.

[67] Thomas Schaffter, Daniel Marbach, and Dario Floreano. Genenetweaver: In silico benchmark generation and performance profiling of network inference methods. *Bioinformatics*, 27(16):2263–2270, 2011.

[68] Alex Kendall, Yarin Gal, and Roberto Cipolla. Multi-task learning using uncertainty to weigh losses for scene geometry and semantics. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2018. arXiv:1705.07115.